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样品承认书

☑ 新物料封样 □ 旧物料更改新封样

品　　号	121002462	承认日期	2026/7/22
品名规格	Charger IC BQ21128RYKR WQFN18 TI RZ专用 G	版　　本	1.0
供 应 商	TI/艾睿ARROW	材　　料	

检验说明:
附：请依样品与图纸检测
“*”尺寸为FAI尺寸，测量工具参照《FAI/CPK尺寸检验报告》

1、必须符合 G-TECH 《有害物质限制标准》 GT-STN-001 要求，以及客人的环保标准。
2、丝印检查：第71页　丝印：BQ128
3、芯片封装检查：第74页

□ 颜色确认
☑ 规格/性能/参数确认
□ 结构确认
□ 尺寸确认
□ Cpk 确认
□ 镭雕确认
□ 表面喷涂确认
□ 信赖性/可靠性测试
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□ 材质证明
☑ RoHS / HF 确认
□ 包装确认
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更改原因:

回收说明:

样品:

平面设计: _____

结构工程师: _____

电子工程师: 梁广燊

审　　核: 陈岳衡

批　　准: 田冬林

珠海市智迪科技股份有限公司RoHS检测报告

No	样品名称	分析组	操作者	测试时间	Cd			Pb			Hg			Cr			Br				
					ppm	sigma	Judgment	ppm	sigma	Judgment	ppm	sigma	Judgment	ppm	sigma	Judgment	ppm	sigma	Judgment	最终判定	送检人
202607100155	TI Charger IC121002462	ScreeningFast不测卤	谢丽梅	2026/7/11	ND	5.2	OK	ND	15.0	OK	ND	ND	OK	ND	20.60	OK	ND	ND	OK	OK	曾泳

说明：

1、当镉（Cd）、汞（Hg）、铅（Pb）EDX-LE检测超标时，需请申请部门送第三方检测机构作进一步检测/分析。

2、Cr含量超标时，由试验员用铬反应试剂（HACH12710-99）进行定性分析检测后作出最终判定。

3、Br 超过限定标准值时， 申请部门/供应商需送第三方检测 PBBs/PBDEs

确认者：谢丽梅

审核：刘福良

BQ21128 I²C Controlled, 1.5A, Maximum 18V Input, Charger with NVDC Power Path Management

1 Features

- High-efficiency, 1.5MHz, synchronous switching mode buck charger for single cell battery
  - >90% efficiency down to 25mA output current from 5V input
  - Charge termination from 5mA to 310mA, 5mA steps . Register option for up to 620mA termination.
  - Flexible JEITA profile for safe charging over temperature
- BATFET control to support shutdown, ship mode and full system reset
  - 1.5µA quiescent current in battery only mode
  - 0.15µA battery leakage current in ship mode
  - 0.1µA battery leakage current in shutdown
- Supports Boost Mode operation to power accesory
  - Boost Mode operation supporting 3.84V to 5.2V output
  - >90% boost efficiency down to 100mA boost current for 5V PMID
- Supports a wide range of input sources
  - 3.9V to 18V wide input operating voltage range with 26V absolute maximum input voltage
  - Maximizes source power with input voltage regulation (VINDPM) and input current regulation (IINDPM)
  - VINDPM threshold automatically tracks battery voltage
- Efficient battery operation with 15mΩ BATFET
- Narrow VDC (NVDC) power path management
  - System instant-on with depleted or no battery
  - Battery supplement when adapter is fully loaded
- Flexible autonomous or I²C-controlled modes
- Out of Audio mode to limit minimum PFM frequency outside of audible range
- Integrated 12-bit ADC for voltage, current, temperature monitoring
- High accuracy
  - ±0.4% charge voltage regulation
  - ±5% charge current regulation
  - ±5% input current regulation
- Safety
  - Thermal regulation and thermal shutdown
  - Input, system, and battery overvoltage protection
  - Battery, and converter overcurrent protection
  - Charging safety timer

2 Applications

- Consumer wearables, Smartwatch
- Portable speakers, TWS earphone
- Hearing aid or TWS charging case

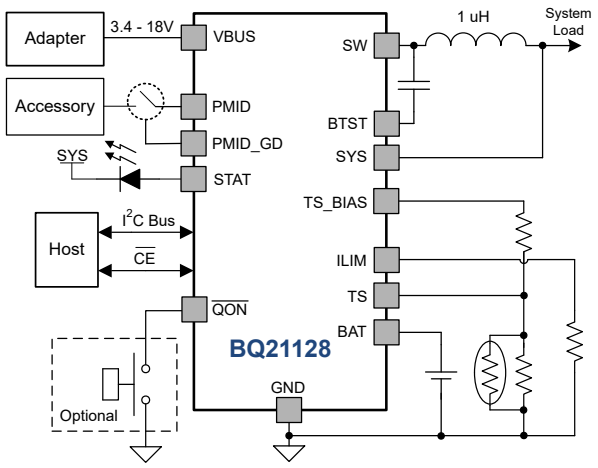
3 Description

The BQ21128 is a highly-integrated switch-mode battery charge management and system power path management devices for single cell Li-Ion and Li-polymer batteries. The solution is highly integrated with built-in current sensing, loop compensation, input reverse-blocking FET (RBFET, Q1), high-side switching FET (HSFET, Q2), low-side switching FET (LSFET, Q3), and battery FET (BATFET, Q4) between system and battery. The device uses narrow VDC power path management, regulating the system slightly above the battery voltage without dropping below a configurable minimum system voltage. The low impedance power path optimizes switch-mode operation efficiency, reduces battery charging time, extends battery life during discharging phase, and the ultra-low 0.15µA ship mode current extends battery shelf life. The I²C serial interface with charging and system settings makes the BQ21128 a truly flexible solution.

Package Information

PART NUMBER	PACKAGE(1)	PACKAGE SIZE(2)
BQ21128	RYK (WQFN 18)	2.50mm × 3.00mm

(1) For all available packages, see the orderable addendum at the end of the data sheet.
(2) The package size (length × width) is a nominal value and includes pins, where applicable.



BQ21128 Simplified Application

An IMPORTANT NOTICE at the end of this data sheet addresses availability, warranty, changes, use in safety-critical applications, intellectual property matters and other important disclaimers. PRODUCTION DATA.

The BQ21128 supports a wide range of input sources, including standard USB host port, USB charging port, and USB compliant high voltage adapter.The BQ21128 has ILIM pin to set the default input current limit and TS_BIAS pin for controlled thermistor bias. The device is compliant with USB 2.0 and USB 3.0 power specifications for input current and voltage regulation up-to 1.5A.

The power path management regulates the system slightly above battery voltage but does not drop below the programmable minimum system voltage with a recommended maximum system load of 2A. With this feature, the system maintains operation even when the battery is completely depleted or removed. When the input current limit or input voltage limit is reached, the power path management automatically reduces the charge current. As the system load continues to increase, the battery starts to discharge until the system power requirement is met. This supplement mode prevents overloading the input source.

The BQ21128 initiates and completes a charging cycle without host control. By sensing the battery voltage, it charges the battery in four different phases: trickle charge, pre-charge, constant current (CC) charge and constant voltage (CV) charge. At the end of the charging cycle, the charger automatically terminates when the charge current is below a preset threshold and the battery voltage is higher than the recharge threshold. Termination is supported for all TS pin temperature zones.

The BQ21128 provides various safety features for battery charging and system operations, including battery negative temperature coefficient thermistor monitoring, charging safety timer and over-voltage and over-current protections. The thermal regulation reduces charge current when the junction temperature exceeds the programmable threshold. The STAT output reports the charging status and any fault conditions. Other safety features include battery temperature sensing for charge mode, thermal shutdown and input UVLO and over-voltage protection. The $\overline{\text{INT}}$ output notifies the host when a fault occurs or status changes.

The BQ21128 is available in a 18-pin, 2.5mm × 3.0mm WQFN package.

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4 Pin Configuration and Functions

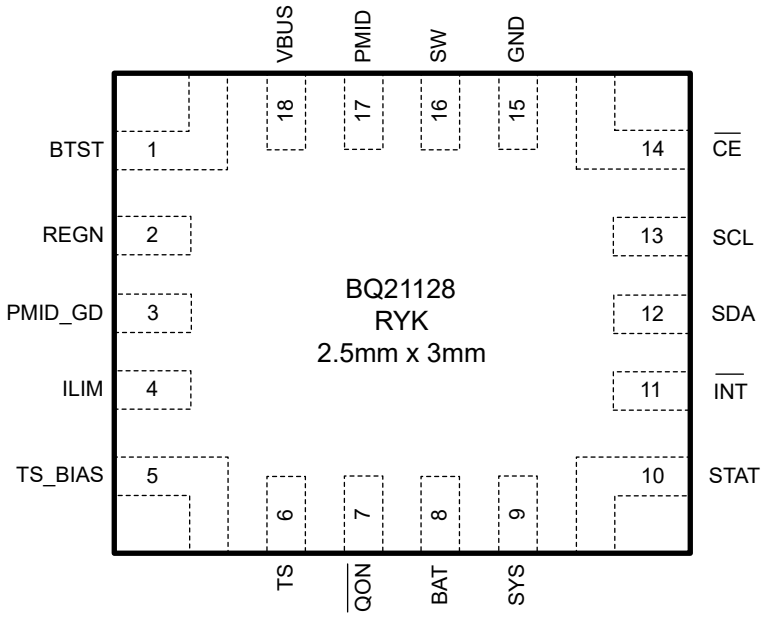


Figure 4-1. BQ21128 Pinout, 18-Pin WQFN Top View

Table 4-1. Pin Functions

NAME	NO.	TYPE[1]	DESCRIPTION
BTST	1	P	**High Side Switching MOSFET Gate Driver Power Supply –** Connect a 10V or higher rating, 47nF ceramic capacitor between SW and BTST as the bootstrap capacitor for driving high side switching MOSFET (Q2).
REGN	2	P	**The Charger Internal Linear Regulator Output –** Internally, REGN is connected to the anode of the boost-strap diode. Connect a 10V or higher rating, 4.7μF ceramic capacitor from REGN to power ground, The capacitor should be placed close to the IC. The REGN LDO output is used for the internal MOSFETs gate driving voltage and for biasing the external TS pin thermistor.
PMID_GD	3	DO	**Open Drain Active High PMID Good Indicator –** Connect to the pull up rail REGN through 10 kΩ resistor. HIGH indicates PMID output voltage is good. This signal can be used to drive external PMOS FET to disconnect the PMID under charging load when boost mode output voltage is too high or output current is too high.
ILIM	4	AIO	**Input Current Limit Setting Input Pin –** ILIM pin sets the input current limit as I_{INREG} = K_{ILIM} / R_{ILIM}, where R_{ILIM} is connected from ILIM pin to GND. The input current is limited to the lower of the two values set by ILIM pin and IINDPM register bits. The ILIM pin can also be used to monitor input current. The input current is proportional to the voltage on ILIM pin and can be calculated by I_{IN} = (K_{ILIM} x V_{ILIM}) / (R_{ILIM} x 0.8). The ILIM pin function is disabled when EN_EXTILIM bit is set to 0.
TS_BIAS	5	P	**Bias for the TS Resistor Voltage Divider –** Provides the bias voltage for the TS resistor voltage divider.
TS	6	AI	**Temperature Qualification Voltage Input –** Connect a negative temperature coefficient thermistor. Program temperature window with a resistor divider from TS pin bias reference (REGN) to TS, then to GND. Charge suspends when TS pin voltage is out of range. Recommend a 103AT-2 10kΩ thermistor.
$\overline{QON}$	7	DI	**BATFET Enable or System Power Reset Control Input –** If the charger is in ship mode, a logic low on this pin with t_{SM_EXIT} duration forces the device to exit ship mode. If the charger is not in ship mode, a logic low on this pin with t_{RST} initiates a full system power reset if either V_{VBUS} < V_{VBUS_UVLO} or BATFET_CTRL_WVBUS = 1. $\overline{QON}$ has no effect during shutdown mode. The pin contains an internal pull-up to maintain default high logic.
BAT	8	P	**The Battery Charging Power Connection –** Connect to the positive terminal of the battery pack. The internal BATFET is connected between SYS and BAT.
SYS	9	P	**The Charger Output Voltage to System –**The Buck converter output connection point to the system. The internal BATFET is connected between SYS and BAT.

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Table 4-1. Pin Functions (continued)

NAME	NO.	TYPE(1)	DESCRIPTION
STAT	10	DO	**Open Drain Charge Status Output –** It indicates various charger operations. Connect to the pull up rail via 10kΩ resistor. LOW indicates charging in progress. HIGH indicates charging completed or charging disabled. When any fault condition occurs, STAT pin blinks at 1Hz. Setting DIS_STAT = 1 disables the STAT pin function, causing the pin to be pulled HIGH. Leave floating if unused.
$\overline{INT}$	11	DO	**Open Drain Interrupt Output. –** Connect to the pull up rail via 10kΩ resistor. The $\overline{INT}$ pin sends an active low, 256µs pulse to the host to report the charger device status and faults.
SDA	12	DIO	**I²C Interface Data –** Connect SDA to the logic rail through a 10 kΩ resistor.
SCL	13	DI	**I²C Interface Clock –** Connect SCL to the logic rail through a 10 kΩ resistor.
$\overline{CE}$	14	DI	**Active Low Charge Enable Pin –** Battery charging is enabled when EN_CHG bit is 1 and $\overline{CE}$ pin is LOW. $\overline{CE}$ pin must be pulled HIGH or LOW, do not leave floating.
GND	15	P	**Ground Return**
SW	16	P	**Switching Node Connecting to Output Inductor –** Internally SW is connected to the source of the n-channel HSFET and the drain of the n-channel LSFET. Connect the 47 nF bootstrap capacitor from SW to BTST.
PMID	17	P	**HSFET Drain Connection –** Internally PMID is connected to the drain of the reverse blocking MOSFET (RBFET) and the drain of HSFET.
VBUS	18	P	**Charger Input Voltage –** The internal n-channel reverse block MOSFET (RBFET) is connected between VBUS and PMID with VBUS on source.

(1) AI = Analog input, AO = Analog Output, AIO = Analog input Output, DI = Digital input, DO = Digital Output, DIO = Digital input Output, P = Power

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5 Specifications

5.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted)(1)

		MIN	MAX	UNIT
Voltage range (with respect to GND)	VBUS (converter not switching)	–2	26	V
	PMID (converter not switching)	–0.3	26	V
	BAT, SYS (converter not switching)	–0.3	6	V
	SW	–2 (50ns)	21	V
	BTST (when converter switching)	–0.3	27	V
	$\overline{CE}$, STAT, SCL, SDA, $\overline{INT}$, REGN, $\overline{QON}$	–0.3	6	V
	ILIM, $\overline{PG}$, TS, TS_BIAS	–0.3	6	V
Output Sink Current	$\overline{INT}$, STAT, $\overline{PG}$		6	mA
Differential Voltage	BTST-SW	–0.3	6	V
	PMID-VBUS	–0.3	6	V
	SYS-BAT	–0.3	6	V
T_J	Junction temperature	–40	150	°C
T_{stg}	Storage temperature	–55	150	°C

(1) Stresses beyond those listed under *Absolute Maximum Rating* may cause permanent damage to the device. These are stress ratings only, which do not imply functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Condition*. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

5.2 ESD Ratings

			VALUE	UNIT
$V_{(ESD)}$	Electrostatic discharge	Human body model (HBM), per ANSI/ESDA/JEDEC JS-001, all pins(1)	±2000	V
		Charged device model (CDM), per JEDEC specification JESD22-C101, all pins(2)	±250	

(1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.
(2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process.

5.3 Recommended Operating Conditions

over operating free-air temperature range (unless otherwise noted)

		MIN	NOM	MAX	UNIT
V_{VBUS}	Input voltage	3.9		18	V
V_{BAT}	Battery voltage			4.8	V
I_{VBUS}	Input current			3.2	A
I_{SW}	Output current (SW)			3.5	A
I_{BAT}	Fast charging current			1.5	A
	RMS discharge current (continuously)			6	A
	Peak discharge current (up to 50ms)			10	A
I_{REGN}	Maximum REGN Current			20	mA
T_A	Ambient temperature	–40		85	°C
T_J	Junction temperature	–40		125	°C
L_{SW}	Inductor for the switching regulator	0.68		2.2	µH
C_{VBUS}	VBUS capacitor (without de-rating)	1			µF
C_{PMID}	PMID capacitor (without de-rating)	10			µF
C_{SYS}	SYS capacitor (without de-rating)	20		500	µF

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5.3 Recommended Operating Conditions (continued)

over operating free-air temperature range (unless otherwise noted)

		MIN	NOM	MAX	UNIT
C_{BAT}	BAT capacitor (without de-rating)	10			µF

5.4 Thermal Information

THERMAL METRIC[1]		BQ21128 RYK (QFN) 18 pins	UNIT
$R_{\theta JA}$	Junction-to-ambient thermal resistance	60.1	°C/W
$R_{\theta JC(top)}$	Junction-to-case (top) thermal resistance	42.1	°C/W
$R_{\theta JB}$	Junction-to-board thermal resistance	13.0	°C/W
Ψ_{JT}	Junction-to-top characterization parameter	1.3	°C/W
Ψ_{JB}	Junction-to-board characterization parameter	12.8	°C/W

(1) For more information about traditional and new thermal metrics, see the Semiconductor and IC Package Thermal Metrics application report.

5.5 Electrical Characteristics

$V_{VBUS_UVLOZ} < V_{VBUS} < V_{VBUS_OVP}$, T_J = -40°C to +125°C, and T_J = 25°C for typical values (unless otherwise noted)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
QUIESCENT CURRENTS						
I_{Q_BAT}	Quiescent battery current (BAT, SYS, SW) when the charger is in the battery only mode, BATFET is enabled, ADC is disabled	VBAT = 4V, No VBUS, BATFET is enabled, I2C enabled, ADC disabled, system is powered by battery. -40 °C < T_J < 60 °C		1.5	3	µA
$I_{Q_BAT_ADC}$	Quiescent battery current (BAT, SYS, SW) when the charger is in the battery only mode, BATFET is enabled, ADC is enabled	VBAT = 4V, No VBUS, BATFET is enabled, I2C enabled, ADC enabled, system is powered by battery. -40 °C < T_J < 60 °C		260		µA
$I_{Q_BAT_SD}$	Quiescent battery current (BAT) when the charger is in shutdown mode, BATFET is disabled, ADC is disabled	VBAT = 4V, No VBUS, BATFET is disabled, I2C disabled, in shutdown mode, ADC disabled, T_J < 60 °C		0.1	0.2	µA
$I_{Q_BAT_SHIP}$	Quiescent battery current (BAT) when the charger is in ship mode, BATFET is disabled, ADC is disabled	VBAT = 4V, No VBUS, BATFET is disabled, I2C disabled, in ship mode, ADC disabled, T_J < 60 °C		0.15	0.5	µA
I_{Q_VBUS}	Quiescent input current (VBUS)	VBUS = 5V, VBAT = 4V, charge disabled, converter switching, ISYS = 0A, PFM enabled		450		µA
$I_{Q_VBUS_HIZ}$	Quiescent input current (VBUS) in HIZ	VBUS = 5V, VBAT = 4V, HIZ mode, ADC disabled		5	20	µA
		VBUS = 15V, VBAT = 4V, HIZ mode, ADC disabled		20	35	µA
VBUS / VBAT SUPPLY						
V_{VBUS_OP}	VBUS operating range		3.9		18	V
V_{VBUS_UVLO}	VBUS falling to turn off I2C, no battery	VBUS falling	3.0	3.15	3.3	V
V_{VBUS_UVLOZ}	VBUS rising for active I2C, no battery	VBUS rising	3.2	3.35	3.5	V
V_{VBUS_OVP}	VBUS overvoltage rising threshold	VBUS rising, VBUS_OVP = 0	6.1	6.4	6.7	V
V_{VBUS_OVPZ}	VBUS overvoltage falling hreshold	VBUS rising, VBUS_OVP = 0	5.8	6.0	6.2	V

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5.5 Electrical Characteristics (continued)

V_{VBUS_UVLOZ} < V_{VBUS} < V_{VBUS_OVP}, T_J = -40°C to +125°C, and T_J = 25°C for typical values (unless otherwise noted)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
V_{VBUS_OVP}	VBUS overvoltage rising threshold	VBUS rising, VBUS_OVP = 1	18.2	18.5	18.8	V
V_{VBUS_OVPZ}	VBUS overvoltage falling threshold	VBUS falling, VBUS_OVP = 1	17.4	17.7	18.0	V
V_{SLEEP}	Enter Sleep mode threshold	(VBUS - VBAT), VBUS falling	9	45	85	mV
V_{SLEEPZ}	Exit Sleep mode threshold	(VBUS - VBAT), VBUS rising	115	220	340	mV
V_{BAT_UVLOZ}	BAT voltage for active I2C, turn on BATFET, no VBUS	VBAT rising	2.3	2.4	2.5	V
V_{BAT_UVLO}	BAT voltage to turnoff I2C, turn off BATFET, no VBUS	VBAT falling, VBAT_UVLO = 0	2.1	2.2	2.3	V
		VBAT falling, VBAT_UVLO = 1	1.7	1.8	1.9	V
$V_{POORSRC}$	Bad adapter detection threshold	VBUS falling	3.6	3.7	3.75	V
$I_{POORSRC}$	Bad adapter detection current source			10		mA
POWER-PATH MANAGEMENT						
$V_{SYS_REG_ACC}$	Typical system voltage regulation	ISYS = 0A, VBAT > VSYSMIN, Charge Disabled. Offset above VBAT		50		mV
		ISYS = 0A, V_{BAT} < VSYSMIN, Charge Disabled. Offset above VSYSMIN		230		mV
V_{SYSMIN_RNG}	VSYSMIN register range		2.56		3.84	V
$V_{SYSMIN_REG_STEP}$	VSYSMIN register step size			80		mV
$V_{SYSMIN_REG_ACC}$	Minimum DC system voltage output	ISYS = 0A, V_{BAT} < VSYSMIN = B00h (3.52V), Charge Disabled	3.52	3.75		V
V_{SYS_SHORT}	VSYS short voltage falling threshold to enter forced PFM			0.9		V
V_{SYS_SHORTZ}	VSYS short voltage rising threshold to exit forced PFM			1.1		V
BATTERY CHARGER						
V_{REG_RANGE}	Typical charge voltage regulation range		3.50		4.80	V
V_{REG_STEP}	Typical charge voltage step			10		mV
V_{REG_ACC}	Charge voltage accuracy	T_J = 25°C	–0.3		0.3	%
		T_J = –10°C - 85°C	–0.4		0.4	%
V_{BAT_SHORTZ}	Battery short voltage rising threshold to start pre-charge	VBAT rising		2.25		V
V_{BAT_SHORT}	Battery short voltage falling threshold to stop pre-charge	VBAT falling, VBAT_UVLO=0		2.05		V
V_{BAT_SHORT}	Battery short voltage falling threshold to stop pre-charge	VBAT falling, VBAT_UVLO=1		1.85		V
V_{BAT_LOWVZ}	Battery voltage rising threshold	Transition from pre-charge to fast charge	2.9	3.0	3.1	V
V_{BAT_LOWV}	Battery voltage falling threshold	Transition from fast charge to pre-charge	2.7	2.8	2.9	V
V_{RECHG}	Battery recharge threshold below V_{REG}	VBAT falling, VRECHG = 0		100		mV
		VBAT falling, VRECHG = 1		200		mV
I_{PMID_LOAD}	PMID discharge load current		20	30		mA
I_{BAT_LOAD}	Battery discharge load current		20	30		mA
I_{SYS_LOAD}	System discharge load current		20	30		mA
BATFET						
R_{BATFET}	MOSFET on resistance from SYS to BAT			15	25	mΩ
BATTERY PROTECTIONS						

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5.5 Electrical Characteristics (continued)

V_{VBUS_UVLOZ} < V_{VBUS} < V_{VBUS_OVP}, T_J = -40°C to +125°C, and T_J = 25°C for typical values (unless otherwise noted)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
V_{BAT_OVP}	Battery overvoltage rising threshold	As percentage of VREG	103	104	105	%
V_{BAT_OVPZ}	Battery overvoltage falling threshold	As percentage of VREG	101	102	103	%
I_{BATFET_OCP}	BATFET over-current rising threshold		6			A
I_{BAT_PK}	Battery discharging peak current rising threshold	IBAT_PK = 10	6			A
		IBAT_PK = 11	12			A
INPUT VOLTAGE / CURRENT REGULATION						
V_{INDPM_RANGE}	Typical input voltage regulation range		3.8		16.8	V
V_{INDPM_STEP}	Typical input voltage regulation step			40		mV
V_{INDPM_ACC}	Input voltage regulation accuracy	VINDPM=4.6V	–4		4	%
		VINDPM=8V	–3		3	%
		VINDPM=16V	–2		2	%
$V_{INDPM_BAT_TRACK}$	Battery tracking VINDPM accuracy	VBAT = 3.9V, VINDPM_BAT_TRACK=1, VINDPM = 4V		4.15		V
I_{INDPM_RANGE}	Typical input current regulation range		0.1		3.2	A
I_{INDPM_STEP}	Typical input current regulation step			20		mA
I_{INDPM_ACC}	Input current regulation accuracy	IINDPM = 500mA, VBUS=5V	450	475	500	mA
		IINDPM = 900mA, VBUS=5V	810	855	900	mA
		IINDPM = 1500mA, VBUS=5V	1350	1425	1500	mA
K_{ILIM}	ILIM Pin Scale Factor, IINREG = K_{ILIM} / R_{ILIM}	INREG = 1.6 A	2250	2500	2750	AΩ
THERMAL REGULATION AND THERMAL SHUTDOWN						
T_{REG}	Junction temperature regulation accuracy	TREG = 1		120		°C
		TREG = 0		60		°C
T_{SHUT}	Thermal Shutdown Rising Threshold	Temperature Increasing		140		°C
T_{SHUT_HYS}	Thermal Shutdown Falling Hysteresis	Temperature Decreasing by T_{SHUT_HYS}		30		°C
THERMISTOR COMPARATORS (CHARGE MODE)						
V_{TS_COLD}	TS pin rising voltage threshold for TH1 comparator to transition from TS_COOL to TS_COLD. Charge suspended above this voltage.	As Percentage to TS pin bias reference (-5°C w/ 103AT), TS_TH1_TH2_TH3 = 100, 101, 110	75.0	75.5	76.0	%
		As Percentage to TS pin bias reference (0°C w/ 103AT), TS_TH1_TH2_TH3 = 000, 001, 010, 011, 111	72.8	73.3	73.8	%
V_{TS_COLDZ}	TS pin falling voltage threshold for TH1 comparator to transition from TS_COLD to TS_COOL. TS_COOL charge settings resume below this voltage.	As Percentage to TS pin bias reference (-2.5°C w/ 103AT), TS_TH1_TH2_TH3 = 100, 101, 110	73.9	74.4	74.9	%
		As Percentage to TS pin bias reference (2.5°C w/ 103AT), TS_TH1_TH2_TH3 = 000, 001, 010, 011, 111	71.7	72.2	72.7	%

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5.5 Electrical Characteristics (continued)

V_{VBUS_UVLOZ} < V_{VBUS} < V_{VBUS_OVP}, T_J = -40°C to +125°C, and T_J = 25°C for typical values (unless otherwise noted)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
V_{TS_COOL}	TS pin rising voltage threshold for TH2 comparator to transition from TS_PRECOOL to TS_COOL. TS_COOL charging settings used above this voltage.	As Percentage to TS pin bias reference (5°C w/ 103AT), TS_TH1_TH2_TH3 = 000, 100	70.6	71.1	71.6	%
		As Percentage to TS pin bias reference (10°C w/ 103AT), TS_TH1_TH2_TH3 = 001, 101, 110, 111	67.9	68.4	68.9	%
		As Percentage to TS pin bias reference (15°C w/ 103AT), TS_TH1_TH2_TH3 = 010	65.0	65.5	66.0	%
		As Percentage to TS pin bias reference (20°C w/ 103AT), TS_TH1_TH2_TH3 = 011	61.9	62.4	62.9	%
V_{TS_COOLZ}	TS pin falling voltage threshold for TH2 comparator to transition from TS_COOL to TS_PRECOOL. TS_PRECOOL charging settings resume below this voltage.	As Percentage to TS pin bias reference (7.5°C w/ 103AT), TS_TH1_TH2_TH3 = 000, 100	69.3	69.8	70.3	%
		As Percentage to TS pin bias reference (12.5°C w/ 103AT), TS_TH1_TH2_TH3 = 001, 101, 110, 111	66.6	67.1	67.6	%
		As Percentage to TS pin bias reference (17.5°C w/ 103AT), TS_TH1_TH2_TH3 = 010	63.7	64.2	64.7	%
		As Percentage to TS pin bias reference (22.5°C w/ 103AT), TS_TH1_TH2_TH3 = 011	60.6	61.1	61.6	%
$V_{TS_PRECOOL}$	TS pin rising voltage threshold for TH3 comparator to transition from TS_NORMAL to TS_PRECOOL. TS_PRECOOL charge settings used above this voltage.	As Percentage to TS pin bias reference (15°C w/ 103AT), TS_TH1_TH2_TH3 = 000, 001, 100, 101	65.0	65.5	66.0	%
		As Percentage to TS pin bias reference (20°C w/ 103AT), TS_TH1_TH2_TH3 = 010, 011, 110, 111	61.9	62.4	62.9	%
$V_{TS_PRECOOLZ}$	TS pin falling voltage threshold for TH3 comparator to transition from TS_PRECOOL to TS_NORMAL. Normal charging resumes below this voltage.	As Percentage to TS pin bias reference (17.5°C w/ 103AT), TS_TH1_TH2_TH3 = 000, 001, 100, 101	63.7	64.2	64.7	%
		As Percentage to TS pin bias reference (22.5°C w/ 103AT), TS_TH1_TH2_TH3 = 010, 011, 110, 111	60.6	61.1	61.6	%
$V_{TS_PREWARM}$	TS pin falling voltage threshold for TH4 comparator to transition from TS_NORMAL to TS_PREWARM. TS_PREWARM charging settings used below this voltage.	As Percentage to TS pin bias reference (35°C w/ 103AT), TS_TH4_TH5_TH6 = 000, 001, 010, 100, 101	51.5	52.0	52.5	%
		As Percentage to TS pin bias reference (40°C w/ 103AT), TS_TH4_TH5_TH6 = 011, 110, 111	47.9	48.4	48.9	%
$V_{TS_PREWARMZ}$	TS pin rising voltage threshold for TH4 comparator to transition from TS_PREWARM to TS_NORMAL. Normal charging resumes above this voltage.	As Percentage to TS pin bias reference (32.5°C w/ 103AT), TS_TH4_TH5_TH6 = 000, 001, 010, 100, 101	53.3	53.8	54.3	%
		As Percentage to TS pin bias reference (37.5°C w/ 103AT), TS_TH4_TH5_TH6 = 011, 110, 111	49.2	49.7	50.2	%

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5.5 Electrical Characteristics (continued)

V_{VBUS_UVLOZ} < V_{VBUS} < V_{VBUS_OVP}, T_J = -40°C to +125°C, and T_J = 25°C for typical values (unless otherwise noted)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
V_{TS_WARM}	TS pin falling voltage threshold for TH5 comparator to transition from TS_PREWARM to TS_WARM. TS_WARM charging settings used below this voltage.	As Percentage to TS pin bias reference (40°C w/ 103AT), TS_TH4_TH5_TH6 = 000, 100	47.9	48.4	48.9	%
		As Percentage to TS pin bias reference (45°C w/ 103AT), TS_TH4_TH5_TH6 = 001, 101, 110	44.3	44.8	45.3	%
		As Percentage to TS pin bias reference (50°C w/ 103AT), TS_TH4_TH5_TH6 = 010, 111	40.7	41.2	41.7	%
		As Percentage to TS pin bias reference (55°C w/ 103AT), TS_TH4_TH5_TH6 = 011	37.2	37.7	38.2	%
V_{TS_WARMZ}	TS pin rising voltage threshold for TH5 comparator to transition from TS_WARM to TS_PREWARM. TS_PREWARM charging settings resume above this voltage.	As Percentage to TS pin bias reference (37.5°C w/ 103AT), TS_TH4_TH5_TH6 = 000, 100	49.2	49.7	50.2	%
		As Percentage to TS pin bias reference (42.5°C w/ 103AT), TS_TH4_TH5_TH6 = 001, 101, 110	45.6	46.1	46.6	%
		As Percentage to TS pin bias reference (47.5°C w/ 103AT), TS_TH4_TH5_TH6 = 010, 111	42.0	42.5	43.0	%
		As Percentage to TS pin bias reference (52.5°C w/ 103AT), TS_TH4_TH5_TH6 = 011	38.5	39	39.5	%
V_{TS_HOT}	TS pin falling voltage threshold for TH6 comparator to transition from TS_WARM to TS_HOT. Charging is suspended below this voltage.	As Percentage to TS pin bias reference (50°C w/ 103AT), TS_TH4_TH5_TH6 = 100 or 101	40.7	41.2	41.7	%
		As Percentage to TS pin bias reference (60°C w/ 103AT), TS_TH4_TH5_TH6 = 000, 001, 010, 011, 110 or 111	33.9	34.4	34.9	%
V_{TS_HOTZ}	TS pin rising voltage threshold for TH6 comparator to transition from TS_HOT to TS_WARM. TS_WARM charging settings resume above this voltage.	As Percentage to TS pin bias reference (47.5°C w/ 103AT), TS_TH4_TH5_TH6 = 100 or 101	42.0	42.5	43.0	%
		As Percentage to TS pin bias reference (57.5°C w/ 103AT), TS_TH4_TH5_TH6 = 000, 001, 010, 011, 110 or 111	35.2	35.7	36.2	%
SWITCHING CONVERTER						
F_{SW}	PWM switching frequency	Oscillator frequency	1.35	1.5	1.65	MHz
MOSFET TURN-ON RESISTANCE						
R_{Q1_ON}	VBUS to PMID on resistance	T_j = –40°C-85°C		26	34	mΩ
R_{Q2_ON}	Buck high-side switching MOSFET turn on resistance between PMID and SW	T_j = –40°C-85°C		55	78	mΩ
R_{Q3_ON}	Buck low-side switching MOSFET turn on resistance between SW and PGND	T_j = –40°C-85°C		60	90	mΩ
REGN LDO						
V_{REGN}	REGN LDO output voltage	V_{VBUS} = 5V, I_{REGN} = 20mA	4.4	4.6		V
		V_{VBUS} = 9V, I_{REGN} = 20mA	4.8	5.0	5.2	V
V_{REGNZ_OK}	REGN not good falling threshold	Converter switching		3.2		V
		Converter not switching		2.3		V
I_{REGN_LIM}	REGN LDO current limit	V_{VBUS} = 5V, VREGN = 4.3V	20			mA

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5.5 Electrical Characteristics (continued)

V_{VBUS_UVLOZ} < V_{VBUS} < V_{VBUS_OVP}, T_J = -40°C to +125°C, and T_J = 25°C for typical values (unless otherwise noted)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
$I_{TS_BIAS_FAULT}$	Rising threshold to transition from TSBIAS good condition to fault condition	REGN=5V; ISINK applied on TS_BIAS pin	2.5	4.5	8	mA
$I_{TS_BIAS_FAULTZ}$	Falling threshold to transition from TSBIAS fault condition to good condition	REGN=5V; ISINK applied on TS_BIAS pin	2	3.85	7	mA
ADC MEASUREMENT ACCURACY AND PERFORMANCE						
t_{ADC_CONV}	Conversion-time, Each Measurement	ADC_SAMPLE = 00		30		ms
		ADC_SAMPLE = 01		15		ms
		ADC_SAMPLE = 10		7.5		ms
		ADC_SAMPLE = 11		3.75		ms
ADC_{RES}	Effective Resolution	ADC_SAMPLE = 00	11	12		bits
		ADC_SAMPLE = 01	10	11		bits
		ADC_SAMPLE = 10	9	10		bits
		ADC_SAMPLE = 11	8	9		bits
ADC MEASUREMENT RANGE AND LSB						
IBUS_ADC	ADC Bus Current Reading	Range	–4		4	A
		LSB		2		mA
VBUS_ADC	ADC VBUS Voltage Reading	Range	0		18.00	V
		LSB		3.97		mV
VPMID_ADC	ADC PMID Voltage Reading	Range	0		18.00	V
		LSB		3.97		mV
VBAT_ADC	ADC BAT Voltage Reading	Range	0		5.572	V
		LSB		1.99		mV
VSYS_ADC	ADC SYS Voltage Reading	Range	0		5.572	V
		LSB		1.99		mV
IBAT_ADC	ADC BAT Current Reading	Range	-7.5		4.0	A
		LSB		4		mA
TS_ADC	ADC TS Voltage Reading	Range as a percent of REGN (–40 °C to 85 °C for 103AT)	20.9		83.2	%
	ADC TS Voltage Reading	LSB		0.0961		%
TDIE_ADC	ADC Die Temperature Reading	Range	–40		140	°C
		LSB		0.5		°C
I2C INTERFACE (SCL, SDA)						
V_{IH}	Input high threshold level, SDA and SCL		0.78			V
V_{IL}	Input low threshold level, SDA and SCL				0.42	V
V_{OL_SDA}	Output low threshold level	Sink current = 5mA, 1.2V VDD			0.3	V
I_{BIAS}	High-level leakage current	Pull up rail 1.8V			1	µA
C_{BUS}	Capacitive load for each bus line				400	pF
LOGIC OUTPUT PIN ($\overline{INT}$, STAT, PMID_GD)						
V_{OL}	Output low threshold level	Sink current = 5mA			0.3	V
I_{OUT_BIAS}	High-level leakage current	Pull up rail 1.8V			1	µA
LOGIC INPUT PIN ($\overline{CE}$, $\overline{QON}$)						
V_{IH_CE}	Input high threshold level, /CE		0.78			V

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5.5 Electrical Characteristics (continued)

V_{VBUS_UVLOZ} < V_{VBUS} < V_{VBUS_OVP}, T_J = -40°C to +125°C, and T_J = 25°C for typical values (unless otherwise noted)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
V_{IL_CE}	Input low threshold level, /CE				0.4	V
$I_{IN_BIAS_CE}$	High-level leakage current, /CE	Pull up rail 1.8V			1	µA
V_{IH_QON}	Input high threshold level, /QON		1.3			V
V_{IL_QON}	Input low threshold level, /QON				0.4	V
V_{QON}	Internal /QON pull up	/QON is pulled up internally. VBUS = 5V, 10MOhm resistance to GND		3.6		V
R_{QON}	Internal /QON pull up resistance			250		kΩ

5.6 Timing Requirements

PARAMETER		TEST CONDITIONS	MIN	NOM	MAX	UNIT
VBUS / VBAT POWER UP						
t_{VBUS_OVP}	VBUS OVP deglitch time to set VBUS_FAULT_STAT and VBUS_FAULT_FLAG			200		µs
$t_{POORSRC}$	Bad adapter detection duration			30		ms
BATTERY CHARGER						
t_{TOP_OFF}	Typical top-off timer accuracy	TOPOFF_TMR = 01	12	17	21	min
		TOPOFF_TMR = 10	24	35	41	min
		TOPOFF_TMR = 11	36	52	61	min
t_{SAFETY_TRKCHG}	Charge safety timer accuracy in trickle charge		0.85	1.25	1.35	hr
t_{SAFETY_PRECHG}	Charge safety timer accuracy in pre-charge	PRECHG_TMR = 0	1.75	2.5	2.75	hr
		PRECHG_TMR = 1	0.43	0.62	0.68	hr
t_{SAFETY}	Charge safety timer accuracy in fast charge	CHG_TMR = 0	10.5	14.5	15.5	hr
		CHG_TMR = 1	21.0	28	31	hr
BATFET CONTROL						
t_{BATFET_DLY}	Time after writing to BATFET_CTRL before BATFET turned off for ship mode or shutdown	BATFET_DLY = 1		12.5		s
		BATFET_DLY = 0		25		ms
t_{SM_EXIT}	Deglitch time for $\overline{QON}$ to be pulled low in order to exit from Ship Mode		0.55	0.8	0.93	s
t_{QON_RST}	Time $\overline{QON}$ is held low to initiate system power reset		9.0	12.5	14.5	s
t_{BATFET_RST}	Duration that BATFET is disabled during system power reset			430		ms
I2C INTERFACE						
f_{SCL}	SCL clock frequency	See Serial Interface section for more details.			1.0	MHz
DIGITAL CLOCK AND WATCHDOG						
t_{LP_WDT}	Watchdog Reset time (EN_HIZ = 1, WATCHDOG = 11)		100	200		s
t_{WDT}	Watchdog Reset time (EN_HIZ = 0, WATCHDOG = 11)		136	200		s

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5.7 Typical Characteristics

C_{VBUS} = 1µF, C_{PMID}= 10µF, C_{SYS}= 20µF, C_{BAT}= 1µF, L= 1µH (unless otherwise specified)

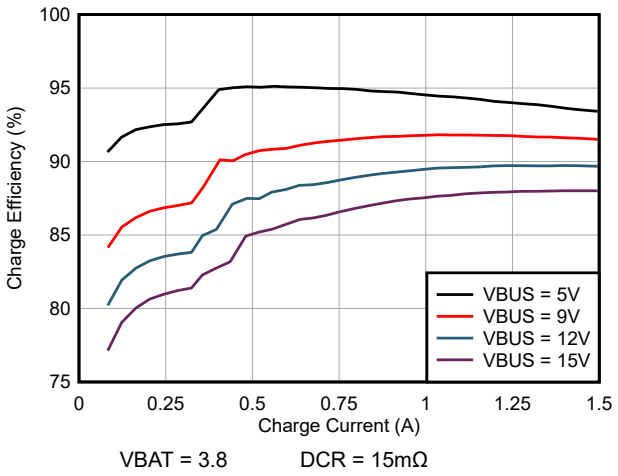


Figure 5-1. Charge Efficiency vs Charge Current

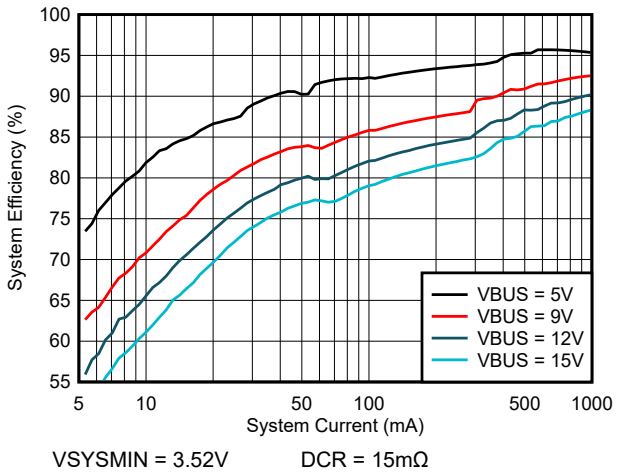


Figure 5-2. System Light Load Efficiency vs System Current

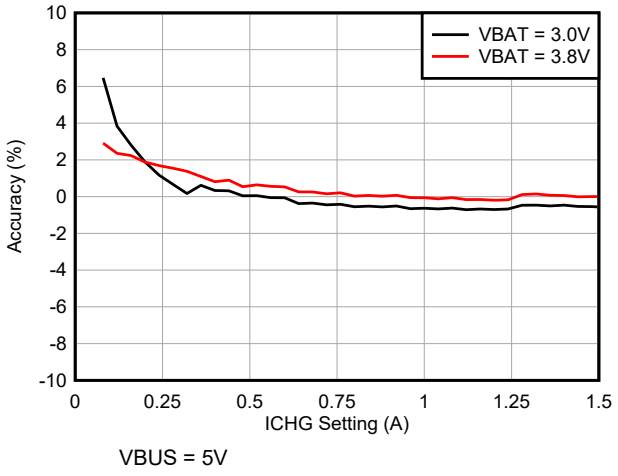


Figure 5-3. Charge Current Accuracy vs ICHG Setting

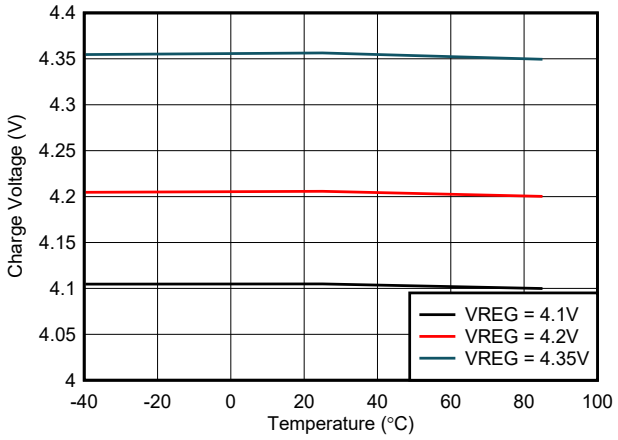


Figure 5-4. Charge Voltage Accuracy vs VREG Setting

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5.7 Typical Characteristics (continued)

C_{VBUS} = 1µF, C_{PMID}= 10µF, C_{SYS}= 20µF, C_{BAT}= 1µF, L= 1µH (unless otherwise specified)

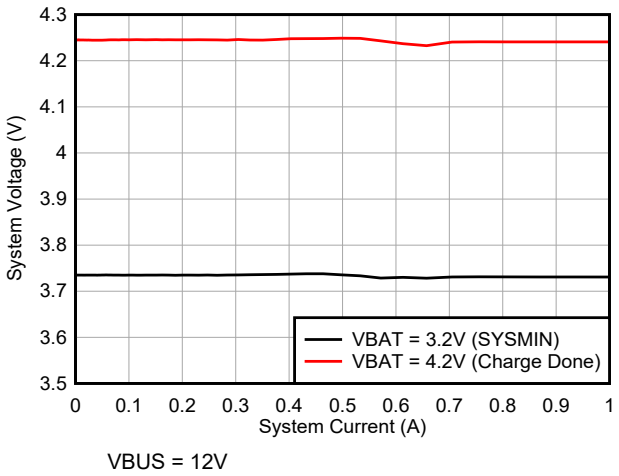


Figure 5-5. System Load Regulation for SYSMIN and After Charge Done

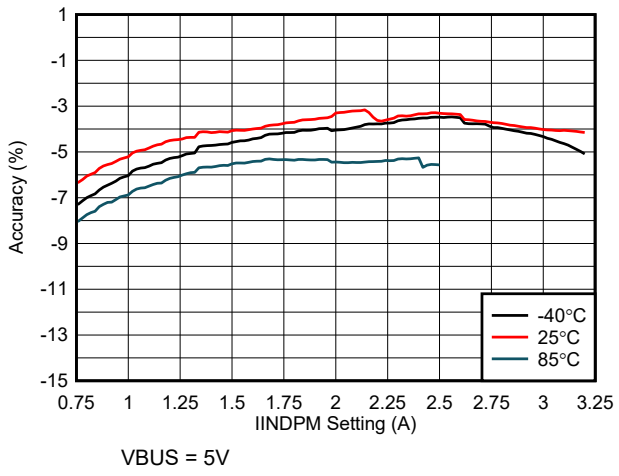


Figure 5-6. Input Current Regulation Accuracy vs IINDPM Setting

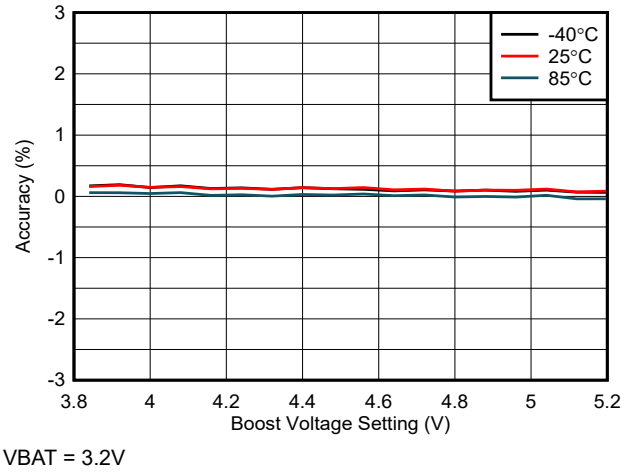


Figure 5-7. Boost Voltage Regulation Accuracy vs Boost Voltage Setting

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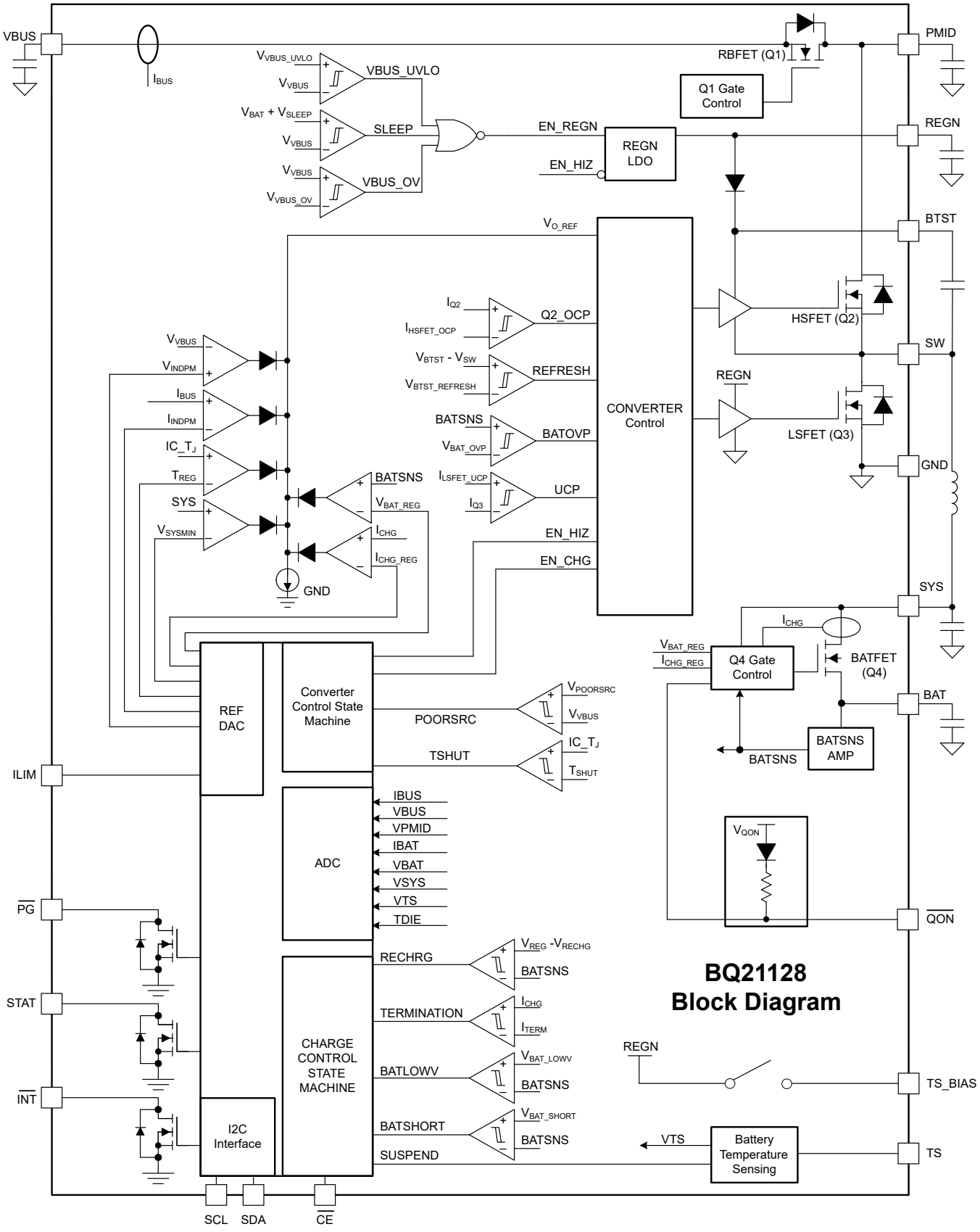
6 Detailed Description

6.1 Overview

BQ21128 is a highly-integrated 1.5A switch-mode battery chargers for single-cell Li-ion and Li-polymer batteries. The device includes input reverse-blocking FET (RBFET, Q1), high-side switching FET (HSFET, Q2), low-side switching FET (LSFET, Q3), battery FET (BATFET, Q4), and bootstrap diode for the high-side gate driver.

6.2 Functional Block Diagram

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BQ21128 Block Diagram

6.3 Feature Description

6.3.1 Power-On-Reset (POR)

BQ21128 powers internal bias circuits from the higher voltage of VBUS and BAT. When either voltage rises above its undervoltage lockout (UVLO) threshold, all registers are reset to their POR values and the I^2C interface is enabled for communication. A non-maskable $\overline{INT}$ pulse is generated, after which the host can access all of the registers.

6.3.2 Device Power Up from Battery

If only the battery is present and the VBAT is above depletion threshold (V_{BAT_UVLOZ}), BQ21128 performs a power-on reset then turns on the BATFET to connect the battery to system. The REGN LDO output remains off to minimize the quiescent current. The low RDSON of BATFET and the low quiescent current on BAT minimize the conduction loss and maximize the battery run time.

6.3.3 Device Power Up from Input Source

When a valid input source is plugged in with VBAT < V_{BAT_UVLOZ}, BQ21128 performs a power-on reset then checks the input source voltage to turn on the REGN LDO and all the bias circuits. It detects and sets the input current limit before the buck converter is started. The power up sequence from input source is as listed:

1. REGN LDO power up (Section 6.3.3.1)
2. Poor source qualification (Section 6.3.3.2)
3. Input voltage limit threshold setting (Section 6.3.3.3)
4. Converter power-up (Section 6.3.3.4)

6.3.3.1 REGN LDO Power Up

The REGN LDO regulator supplies internal bias circuits as well as the HSFET and LSFET gate drive. The REGN LDO also provides bias rail to TS external resistors. The pull-up rail of STAT can be connected to REGN as well. The REGN is enabled when all the below conditions are valid:

- VBUS above V_{VBUS_UVLOZ}
- VBUS above V_{BAT} + V_{SLEEPZ}
- EN_HIZ = 0
- After 220-ms delay is completed

If any one of the above conditions is not valid, the REGN LDO and the converter power stage remain off with the converter disabled. In this state, the battery supplies power to the system.

6.3.3.2 Poor Source Qualification

After the REGN LDO powers up, the device checks the current capability of the input source. The input source has to meet the following requirements in order to move forward to the next power on steps.

1. VBUS voltage below V_{VBUS_OVP}
2. VBUS voltage above $V_{POORSRC}$ when pulling $I_{POORSRC}$

Once these conditions are met, BQ21128 sets VBUS_STAT to 0x4 and VBUS_FLAG transitions to 1.

6.3.3.3 Input Voltage Limit Threshold Setting (VINDPM Threshold)

BQ21128 supports a wide range of input voltage limit (3.8 V – 16.8V). Its POR default VINDPM threshold is set at 4.6V. BQ21128 also supports dynamic VINDPM tracking, which tracks the battery voltage to ensure a sufficient margin between input and battery voltages for proper operation of the buck converter. This function is enabled via the VINDPM_BAT_TRACK register bit. When enabled, the actual input voltage limit is the higher of the VINDPM register or $V_{INDPM_BAT_TRACK}$ (VBAT + 400 mV typical offset.)

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6.3.3.4 Converter Power-Up

After the input current and voltage limits are set, the converter is enabled and the HSFET and LSFET start switching. If battery charging is disabled, the BATFET turns off. Otherwise, the BATFET stays on to charge the battery. Converter startup requires the following conditions:

- VBUS has passed poor source qualification (Section 6.3.3.2)
- VBUS > V_{BAT} + V_{SLEEPZ}
- V_{VBUS} < V_{VBUS_OVP}
- EN_HIZ = 0
- V_{SYS} < V_{SYS_OVP}
- T_J < T_{SHUT}

BQ21128 provides soft start when the system rail is ramped up by setting IINDPM to its lowest programmable value and stepping up through each available setting until reaching the value set by IINDPM register. Concurrently, the system short protection limits the output current to approximately 0.5A when the system rail is below V_{SYS_SHORT}.

The device uses a highly efficient 1.5 MHz, fixed frequency pulse width modulated (PWM) step-down switching regulator. The internally compensated feedback loop keep tight control of the switching frequency under all conditions of input voltage, battery voltage, charge current and temperature, simplifying output filter design.

The device switches to pulse frequency modulation (PFM) control at light load condition. The PFM_FWD_DIS bits can be used to disable the PFM operation in buck and boost respectively.The minimum frequency in PFM mode can be limited to 25 kHz when the Out of Audio (OOA) feature is enabled (EN_OOA=1b).

6.3.4 Power Path Management

BQ21128 accommodates a wide range of input sources from USB, wall adapter, wireless charger, to car charger. They provide automatic power path selection to supply the system from input source, battery, or both.

6.3.4.1 Narrow VDC Architecture

BQ21128 uses the Narrow VDC architecture (NVDC) with BATFET separating system from battery. The minimum system voltage is set by VSYSMIN register setting. Even with a fully depleted battery, the system is regulated to the minimum system voltage. If charging is enabled, the BATFET operates in linear mode (LDO mode). The default minimum system voltage at POR is 3.52 V.

As the battery voltage rises above the minimum system voltage, the BATFET is turned fully on. When battery charging is disabled and V_{BAT} is above the minimum system voltage setting, or charging is terminated, the system is regulated 50mV (typical) above battery voltage.

6.3.4.2 Dynamic Power Management

To meet the USB maximum current limit and avoid overloading the adapter, the device features Dynamic Power Management (DPM), which continuously monitors the input current and input voltage. When the input source is overloaded, either the current exceeds the input current limit (IINDPM) or the voltage falls below the input voltage limit (VINDPM). The device then reduces the charge current until the input current falls below the input current limit and the input voltage rises above the input voltage limit.

When the charge current is reduced to zero, but the input source is still overloaded, the system voltage starts to drop. Once the system voltage falls below the battery voltage by V_{SUPP}, the device automatically enters the supplement mode where the BATFET turns on and the battery starts discharging so that the system is supported from both the input source and battery.

6.3.4.3 High Impedance Mode

The host may place BQ21128 into high impedance mode by writing EN_HIZ = 1. In high impedance mode, RBFET (Q1), HSFET (Q2) and LSFET (Q3) are turned off. The RBFET and HSFET block current flow to and from VBUS, putting the VBUS pin into a high impedance state. The BATFET (Q4) is turned on to connect the BAT to SYS. During high impedance mode, REGN is disabled and the digital clock is slowed to conserve power.

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BQ21128 drives PMID_GD low in high-impedance mode.

6.3.5 Battery Charging Management

BQ21128 charges a 1-cell Li-Ion battery with up to 1.5 A charge current. The 15 mΩ BATFET improves charging efficiency and minimizes the voltage drop during discharging.

6.3.5.1 Autonomous Charging Cycle

When battery charging is enabled (EN_CHG bit = 1 and $\overline{CE}$ pin is LOW), BQ21128 autonomously completes a charging cycle without host involvement. The device default charging parameters are listed in Table 6-1. The host can always control the charging operations and optimize the charging parameters by writing to the corresponding registers through I^2C.

Table 6-1. Charging Parameter Default Setting

	VREG	VRECHG	ITRICKLE	IPRECHG	ICHG	ITERM	TOPOFF TIMER
BQ21128	4.2V	VREG - 100 mV	10 mA	30 mA	320 mA	20 mA	Disabled

A new charge cycle starts when the following conditions are valid:

- Converter starts per the conditions in Section 6.3.3.4
- EN_CHG = 1
- $\overline{CE}$ pin is low
- No thermistor fault on TS
- No safety timer fault

BQ21128 automatically terminates the charging cycle when the charging current is below termination threshold, battery voltage is above recharge threshold, and device not is in DPM or thermal regulation. When a fully charged battery is discharged below VRECHG, the device automatically starts a new charging cycle. After charging terminates, toggling $\overline{CE}$ pin or EN_CHG bit also initiates a new charging cycle.

The STAT output indicates the charging status. Refer to Section 6.3.7.2 for details of STAT pin operation. In addition, the status register (CHG_STAT) indicates the different charging phases: 00-charging disabled or terminated, 01-constant current, 10 constant voltage, 11-topoff charging.

6.3.5.2 Battery Charging Profile

BQ21128 charges the battery in five phases: trickle charge, pre-charge, constant current, constant voltage and an optional top-off charging phase. At the beginning of a charging cycle, the device checks the battery voltage and regulates current and voltage accordingly.

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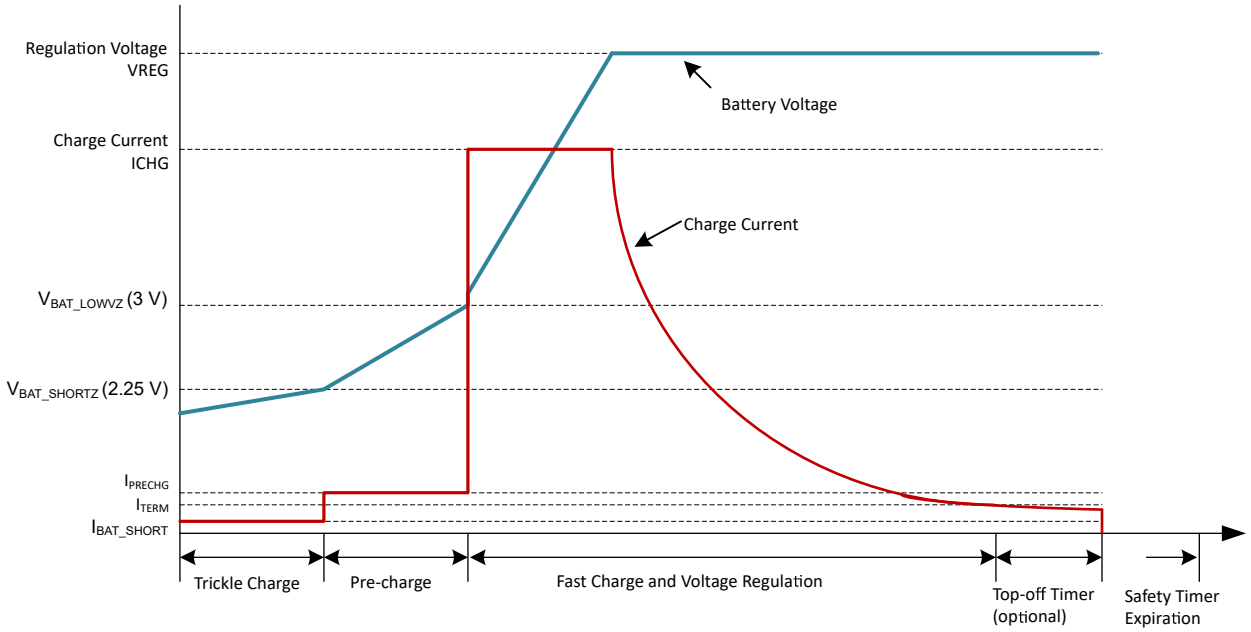


Figure 6-1. Battery Charging Profile

6.3.5.3 Charging Termination

BQ21128 terminates a charge cycle when the battery voltage is above recharge threshold and the current is below ITERM. To avoid early termination, BQ21128 does not terminate while IINDPM, VINDPM or thermal regulation loops are active. After the charging cycle is completed, the BATFET turns off. The converter keeps running to power the system, and the BATFET can turn on again to engage supplement mode. Termination can be permanently disabled by writing 0 to EN_TERM bit prior to charge termination. The termination current range can be increased using ITERM_2X. If ITERM_2X=1b, the actual termination current is set to the double the ITERM register setting.

At low termination currents, due to the comparator offset, the actual termination current may be 10 mA-20 mA higher than the termination target. in order to compensate for comparator offset, a programmable top-off timer can be applied after termination is detected. The top-off timer follows safety timer constraints, such that if the safety timers suspend, so does the top-off timer. Similarly, if the safety timers count at half-clock rate, so does the top-off timer. Refer to Section 6.3.5.5 for the list of conditions. The host can read CHG_STAT to find out the termination status.

Top-off timer gets reset by any of the following conditions:

1. Charging cycle stop and restart (toggle CE pin, toggle EN_CHG bit, charged battery falls below recharge threshold or adapter removed and replugged)
2. Termination status low to high
3. REG_RST register bit is set

The top-off timer settings are read in after is detected by the charger. Programming a top-off timer value after termination has no effect unless a recharge cycle is initiated. CHG_FLAG is set to 1 when entering top-off timer segment and again when the top-off timer expires.

6.3.5.4 Thermistor Qualification

BQ21128 provides a single thermistor input for battery temperature monitoring. The TS pin input of the battery temperature can be ignored by the charger if TS_IGNORE = 1. When the TS pin feedback is ignored, the charger considers the TS to always be valid for charging and OTG modes, and TS_STAT always reports 000. The TS pin may be left floating if TS_IGNORE is set to 1.

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When TS_IGNORE=1, the TS_ADC channel is disabled, with TS_ADC_DIS forced to 1; Attempting to write to 0 is ignored.

When TS_IGNORE = 0, the charger adjusts the charging profile based on the TS pin feedback information according to the configurable profile described in Section 6.3.5.4.1. When the battery temperature crosses from one temperature range to another, TS_STAT is updated accordingly, and the charger sets the FLAG bit for the newly-entered temperature range. If TS_MASK is set to 0, any change to TS_STAT, including a transition to TS_NORMAL, generates an $\overline{INT}$ pulse.

6.3.5.4.1 Advanced Temperature Profile in Charge Mode

To improve the safety of charging Li-ion batteries, JEITA guideline was released on April 20, 2007. The guideline emphasized the importance of avoiding a high charge current and high charge voltage at certain low and high temperature ranges. As battery technology continues to evolve, battery manufacturers have released temperature safety specifications that extend beyond the JEITA standard. BQ21128 features a highly flexible temperature-based charging profile to meet these advanced specifications while remaining backwards compatible with the original JEITA standard. Figure 8-3 shows the programmability for charger behavior under different battery temperature (TS) operating regions.

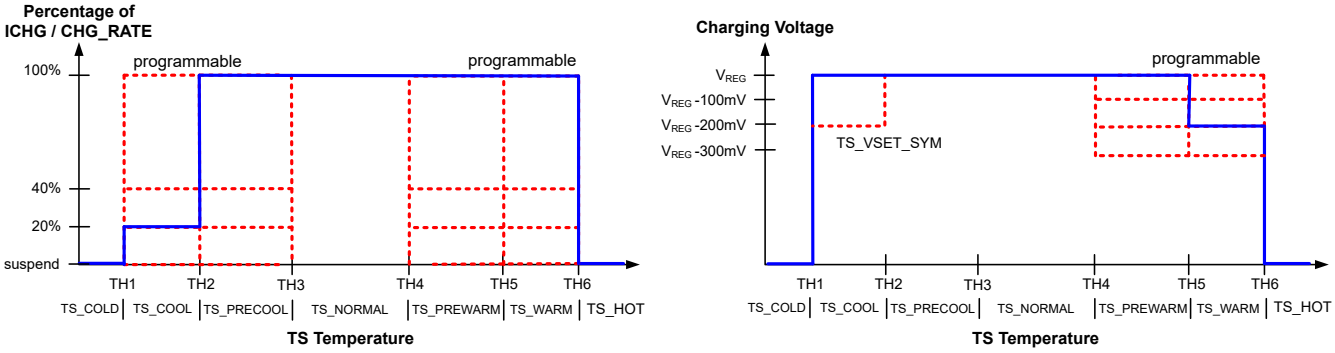


Figure 6-2. TS Charging Values

Charging safety timer is adjusted within the temperature zones to reflect changes to the charging current. When IPRECHG and ICHG are reduced to 20% or 40% in the cool or warm temperature zones, the charging safety timer counts at half rate. If charging is suspended, the safety timer is suspended, the STAT pin blinks and CHG_STAT is set to 00 (not charging or charge terminated.)

6.3.5.4.2 TS Pin Thermistor Configuration

The typical TS resistor network is illustrated below.

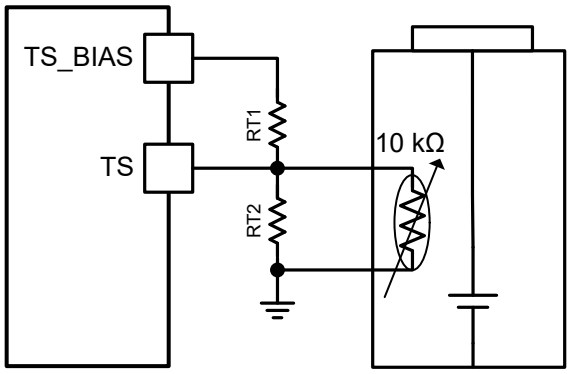


Figure 6-3. TS Resistor Network

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The value of RT1 and RT2 are determined from the resistance of the thermistor at 0 and 60 ºC (RTH_{0degC} and RTH_{60degC}) and the corresponding voltage thresholds V_{TS_0degC} and V_{TS_60degC} (expressed as percentage of REGN with value between 0 and 1.) For the most accurate thermistor curve fitting, use the rising threshold for V_{TS_COLD} at 0 ºC and the falling threshold for V_{TS_HOT} at 60 ºC, regardless of the actual register settings for TS_TH1_TH2_TH3 and TS_TH4_TH5_TH6.

$$RT2 = \frac{RTH_{0degC} \times RTH_{60degC} \times \left(\frac{1}{V_{TS_0degC}} - \frac{1}{V_{TS_60degC}}\right)}{RTH_{60degC} \times \left(\frac{1}{V_{TS_60degC}} - 1\right) - RTH_{0degC} \times \left(\frac{1}{V_{TS_0degC}} - 1\right)} \quad (1)$$

$$RT1 = \frac{\frac{1}{V_{TS_0degC}} - 1}{\frac{1}{R_{T2}} + \frac{1}{RTH_{0degC}}} \quad (2)$$

Assuming a 103AT NTC thermistor on the battery pack, the RT1 and RT2 are calculated to be 5.23 kΩ and 30.1 kΩ respectively.

If the thermistor is biased from TS_BIAS, the maximum current should be checked against $I_{TS_BIAS_FAULT}$. For the worst-case condition of thermistor at 0 Ω impedance (very hot), the bias current is :

$$I_{BIAS_MAX} = \frac{V_{REGN}}{RT1} \quad (3)$$

For 5.23 kΩ RT1, this has a maximum I_{BIAS} of 0.94 mA, which is well below the minimum $I_{TS_BIAS_FAULT}$ threshold. The 103AT NTC thermistor is the recommended thermistor and has 10 kΩ nominal impedance. Using a lower impedance thermistor will change the value of R1 and may produce a bias current that exceeds the TS_BIAS pin fault threshold. TS_STAT[2:0] is set to 111.

6.3.5.4.3 JEITA Charge Rate Scaling

The TS_ISET_PRECOOL, TS_ISET_COOL, TS_ISET_PREWARM and TS_ISET_WARM cool and warm charge current fold backs are based on a 1C charging rate. The 1C rate is the battery capacity in mA-hours divided by 1 hour, so that a 500 mA-hour battery would have a 1C charging rate of 500mA. The same battery would have a 2C charging rate of 1,000mA. In order to convert the charging foldback, the host must set the CHG_RATE register to the C rate for the battery. This scales the fold back accordingly.

When TS_ISET_PRECOOL, TS_ISET_COOL, TS_ISET_PREWARM or TS_ISET_WARM is set to either 00 (suspend) or 11 (unchanged), the CHG_RATE setting has no effect. A summary is provided in Table 6-2.

Table 6-2. ICHG Fold Back

TS_ISET_PRECOOL, TS_ISET_COOL, TS_ISET_PREWARM or TS_ISET_WARM	CHG_RATE	FOLD-BACK CURRENT AS PERCENTAGE OF ICHG
00	Any	0% (Suspended)
01 (20%)	00 (1C)	20%
	01 (2C)	10%
	10 (4C)	5%
	11 (6C)	3.3%
10 (40%)	00 (1C)	40%
	01 (2C)	20%
	10 (4C)	10%
	11 (6C)	6.6%
11	Any	100%

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6.3.5.5 Charging Safety Timers

BQ21128 has three built-in safety timers to prevent extended charging cycle due to abnormal battery conditions. The fast charge safety timer and pre-charge safety timers are set through I²C CHG_TMR and PRECHG_TMR fields, respectively. The trickle charge timer is fixed at 1 hour.

The trickle charging, pre-charging and fast charging safety timers can be disabled by setting EN_SAFETY_TMRS = 0. EN_SAFETY_TMRS can be enabled anytime regardless of which charging stage the charger is in. Each timer starts to count as soon as the following two conditions are simultaneously true: EN_SAFETY_TMRS=1 and the corresponding charging stage is active.

When either the fast charging, trickle charging or pre-charging safety timer expires, the SAFETY_TMR_STAT and SAFETY_TMR_FLAG bits are set to 1.

Events that cause a reduction in charging current also cause the charging safety timer to count at half-clock rate if TMR2X_EN bit is set.

During faults which suspend charging, the charge, pre-charge and trickle safety timers are also suspended, regardless of the state of the TMR2X_EN bit. Once the fault goes away, charging resumes and the safety timer resumes from where it stopped.

The charging safety timer and the charging termination can be disabled at the same time. Under this condition, the charging keeps running until it is disabled by the host.

6.3.6 Integrated 12-Bit ADC for Monitoring

BQ21128 provides an integrated 12-bit ADC for the host to monitor various system parameters.

To enable the ADC, the ADC_EN bit must be set to ‘1’. The ADC is disabled by default (ADC_EN=0) to conserve power. The ADC is allowed to operate if either VBUS > $V_{POORSRC}$ or VBAT > V_{BAT_LOWV} is valid. If ADC_EN is set to ‘1’ before VBUS or VBAT reach their respective valid thresholds, then ADC_EN stays '0'. When the charger enters HIZ mode, the ADC is temporarily suspended.

At battery only condition, if the TS_ADC channel is enabled, the ADC only operates when the battery voltage is higher than 3.2V (the minimal value to turn on REGN), otherwise, the ADC operates when the battery voltage is higher than V_{BAT_LOWV}.

The ADC_DONE_STAT, ADC_DONE_FLAG bits are set when a conversion is complete in one-shot mode only. During continuous conversion mode, the ADC_DONE_STAT, ADC_DONE_FLAG bits have no meaning and remain at 0. In one-shot mode, the ADC_EN bit is set to 0 at the completion of the conversion, at the same time as the ADC_DONE_FLAG bit is set. In continuous mode, the ADC_EN bit remains at 1 until the user disables the ADC by setting it to 0.

6.3.7 Status Outputs ($\overline{PG}$, STAT, $\overline{INT}$)

6.3.7.1 Interrupts and Status, Flag and Mask Bits

BQ21128 incorporates an interrupt pin ($\overline{INT}$) to inform a host microcontroller of status changes without requiring microcontroller polling. Each reported event has a status field, a flag bit and a mask bit. The status field reports the status at the time that it is read. The flag bit is latched and, once set to 1, will remain at 1 until the host reads the bit, which will clear it to 0. The mask bit determines whether or not an interrupt pulse will be generated when the flag bit is set.

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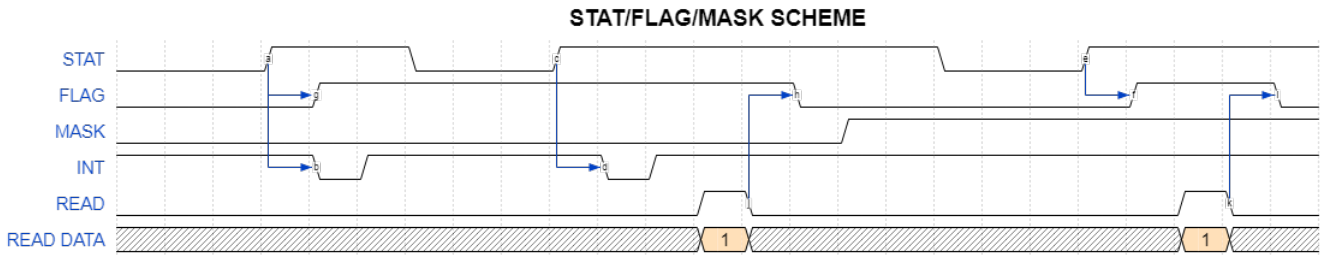


Figure 6-4. Relationship between STAT, FLAG and MASK

The flag bit is set upon certain transitions of the status field. These transitions also generate an $\overline{INT}$ pulse if the associated mask bit is set to 0. Because the $\overline{INT}$ is generated from the status field transition and not the flag bit, an $\overline{INT}$ pulse is sent to the host even if the associated flag is already set to 1 when the status transition occurs. Details of this behavior are shown in Figure 6-4.

The default behavior is to generate a 256-µs $\overline{INT}$ pulse when any flag bit is set to 1. These pulses may be masked out on a flag-by-flag basis by setting a flag's mask bit to 1. Setting the mask bit does not affect the transition of the flag bit from 0 to 1, only the generation of the 256-µs $\overline{INT}$ pulse.

6.3.7.2 Charging Status Indicator (STAT)

BQ21128 indicates charging state on the open drain STAT pin. The STAT pin can drive an LED. The STAT pin function can be disabled via the DIS_STAT bit.

Table 6-3. STAT Pin State

CHARGING STATE	STAT INDICATOR
Charging in progress (including recharge)	LOW
Not charging, no fault detected. (Includes charging complete, Charge Disabled, no adapter present)	HIGH
Charge suspend	Blinking at 1 Hz

6.3.7.3 Interrupt to Host ($\overline{INT}$)

In many applications, the host does not continually poll the charger status registers. Instead, the INT pin may be used to notify the host of a status change with a 256-µs $\overline{INT}$ pulse. Upon receiving the interrupt pulse, the host may read the flag registers (Charger_Flag_X and FAULT_Flag_X) to determine the event that caused the interrupt, and for each flagged event, read the corresponding status registers (Charger_Status_X and FAULT_Status_X) to determine the current state. Once set to 1, the flag bits remain latched at 1 until they are read by the host, which clears them. The status bits, however, are updated whenever there is a change to status and always represent the current state of the system.

All of the $\overline{INT}$ events can be masked off to prevent $\overline{INT}$ pulses from being sent out when they occur, with the exception of the initial power-up interrupt. Interrupt events are masked by setting their mask bit in registers (Charger_Mask_X and FAULT_Mask_X.) Events always cause the corresponding flag bit to be set to 1, regardless of whether or not the interrupt pulse has been masked.

6.3.8 BATFET Control

BQ21128 has an integrated, bi-directionally blocking BATFET that can be turned off to remove leakage current from the battery to the system. The BATFET is controlled by the BATFET_CTRL register bits, and supports shutdown mode, ship mode and system power reset.

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Table 6-4. BATFET Control Modes

MODE	BATFET	I²C	ENTRY, NO ADAPTER	ENTRY, WITH ADAPTER, BATFET_CTRL_WVBUS = 0	ENTRY, WITH ADAPTER, BATFET_CTRL_WVBUS = 1	EXIT
Normal	On	Active	N/A	N/A	N/A	N/A
Ship mode	Off	Off	Writing BATFET_CTRL = 10 turns off BATFET after BATFET_DLY and enters ship mode.	Writing BATFET_CTRL = 10 has no effect while adapter is present. When both BATFET_DLY has expired and the adapter is removed, the device turns off BATFET and enters ship mode. Writing BATFET_CTRL = 00 before adapter is removed aborts ship mode.	Writing BATFET_CTRL = 10 turns off BATFET after BATFET_DLY. When both BATFET_DLY has expired and adapter is removed, the device enters ship mode. Writing BATFET_CTRL = 00 before adapter is removed turns BATFET on and aborts ship mode.	$\overline{QON}$ or adapter plug-in
System reset	On to Off to On	Active	Writing BATFET_CTRL = 11 initiates system reset after BATFET_DLY. Holding $\overline{QON}$ low for t_{QON_RST} initiates immediate reset (BATFET_DLY is not applied.)	Writing BATFET_CTRL = 11 is ignored and BATFET_CTRL resets to 00. Holding $\overline{QON}$ low for t_{QON_RST} is ignored.	Writing BATFET_CTRL = 11 initiates system reset after BATFET_DLY. Holding $\overline{QON}$ low for t_{QON_RST} initiates immediate reset. Converter is placed in HIZ during system reset and exits HIZ when system reset completes.	N/A
Shutdown mode	Off	Off	Writing BATFET_CTRL = 01 turns off BATFET after BATFET_DLY and enters shutdown.	Writing BATFET_CTRL = 01 with adapter present is ignored, regardless of BATFET_CTRL_WVBUS setting, and BATFET_CTRL is reset to 00.		Adapter plug-in

6.3.8.1 Shutdown Mode

For the lowest battery leakage current, the host can shut down BQ21128 by setting the register bits BATFET_CTRL to 01. In this mode, the BATFET is turned off to prevent the battery from powering the system, the I2C is disabled and the charger is totally shut down. BQ21128 can only be woken up by plugging in an adapter. When the adapter is plugged in, BQ21128 starts back up with all register settings in their POR default.

After the host sets BATFET_CTRL to 01, the BATFET turns off after waiting either 25 ms or 12.5 s as configured by BATFET_DLY register bit. Shutdown mode can only be entered when $V_{VBUS} < V_{VBUS_UVLO}$, regardless of the BATFET_CTRL_WVBUS setting, which has no effect on shutdown mode entry. If the host writes BATFET_CTRL = 01 with $V_{VBUS} > V_{VBUS_UVLOZ}$, the request is ignored and the BATFET_CTRL bits are set back to 00.

$\overline{QON}$ has no effect during shutdown mode. The internal pull-up on the $\overline{QON}$ pin is disabled during shutdown to prevent leakage through the pin.

6.3.8.2 Ship Mode

The host may place BQ21128 into ship mode by setting BATFET_CTRL = 10. In ship mode, the BATFET is turned off to prevent the battery from powering the system, and the I2C is disabled. Ship mode has slightly higher quiescent current than shutdown mode, but $\overline{QON}$ may be used to exit from ship mode. BQ21128 is taken out of ship mode by either of these methods:

- Pulling the $\overline{QON}$ pin low for t_{SM_EXIT}
- $V_{VBUS} > V_{VBUS_UVLOZ}$ (adapter plug-in)

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When BQ21128 exits from ship mode, the registers are reset to their POR values.

Ship mode is only entered when the adapter is not present. Setting BATFET_CTRL = 10 while V_{VBUS} > V_{VBUS_UVLOZ} (adapter present) either disables the BATFET or has no immediate effect depending on the setting of BATFET_CTRL_WVBUS.

6.3.8.3 System Power Reset

The BATFET functions as a load switch between battery and system when the converter is not running. By changing the state of BATFET from on to off, systems connected to SYS can be power cycled. Any of the following conditions initiates a system power reset:

- BATFET_CTRL_WVBUS = 1 and $\overline{QON}$ is pulled low for tQON_RST
- BATFET_CTRL_WVBUS = 1 and BATFET_CTRL = 11
- BATFET_CTRL_WVBUS = 0 and VBUS < V_{VBUS_UVLO} simultaneously with $\overline{QON}$ pulled low for tQON_RST
- BATFET_CTRL_WVBUS = 0 and VBUS < V_{VBUS_UVLO} and BATFET_CTRL = 11

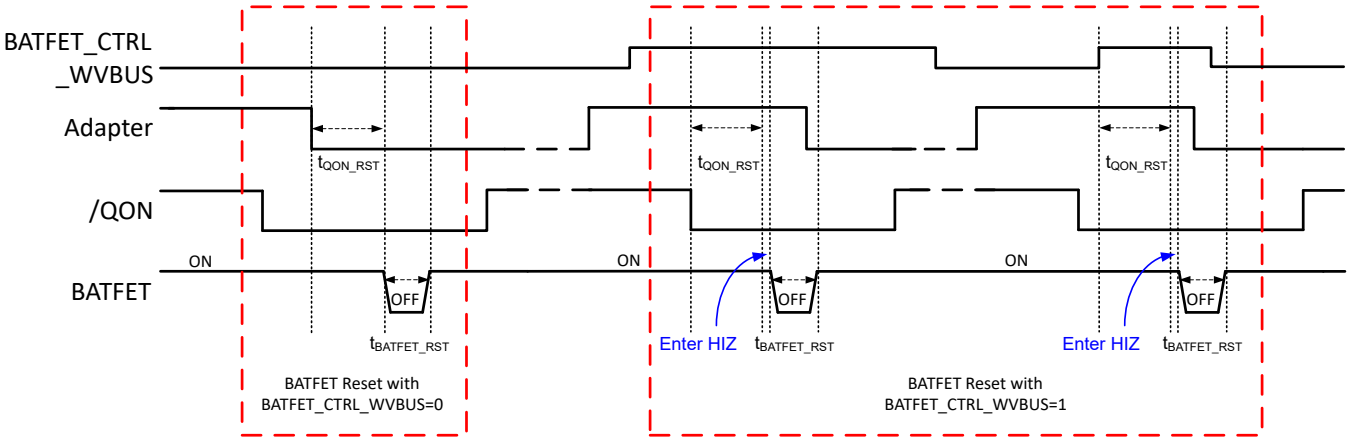


Figure 6-5. System Power Reset Timing

When BATFET_CTRL_WVBUS is set to 1, system power reset proceeds if either BATFET_CTRL is set to 11 or $\overline{QON}$ is pulled low for t_{QON_RST}, regardless of whether VBUS is present or not . There is a delay of t_{BATFET_DLY} before initiating the system power reset. If $\overline{QON}$ is pulled low, there is no delay after the tQON_RST completes, regardless of BATFET_DLY setting.

6.3.9 Protections

6.3.9.1 Voltage and Current Monitoring in Battery Only and HIZ Modes

BQ21128 monitors a reduced set of voltages and currents when operating from battery without an adapter or when operating from battery in high impedance mode.

6.3.9.1.1 Battery Undervoltage Lockout

In battery-only mode, BQ21128 will disable the BATFET if V_{BAT} falls below V_{BAT_UVLO}, separating the system from the battery. I^2C is disabled as well. Upon exit from the undervoltage lockout condition when either V_{BAT} rises above V_{BAT_UVLOZ} or V_{VBUS} rises above V_{VBUS_UVLOZ}, I^2C will be re-enabled and the registers are reset to their POR values.

6.3.9.1.2 Battery Overcurrent Protection

BQ21128 has a two-level battery overcurrent protection. The I_{BAT_PK} threshold is set by IBAT_PK and provides a fast (100 µs) protection for the battery discharging. I_{BATFET_OCP} provides a slower (50 ms), fixed-threshold protection for the BATFET. If the battery discharge current becomes higher than either threshold for its deglitch time, the BAT_FAULT_STAT and BAT_FAULT_FLAG fault register bits are set to 1, and the BATFET enters hiccup mode with 100ms off-time and ~1% on-time. The BAT_FAULT_STAT will return to 0 once the BATFET is

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disabled for the hiccup mode. Once the BATFET is turned back on, the I_{BAT_PK} and I_{BATFET_OCP} thresholds are re-evaluated.

6.3.9.2 Voltage and Current Monitoring in Buck Mode

6.3.9.2.1 Input Overvoltage

If VBUS voltage rises above V_{VBUS_OVP}, the converter stops switching to protect the internal power MOSFETs and I_{PMID_LOAD} discharge current is applied to bring down VBUS voltage. VBUS_FAULT_FLAG and VBUS_FAULT_STAT bit transition to 1. When VBUS falls back below V_{VBUS_OVPZ}, VBUS_FAULT_STAT will transition to 0 and the converter will resume switching.

6.3.9.2.2 System Overvoltage Protection (SYSOVP)

When VSYS rises above V_{SYS_OVP} in forward converter operation, the converter stops switching to limit voltage overshoot and applies I_{SYS_LOAD} to pull down the system voltage. VSYS_FAULT_FLAG and VSYS_FAULT_STAT transition to 1. Once VSYS drops below V_{SYS_OVP}, the converter resumes switching, the 30 mA discharge current is removed and VSYS_FAULT_STAT transitions to 0.

6.3.9.2.3 Forward Converter Cycle-by-Cycle Current Limit

The converter has cycle-by-cycle peak overcurrent protection in the switching MOSFETs. In forward mode, if the current through Q2 exceeds I_{Q2_CBC}, the converter will immediately turn off the high-side gate drive for the remainder of the switching cycle. Normal switching resumes on the next switching cycle.

6.3.9.2.4 System Short

When the SYS voltage falls below V_{SYS_SHORT}, the charger enters PFM operation to limit the output current to approximately 0.5A or less.SYS_FAULT_STAT and SYS_FAULT_FLAG bits are set to 1. If V_{SYS} rises above V_{SYS_SHORTZ}, the converter exits forced PFM mode, and the SYS_FAULT_STAT bit is set to 0.

6.3.9.2.5 Battery Overvoltage Protection (BATOVP)

When V_{BAT} transitions above V_{BAT_OVP}, BQ21128 disables charging by disabling the BATFET and applies I_{BAT_LOAD} current source to discharge excess BAT voltage. If battery voltage remains above the threshold, BAT_FAULT_FLAG is set to 1 and BAT_FAULT_STAT transitions to 1. Once V_{BAT} falls below V_{BAT_OVPZ}, charging resumes and BAT_FAULT_STAT transitions back to 0.

6.3.9.2.6 Sleep and Poor Source Comparators

The sleep comparator is used to suspend the converter if the adapter voltage is insufficient to maintain buck converter operation while charging the battery. If V_{VBUS} falls below V_{BAT} + V_{SLEEP}, the converter stops switching, , and VBUS_FAULT_STAT and VBUS_FAULT_FLAG are set to 1. If V_{VBUS} rises back above V_{BAT} + V_{SLEEPZ}, the converter restarts.

If V_{VBUS} falls below $V_{POORSRC}$, the converter stops switching (if not already suspended and high due to the sleep comparator), and the VBUS_STAT transitions to 000 and the device transitions to battery-only mode. If V_{VBUS} rises above $V_{POORSRC}$, it is a new adapter attach, and poor source qualification will be run. VBUS_STAT will be determined by the adapter attach sequence as outlined in Section 6.3.3.

6.3.9.3 Thermal Regulation and Thermal Shutdown

6.3.9.3.1 Thermal Protection in Buck Mode

The BQ21128 monitors the internal junction temperature T_J to avoid overheating the chip and limits the IC junction temperature in buck mode. When the internal junction temperature exceeds the T_{REG} thermal regulation limit (TREG register configuration), the device lowers the charging current. During thermal regulation, the safety timer runs at half the clock rate, and the TREG_FLAG and TREG_STAT bits are set to 1. Additionally, the device has thermal shutdown to turn off the converter and BATFET when IC junction temperature exceeds T_{SHUT}. The fault bit TSHUT_FLAG is set to 1 and TSHUT_STAT transitions to 1. The BATFET and converter are re-enabled when IC temperature is T_{SHUT_HYS} below T_{SHUT}, and TSHUT_STAT transitions to 0.

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6.3.9.3.2 Thermal Protection in Battery-only Mode

BQ21128 monitors the internal junction temperature T_J to avoid overheating the chip and limits the IC junction temperature in battery-only mode. The device has thermal shutdown to turn off the BATFET when IC junction temperature exceeds T_{SHUT}. The fault bit TSHUT_FLAG is set to 1 and TSHUT_STAT transitions to 1. The BATFET is re-enabled when IC temperature is T_{SHUT_HYS} below T_{SHUT}, and TSHUT_STAT transitions to 0.

6.4 Device Functional Modes

6.4.1 Host Mode and Default Mode

The device is a host controlled charger, but it can operate in default mode without host management. In default mode, the device can be used as an autonomous charger with no host or while host is in sleep mode. When the charger is in default mode, WD_STAT bit becomes HIGH, WD_FLAG is set to 1, and an $\overline{INT}$ is asserted low to alert the host (unless masked by WD_MASK). The WD_FLAG bit would read as 1 upon the first read and then 0 upon subsequent reads. When the charger is in host mode, WD_STAT bit is LOW.

After power-on-reset, the device starts in default mode with watchdog timer expired. All the registers are in the default settings.

In default mode, the device keeps charging the battery with default 1-hour trickle charging safety timer, 2-hour pre-charging safety timer and the 12-hour fast charging safety timer. At the end of the 1-hour or 2-hour or 12-hour timer expired, the charging is stopped and the buck converter continues to operate to supply system load.

A write to any I^2C register transitions the charger from default mode to host mode, and initiates the watchdog timer. All the device parameters can be programmed by the host. To keep the device in host mode, the host has to reset the watchdog timer by writing 1 to WD_RST bit before the watchdog timer expires (WD_STAT bit is set), or disable watchdog timer by setting WATCHDOG bits = 00.

When the watchdog expires, the device returns to default mode. The ICHG value is divided in half when the watchdog timer expires, and a number of other fields are reset to their POR default values as shown in the notes column of the register tables in Section 6.6. When watchdog timer expires, WD_STAT and WD_FLAG is set to 1, and an $\overline{INT}$ is asserted low to alert the host (unless masked by WD_MASK).

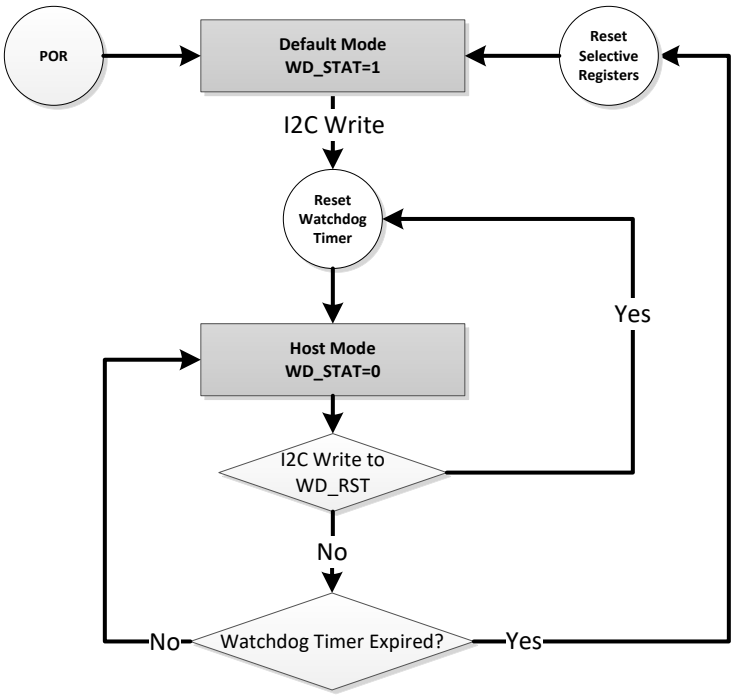


Figure 6-6. Watchdog Timer Flow Chart

6.4.2 Register Bit Reset

Beside the register reset by the watchdog timer in the default mode, the register and the timer could be reset to the default value by writing the REG_RST bit to 1. The register bits, which can be reset by the REG_RST bit, are noted in the Register Map section. After the register reset, the REG_RST bit goes back from 1 to 0 automatically.

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6.5 Programming

6.5.1 Serial Interface

BQ21128 uses I^{2}C compatible interface for flexible charging parameter programming and instantaneous device status reporting. I^{2}C is a bi-directional 2-wire serial interface. Only two open-drain bus lines are required: a serial data line (SDA), and a serial clock line (SCL).

The device has 7-bit I^{2}C address 0x6A , receiving control inputs from a host device such as a micro-controller or digital signal processor through register addresses 0x02 – 0x38. The host device initiates all transfers and the charger responds. Register reads outside of these addresses return 0xFF. When the bus is free, both SDA and SCL lines are HIGH.

The I^{2}C interface supports standard mode (up to 100 kbits/s), fast mode (up to 400 kbits/s) and fast mode plus (up to 1 Mbits/s.) These lines are pulled up to a reference voltage via pull-up resistor. The device I^{2}C detection thresholds support a communication reference voltage from 1.2 V to 5 V.

6.5.1.1 Data Validity

The data on the SDA line must be stable during the HIGH period of the clock. The HIGH or LOW state of the data line can only change when the clock signal on the SCL line is LOW. One clock pulse is generated for each data bit transferred.

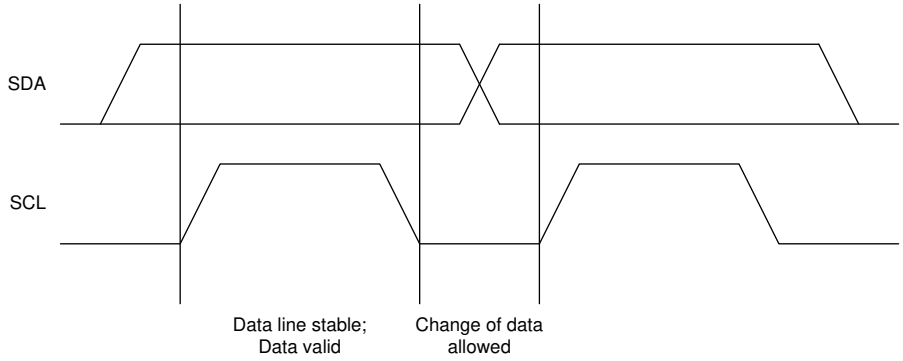


Figure 6-7. Bit Transfer on the I^{2}C Bus

6.5.1.2 START and STOP Conditions

All transactions begin with a START (S) and are terminated with a STOP (P). A HIGH to LOW transition on the SDA line while SCL is HIGH defines a START condition. A LOW to HIGH transition on the SDA line when the SCL is HIGH defines a STOP condition.

START and STOP conditions are always generated by the host. The bus is considered busy after the START condition, and free after the STOP condition.

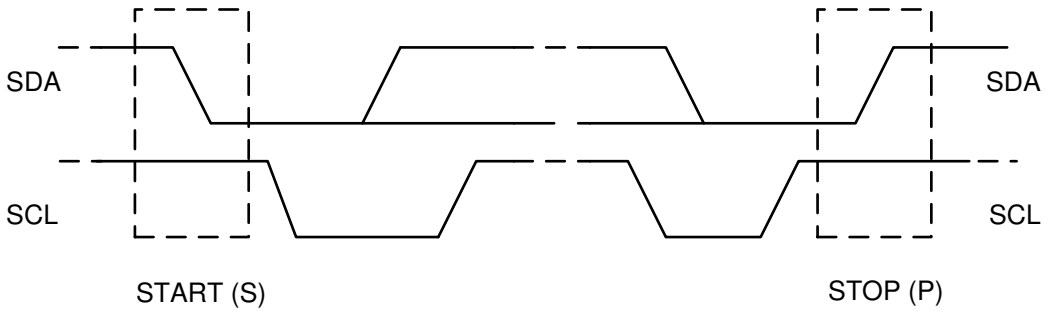


Figure 6-8. START and STOP Conditions on the I^{2}C Bus

6.5.1.3 Byte Format

Every byte on the SDA line must be 8 bits long. The number of bytes to be transmitted per transfer is unrestricted. Each byte has to be followed by an ACKNOWLEDGE (ACK) bit. Data is transferred with the Most Significant Bit (MSB) first. If target cannot receive or transmit another complete byte of data until it has performed some other function, it can hold the SCL line low to force the host into a wait state (clock stretching). Data transfer then continues when the target is ready for another byte of data and releases the SCL line.

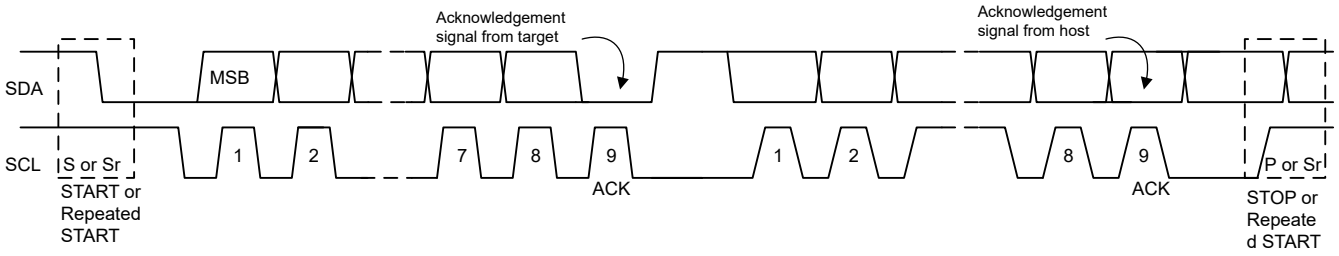


Figure 6-9. Data Transfer on the I²C Bus

6.5.1.4 Acknowledge (ACK) and Not Acknowledge (NACK)

The ACK signaling takes place after each transmitted byte. The ACK bit allows the target to signal the host that the byte was successfully received and another byte may be sent. All clock pulses, including the acknowledge 9th clock pulse, are generated by the host.

The host releases the SDA line during the acknowledge clock pulse so the target can pull the SDA line LOW and it remains stable LOW during the HIGH period of this 9th clock pulse.

A NACK is signaled when the SDA line remains HIGH during the 9th clock pulse. The host can then generate either a STOP to abort the transfer or a repeated START to start a new transfer.

6.5.1.5 Target Address and Data Direction Bit

After the START signal, a target address is sent. This address is 7 bits long, followed by the 8 bit as a data direction bit (bit R/ $\overline{W}$). A zero indicates a transmission (WRITE) and a one indicates a request for data (READ). The device 7-bit address is defined as

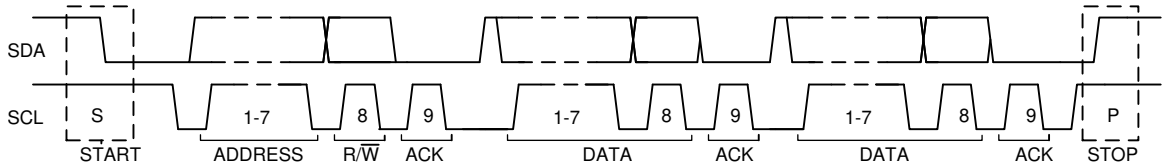


Figure 6-10. Complete Data Transfer on the I²C Bus

6.5.1.6 Single Write and Read

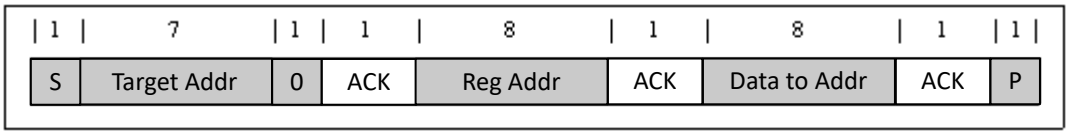


Figure 6-11. Single Write

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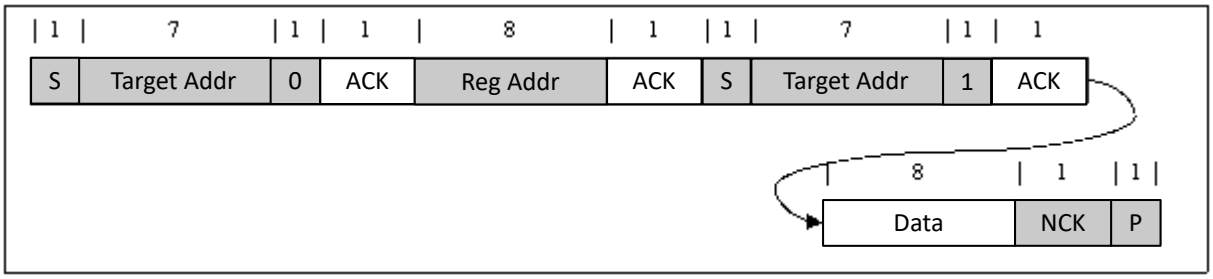


Figure 6-12. Single Read

If the register address is not defined, the charger IC sends back NACK and returns to the idle state.

6.5.1.7 Multi-Write and Multi-Read

The charger device supports multi-byte read and multi-byte write of all registers. These multi-byte operations are allowed to cross register boundaries. For instance, the entire register map may be read in a single operation with a 39-byte read that starts at register address 0x01.

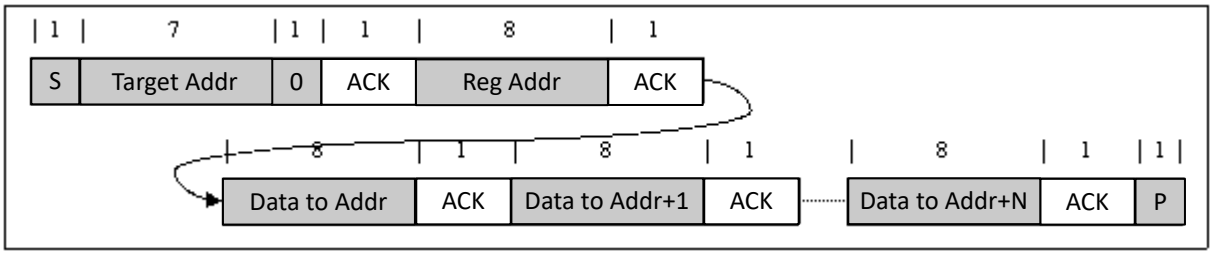


Figure 6-13. Multi-Write

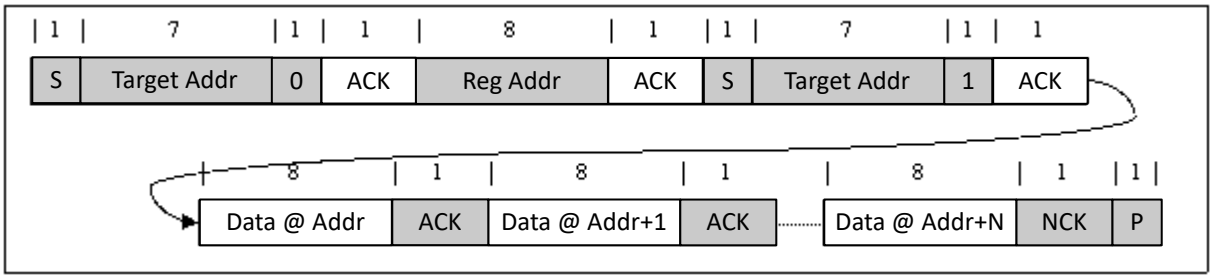


Figure 6-14. Multi-Read

6.6 Register Maps

I^{2}C Device Address: 0x6A.

6.6.1 Register Programming

The BQ21128 contains 8-bit and 16-bit registers. When writing to 16-bit registers, I^{2}C transactions follow the little endian format, starting at the address of the least significant byte and writing both register bytes in a single 16-bit transaction.

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6.6.2 BQ21128 Registers

Table 6-5 lists the memory-mapped registers for the BQ21128 registers. All register offset addresses not listed in Table 6-5 should be considered as reserved locations and the register contents should not be modified.

Table 6-5. BQ21128 Registers

Address	Acronym	Register Name	Section
0x2	REG0x02_Charge_Current_Limit	Charge Current Limit	Go
0x4	REG0x04_Charge_Voltage_Limit	Charge Voltage Limit	Go
0x6	REG0x06_Input_Current_Limit	Input Current Limit	Go
0x8	REG0x08_Input_Voltage_Limit	Input Voltage Limit	Go
0xE	REG0x0E_Minimal_System_Voltage	Minimal System Voltage	Go
0x10	REG0x10_Pre-charge_Control	Pre-charge Control	Go
0x12	REG0x12_Termination_Control	Termination Control	Go
0x14	REG0x14_Charge_Control	Charge Control	Go
0x15	REG0x15_Charge_Timer_Control	Charge Timer Control	Go
0x16	REG0x16_Charger_Control_0	Charger Control 0	Go
0x17	REG0x17_Charger_Control_1	Charger Control 1	Go
0x18	REG0x18_Charger_Control_2	Charger Control 2	Go
0x19	REG0x19_Charger_Control_3	Charger Control 3	Go
0x1A	REG0x1A_NTC_Control_0	NTC Control 0	Go
0x1B	REG0x1B_NTC_Control_1	NTC Control 1	Go
0x1C	REG0x1C_NTC_Control_2	NTC Control 2	Go
0x1D	REG0x1D_Charger_Status_0	Charger Status 0	Go
0x1E	REG0x1E_Charger_Status_1	Charger Status 1	Go
0x1F	REG0x1F_FAULT_Status_0	FAULT Status 0	Go
0x20	REG0x20_Charger_Flag_0	Charger Flag 0	Go
0x21	REG0x21_Charger_Flag_1	Charger Flag 1	Go
0x22	REG0x22_FAULT_Flag_0	FAULT Flag 0	Go
0x23	REG0x23_Charger_Mask_0	Charger Mask 0	Go
0x24	REG0x24_Charger_Mask_1	Charger Mask 1	Go
0x25	REG0x25_FAULT_Mask_0	FAULT Mask 0	Go
0x26	REG0x26_ADC_Control	ADC Control	Go
0x27	REG0x27_ADC_Function_Disable_0	ADC Function Disable 0	Go
0x28	REG0x28_IBUS_ADC	IBUS ADC	Go
0x2A	REG0x2A_IBAT_ADC	IBAT ADC	Go
0x2C	REG0x2C_VBUS_ADC	VBUS ADC	Go
0x2E	REG0x2E_VPMID_ADC	VPMID ADC	Go
0x30	REG0x30_VBAT_ADC	VBAT ADC	Go
0x32	REG0x32_VSYS_ADC	VSYS ADC	Go
0x34	REG0x34_TS_ADC	TS ADC	Go
0x36	REG0x36_TDIE_ADC	TDIE ADC	Go
0x38	REG0x38_Part_Information	Part Information	Go

Complex bit access types are encoded to fit into small table cells. Table 6-6 shows the codes that are used for access types in this section.

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Table 6-6. BQ21128 Access Type Codes

Access Type	Code	Description
Read Type		
R	R	Read
Write Type		
W	W	Write
Reset or Default Value		
-n		Value after reset or the default value

6.6.2.1 REG0x02_Charge_Current_Limit Register (Address = 0x2) [Reset = X]

REG0x02_Charge_Current_Limit is shown in Figure 6-15 and described in Table 6-7.

Return to the Summary Table.

Charge Current Limit

Figure 6-15. REG0x02_Charge_Current_Limit Register

15	14	13	12	11	10	9	8
RESERVED					ICHG		
R-0x0					R/W-X		
7	6	5	4	3	2	1	0
ICHG			RESERVED				
R/W-X			R-0x0				

Table 6-7. REG0x02_Charge_Current_Limit Register Field Descriptions

Bit	Field	Type	Reset	Notes	Description
15:12	RESERVED	R	0x0		Reserved
11:5	ICHG	R/W	0x8	WATCHDOG Timer Expiration sets ICHG to 1/2 its previous value (rounded down) Reset by: REG_RESET	Charge Current Regulation Limit: This 16-bit register follows the little-endian convention. ICHG[5:3] falls in REG0x03[2:0], and ICHG[2:0] falls in REG0x02[7:5]. POR: 320mA (8h) Range: 40mA-1520mA (1h-26h) Clamped Low Clamped High Bit Step: 40mA (1h) NOTE: When Q4_FULLON=1, this register has a minimum value of 80mA
4:0	RESERVED	R	0x0		Reserved

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6.6.2.2 REG0x04_Charge_Voltage_Limit Register (Address = 0x4) [Reset = 0x0D20]

REG0x04_Charge_Voltage_Limit is shown in Figure 6-16 and described in Table 6-8.

Return to the Summary Table.

Charge Voltage Limit

Figure 6-16. REG0x04_Charge_Voltage_Limit Register

15	14	13	12	11	10	9	8
RESERVED				VREG			
R-0x0				R/W-0x1A4			
7	6	5	4	3	2	1	0
VREG					RESERVED		
R/W-0x1A4					R-0x0		

Table 6-8. REG0x04_Charge_Voltage_Limit Register Field Descriptions

Bit	Field	Type	Reset	Notes	Description
15:12	RESERVED	R	0x0		Reserved
11:3	VREG	R/W	0x1A4	Reset by: REG_RESET	Battery Voltage Regulation Limit: This 16-bit register follows the little-endian convention. VREG[8:5] falls in REG0x05[3:0], and VREG[4:0] falls in REG0x04[7:3]. POR: 4200mV (1A4h) Range: 3500mV-4800mV (15Eh-1E0h) Clamped Low Clamped High Bit Step: 10mV
2:0	RESERVED	R	0x0		Reserved

6.6.2.3 REG0x06_Input_Current_Limit Register (Address = 0x6) [Reset = 0x0A00]

REG0x06_Input_Current_Limit is shown in Figure 6-17 and described in Table 6-9.

Return to the Summary Table.

Input Current Limit

Figure 6-17. REG0x06_Input_Current_Limit Register

15	14	13	12	11	10	9	8
RESERVED				IINDPM			
R-0x0				R/W-0xA0			
7	6	5	4	3	2	1	0
IINDPM				RESERVED			
R/W-0xA0				R-0x0			

Table 6-9. REG0x06_Input_Current_Limit Register Field Descriptions

Bit	Field	Type	Reset	Notes	Description
15:12	RESERVED	R	0x0		Reserved
11:4	IINDPM	R/W	0xA0	Reset by: REG_RESET Adapter Removal	Input Current Regulation Limit: This 16-bit register follows the little-endian convention. IINDPM[7:4] falls in REG0x07[3:0], and IINDPM[3:0] falls in REG0x06[7:4]. POR: 3200mA (A0h) Range: 100mA-3200mA (5h-A0h) Clamped Low Clamped High Bit Step: 20mA When the adapter is removed, IINDPM is reset to its POR value of 3.2 A.

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Table 6-9. REG0x06_Input_Current_Limit Register Field Descriptions (continued)

Bit	Field	Type	Reset	Notes	Description
3:0	RESERVED	R	0x0		Reserved

6.6.2.4 REG0x08_Input_Voltage_Limit Register (Address = 0x8) [Reset = 0x0E60]

REG0x08_Input_Voltage_Limit is shown in Figure 6-18 and described in Table 6-10.

Return to the Summary Table.

Input Voltage Limit

Figure 6-18. REG0x08_Input_Voltage_Limit Register

15	14	13	12	11	10	9	8
RESERVED				VINDPM			
R-0x0				R/W-0x73			
7	6	5	4	3	2	1	0
VINDPM				RESERVED			
R/W-0x73				R-0x0			

Table 6-10. REG0x08_Input_Voltage_Limit Register Field Descriptions

Bit	Field	Type	Reset	Notes	Description
15:14	RESERVED	R	0x0		Reserved
13:5	VINDPM	R/W	0x73		Absolute Input Voltage Regulation Limit: This 16-bit register follows the little-endian convention. VINDPM[8:3] falls in REG0x09[5:0], and VINDPM[2:0] falls in REG0x08[7:5]. POR: 4600mV (73h) Range: 3800mV-16800mV (5Fh-1A4h) Clamped Low Clamped High Bit Step: 40mV
4:0	RESERVED	R	0x0		Reserved

6.6.2.6 REG0x0E_Minimal_System_Voltage Register (Address = 0xE) [Reset = 0x0B00]

REG0x0E_Minimal_System_Voltage is shown in Figure 6-19 and described in Table 6-11.

Return to the Summary Table.

Minimal System Voltage

Figure 6-19. REG0x0E_Minimal_System_Voltage Register

15	14	13	12	11	10	9	8
RESERVED				VSYSMIN			
R-0x0				R/W-0x2C			
7	6	5	4	3	2	1	0
VSYSMIN				RESERVED			
R/W-0x2C				R-0x0			

Table 6-11. REG0x0E_Minimal_System_Voltage Register Field Descriptions

Bit	Field	Type	Reset	Notes	Description
15:12	RESERVED	R	0x0		Reserved

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Table 6-11. REG0x0E_Minimal_System_Voltage Register Field Descriptions (continued)

Bit	Field	Type	Reset	Notes	Description
11:6	VSYSMIN	R/W	0x2C	Reset by: REG_RESET	Minimal System Voltage: This 16-bit register follows the little-endian convention. VSYSMIN[5:2] falls in REG0x0F[3:0], and VSYSMIN[1:0] falls in REG0x0E[7:6]. POR: 3520mV (2Ch) Range: 2560mV-3840mV (20h-30h) Clamped Low Clamped High Bit Step: 80mV
5:0	RESERVED	R	0x0		Reserved

6.6.2.7 REG0x10_Pre-charge_Control Register (Address = 0x10) [Reset = 0x0018]

REG0x10_Pre-charge_Control is shown in Figure 6-20 and described in Table 6-12.

Return to the Summary Table.

Pre-charge Control

Figure 6-20. REG0x10_Pre-charge_Control Register

15	14	13	12	11	10	9	8
RESERVED							IPRECHG
R-0x0							R/W-0x3
7	6	5	4	3	2	1	0
IPRECHG						RESERVED	
R/W-0x3						R-0x0	

Table 6-12. REG0x10_Pre-charge_Control Register Field Descriptions

Bit	Field	Type	Reset	Notes	Description
15:9	RESERVED	R	0x0		Reserved
8:3	IPRECHG	R/W	0x3	Reset by: REG_RESET	Pre-charge current regulation limit: This 16-bit register follows the little-endian convention. IPRECHG[4:0] falls in REG0x10[7:3] POR: 30mA (3h) Range: 10mA-310mA (1h-1Fh) Clamped Low Bit Step: 10mA (1h) NOTE: When Q4_FULLON=1, this register has a minimum value of 80mA, so Reset value becomes 80mA in this case
2:0	RESERVED	R	0x0		Reserved

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6.6.2.8 REG0x12_Termination_Control Register (Address = 0x12) [Reset = 0x0010]

REG0x12_Termination_Control is shown in Figure 6-21 and described in Table 6-13.

Return to the Summary Table.

Termination Control

Figure 6-21. REG0x12_Termination_Control Register

15	14	13	12	11	10	9	8
RESERVED							ITERM
R-0x0							R/W-0x4
7	6	5	4	3	2	1	0
ITERM						RESERVED	
R/W-0x4						R-0x0	

Table 6-13. REG0x12_Termination_Control Register Field Descriptions

Bit	Field	Type	Reset	Notes	Description
15:9	RESERVED	R	0x0		Reserved
8:2	ITERM	R/W	0x4	Reset by: REG_RESET	Termination Current Threshold: This 16-bit register follows the little-endian convention. ITERM[5:0] falls in REG0x12[7:2]. POR: 20mA (4h) Range: 5mA-310mA (1h-3Eh) Clamped Low Bit Step: 5mA (1h) NOTE: When Q4_FULLON=1, this register has a minimum value of 60mA, so Reset value becomes 60mA in this case. If ITERM_2X = 1, termination current = 2x ITERM
1:0	RESERVED	R	0x0		Reserved

6.6.2.9 REG0x14_Charge_Control Register (Address = 0x14) [Reset = 0x06]

REG0x14_Charge_Control is shown in Figure 6-22 and described in Table 6-14.

Return to the Summary Table.

Charge Control

Figure 6-22. REG0x14_Charge_Control Register

7	6	5	4	3	2	1	0
Q1_FULLON	Q4_FULLON	ITRICKLE	TOPOFF_TMR		EN_TERM	VINDPM_BAT_TRACK	VRECHG
R/W-0x0	R/W-0x0	R/W-0x0	R/W-0x0		R/W-0x1	R/W-0x1	R/W-0x0

Table 6-14. REG0x14_Charge_Control Register Field Descriptions

Bit	Field	Type	Reset	Notes	Description
7	Q1_FULLON	R/W	0x0		Forces RBFET (Q1) into low resistance state (26 mΩ) , regardless of IINDPM setting. 0x0 = RBFET RDSON determined by IINDPM setting (default) 0x1 = RBFET RDSON is always 26 mΩ
6	Q4_FULLON	R/W	0x0		Forces BATFET (Q4) into low resistance state (15 mΩ), regardless of ICHG setting (Only applies when VBAT > VSYSMIN). 0x0 = BATFET RDSON determined by charge current (default) 0x1 = BATFET RDSON is always 15 mΩ

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Table 6-14. REG0x14_Charge_Control Register Field Descriptions (continued)

Bit	Field	Type	Reset	Notes	Description
5	ITRICKLE	R/W	0x0	Reset by: REG_RESET	Trickle charging current setting: 0b = 10mA (default) 1b = 40mA
4:3	TOPOFF_TMR	R/W	0x0	Reset by: REG_RESET	Top-off timer control: 0x0 = Disabled (default) 0x1 = 17 mins 0x2 = 35 mins 0x3 = 52 mins
2	EN_TERM	R/W	0x1	Reset by: REG_RESET WATCHDOG	Enable termination: 0x0 = Disable 0x1 = Enable (default)
1	VINDPM_BAT_TRACK	R/W	0x1	Reset by: REG_RESET	Sets VINDPM to track BAT voltage. Actual VINDPM is higher of the VINDPM register value and VBAT + VINDPM_BAT_TRACK. 0x0 = Disable function (VINDPM set by register) 0x1 = VBAT + 400 mV (default)
0	VRECHG	R/W	0x0	Reset by: REG_RESET	Battery Recharge Threshold Offset (Below VREG) 0x0 = 100mV (default) 0x1 = 200mV

6.6.2.10 REG0x15_Charge_Timer_Control Register (Address = 0x15) [Reset = 0x5C]

REG0x15_Charge_Timer_Control is shown in Figure 6-23 and described in Table 6-15.

Return to the Summary Table.

Charge Timer Control

Figure 6-23. REG0x15_Charge_Timer_Control Register

7	6	5	4	3	2	1	0
DIS_STAT	RESERVED			TMR2X_EN	EN_SAFETY_TMRS	PRECHG_TMR	CHG_TMR
R/W-0x0	R-0x0			R/W-0x1	R/W-0x1	R/W-0x0	R/W-0x0

Table 6-15. REG0x15_Charge_Timer_Control Register Field Descriptions

Bit	Field	Type	Reset	Notes	Description
7	DIS_STAT	R/W	0x0	Reset by: REG_RESET	Disable the STAT pin output 0x0 = Enable (default) 0x1 = Disable
6-4	RESERVED	R	0x0		Reserved
3	TMR2X_EN	R/W	0x1	Reset by: REG_RESET	2X charging timer control 0x0 = Trickle charge, pre-charge and fast charge timer not slowed by 2X during input DPM or thermal regulation. 0x1 = Trickle charge, pre-charge and fast charge timer slowed by 2X during input DPM or thermal regulation (default)
2	EN_SAFETY_TMRS	R/W	0x1	Reset by: REG_RESET WATCHDOG	Enable fast charge, pre-charge and trickle charge timers 0x0 = Disable 0x1 = Enable (default)
1	PRECHG_TMR	R/W	0x0	Reset by: REG_RESET	Pre-charge safety timer setting 0x0 = 2.5 hrs (default) 0x1 = 0.62 hrs
0	CHG_TMR	R/W	0x0	Reset by: REG_RESET	Fast charge safety timer setting 0x0 = 14.5 hrs (default) 0x1 = 28 hrs

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6.6.2.11 REG0x16_Charger_Control_0 Register (Address = 0x16) [Reset = 0xA1]

REG0x16_Charger_Control_0 is shown in Figure 6-24 and described in Table 6-16.

Return to the Summary Table.

Charger Control 0

Figure 6-24. REG0x16_Charger_Control_0 Register

7	6	5	4	3	2	1	0
EN_AUTO_IBATDIS	FORCE_IBATDIS	EN_CHG	EN_HIZ	FORCE_PMID_DIS	WD_RST	WATCHDOG	
R/W-0x1	R/W-0x0	R/W-0x1	R/W-0x0	R/W-0x0	R/W-0x0	R/W-0x1	

Table 6-16. REG0x16_Charger_Control_0 Register Field Descriptions

Bit	Field	Type	Reset	Notes	Description
7	EN_AUTO_IBATDIS	R/W	0x1	Reset by: REG_RESET	Enable the auto battery discharging during the battery OVP fault 0x0 = The charger does NOT apply a discharging current on BAT during battery OVP triggered 0x1 = The charger does apply a discharging current on BAT during battery OVP triggered (default)
6	FORCE_IBATDIS	R/W	0x0	Reset by: REG_RESET WATCHDOG	Force a battery discharging current (~30mA) 0x0 = IDLE (default) 0x1 = Force the charger to apply a discharging current on BAT
5	EN_CHG	R/W	0x1	Reset by: REG_RESET WATCHDOG	Charger enable configuration 0x0 = Charge Disable 0x1 = Charge Enable (default)
4	EN_HIZ	R/W	0x0	Reset by: REG_RESET WATCHDOG Adapter Plug In	Enable HIZ mode. 0x0 = Disable (default) 0x1 = Enable
3	FORCE_PMID_DIS	R/W	0x0	Reset by: REG_RESET WATCHDOG	Force a PMID discharge current (~30mA.) 0x0 = Disable (default) 0x1 = Enable
2	WD_RST	R/W	0x0	Reset by: REG_RESET	I2C watch dog timer reset 0x0 = Normal (default) 0x1 = Reset (this bit goes back to 0 after timer reset)
1:0	WATCHDOG	R/W	0x1	Reset by: REG_RESET	Watchdog timer setting 0x0 = Disable 0x1 = 50s (default) 0x2 = 100s 0x3 = 200s

6.6.2.12 REG0x17_Charger_Control_1 Register (Address = 0x17) [Reset = 0x4F]

REG0x17_Charger_Control_1 is shown in Figure 6-25 and described in Table 6-17.

Return to the Summary Table.

Charger Control 1

Figure 6-25. REG0x17_Charger_Control_1 Register

7	6	5	4	3	2	1	0
REG_RST	TREG	SET_CONV_FREQ		SET_CONV_STRN		RESERVED	VBUS_OVP
R/W-0x0	R/W-0x1	R/W-0x0		R/W-0x3		R/W-0x1	R/W-0x1

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Table 6-17. REG0x17_Charger_Control_1 Register Field Descriptions

Bit	Field	Type	Reset	Notes	Description
7	REG_RST	R/W	0x0	REG_RESET	Reset registers to default values and reset timer Value resets to 0 after reset completes. 0x0 = Not reset (default) 0x1 = Reset
6	TREG	R/W	0x1	Reset by: REG_RESET	Thermal regulation thresholds. 0x0 = 60C 0x1 = 120C (default)
5:4	SET_CONV_FREQ	R/W	0x0	Reset by: REG_RESET	Adjust switching frequency of the converter 0x0 = Nominal, 1.5 MHz (default) 0x1 = -10%, 1.35 MHz 0x2 = +10%, 1.65 MHz 0x3 = RESERVED
3:2	SET_CONV_STRN	R/W	0x3	Reset by: REG_RESET	Adjust the high side and low side drive strength of the converter to adjust efficiency versus EMI. 0x0 = weak 0x1 = normal 0x2 = RESERVED 0x3 = strong
1	RESERVED	R/W	0x1		Reserved
0	VBUS_OVP	R/W	0x1	Reset by: REG_RESET	Sets VBUS overvoltage protection threshold 0x0 = 6.3 V 0x1 = 18.5 V (default)

6.6.2.13 REG0x18_Charger_Control_2 Register (Address = 0x18) [Reset = 0x04]

REG0x18_Charger_Control_2 is shown in Figure 6-26 and described in Table 6-18.

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Charger Control 2

Figure 6-26. REG0x18_Charger_Control_2 Register

7	6	5	4	3	2	1	0
RESERVED			PFM_FWD_DIS	BATFET_CTRL_WVBUS	BATFET_DLY	BATFET_CTRL	
R-0x0			R/W-0x0	R/W-0x0	R/W-0x1	R/W-0x0	

Table 6-18. REG0x18_Charger_Control_2 Register Field Descriptions

Bit	Field	Type	Reset	Notes	Description
7:5	RESERVED	R	0x0		Reserved
4	PFM_FWD_DIS	R/W	0x0	Reset by: REG_RESET	Disable PFM in forward buck mode 0x0 = Enable (Default) 0x1 = Disable
3	BATFET_CTRL_WVBUS	R/W	0x0		Optionally allows BATFET off or system power reset with adapter present. 0x0 = 0x0 0x1 = 0x1
2	BATFET_DLY	R/W	0x1	Reset by: REG_RESET	Delay time added to the taking action in bits [1:0] of the BATFET_CTRL 0x0 = Add 25 ms delay time 0x1 = Add 12.5s delay time (default)

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Table 6-18. REG0x18_Charger_Control_2 Register Field Descriptions (continued)

Bit	Field	Type	Reset	Notes	Description
1:0	BATFET_CTRL	R/W	0x0	Reset by: REG_RESET	BATFET control The control logic of the BATFET to force the device enter different modes. 0x0 = Normal (default) 0x1 = Shutdown Mode 0x2 = Ship Mode 0x3 = System Power Reset

6.6.2.14 REG0x19_Charger_Control_3 Register (Address = 0x19) [Reset = 0xC4]

REG0x19_Charger_Control_3 is shown in Figure 6-27 and described in Table 6-19.

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Charger Control 3

Figure 6-27. REG0x19_Charger_Control_3 Register

7	6	5	4	3	2	1	0
IBAT_PK		VBAT_UVLO	RESERVED		RESERVED	CHG_RATE	
R/W-0x3		R/W-0x0	R-0x0		R/W-0x1	R/W-0x0	

Table 6-19. REG0x19_Charger_Control_3 Register Field Descriptions

Bit	Field	Type	Reset	Notes	Description
7:6	IBAT_PK	R/W	0x3	Reset by: REG_RESET	Battery discharging peak current protection threshold setting 0x0 = RESERVED 0x1 = RESERVED 0x2 = 6A 0x3 = 12A (default)
5	VBAT_UVLO	R/W	0x0	Reset by: REG_RESET	Select the VBAT_UVLO falling threshold and VBAT_SHORT threshold 0x0 = VBAT_UVLO 2.2V, VBAT_SHORT 2.05V (default) 0x1 = VBAT_UVLO 1.8V, VBAT_SHORT 1.85V
4:3	RESERVED	R	0x0		Reserved
2	EN_EXTILIM	R/W	0x1		Enable the external ILIM pin input current regulation 0x0 = Disabled 0x1 = Enabled (default)
1:0	CHG_RATE	R/W	0x0	Reset by: REG_RESET	The charge rate definition for the fast charge stage. The charging current fold back value is equal to ICHG register setting times the fold back ratio, then divided by the charge rate. 0x0 = 1C (default) 0x1 = 2C 0x2 = 4C 0x3 = 6C

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6.6.2.15 REG0x1A_NTC_Control_0 Register (Address = 0x1A) [Reset = 0x3D]

REG0x1A_NTC_Control_0 is shown in Figure 6-28 and described in Table 6-20.

Return to the Summary Table.

NTC Control 0

Figure 6-28. REG0x1A_NTC_Control_0 Register

7	6	5	4	3	2	1	0
TS_IGNORE	RESERVED			TS_ISET_WARM		TS_ISET_COOL	
R/W-0x0	R-0x0			R/W-0x3		R/W-0x1	

Table 6-20. REG0x1A_NTC_Control_0 Register Field Descriptions

Bit	Field	Type	Reset	Notes	Description
7	TS_IGNORE	R/W	0x0	Reset by: REG_RESET WATCHDOG	Ignore the TS feedback: the charger considers the TS is always good to allow charging, TS_STAT reports TS_NORMAL condition. 0x0 = Not ignore (Default) 0x1 = Ignore
6:4	RESERVED	R	0x0		Reserved
3:2	TS_ISET_WARM	R/W	0x3	Reset by: REG_RESET	TS_WARM Current Setting 0x0 = Charge Suspend 0x1 = Set ICHG to 20% 0x2 = Set ICHG to 40% 0x3 = ICHG unchanged (default)
1:0	TS_ISET_COOL	R/W	0x1	Reset by: REG_RESET	TS_COOL Current Setting 0x0 = Charge Suspend 0x1 = Set ICHG to 20% (default) 0x2 = Set ICHG to 40% 0x3 = ICHG unchanged

6.6.2.16 REG0x1B_NTC_Control_1 Register (Address = 0x1B) [Reset = 0x25]

REG0x1B_NTC_Control_1 is shown in Figure 6-29 and described in Table 6-21.

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NTC Control 1

Figure 6-29. REG0x1B_NTC_Control_1 Register

7	6	5	4	3	2	1	0
TS_TH1_TH2_TH3			TS_TH4_TH5_TH6			TS_VSET_WARM	
R/W-0x1			R/W-0x1			R/W-0x1	

Table 6-21. REG0x1B_NTC_Control_1 Register Field Descriptions

Bit	Field	Type	Reset	Notes	Description
7:5	TS_TH1_TH2_TH3	R/W	0x1	Reset by: REG_RESET	TH1, TH2 and TH3 comparator falling temperature thresholds when a 103AT NTC thermistor is used, RT1=5.24kΩ and RT2=30.31kΩ. 0x0 = TH1 is 0°C, TH2 is 5°C, TH3 is 15°C 0x1 = TH1 is 0°C, TH2 is 10°C, TH3 is 15°C (default) 0x2 = TH1 is 0°C, TH2 is 15°C, TH3 is 20°C 0x3 = TH1 is 0°C, TH2 is 20°C, TH3 20°C 0x4 = TH1 is -5°C, TH2 is 5°C, TH3 is 15°C 0x5 = TH1 is -5°C, TH2 is 10°C, TH3 is 15°C 0x6 = TH1 is -5°C, TH2 is 10°C, TH3 is 20°C 0x7 = TH1 is 0°C, TH2 is 10°C, TH3 is 20°C

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Table 6-21. REG0x1B_NTC_Control_1 Register Field Descriptions (continued)

Bit	Field	Type	Reset	Notes	Description
4:2	TS_TH4_TH5_TH6	R/W	0x1	Reset by: REG_RESET	TH4, TH5 and TH6 comparator rising temperature thresholds when a 103AT NTC thermistor is used, RT1=5.24kΩ and RT2=30.31kΩ. 0x0 = TH4 is 35°C, TH5 is 40°C, TH6 is 60°C 0x1 = TH4 is 35°C, TH5 is 45°C, TH6 is 60°C (default) 0x2 = TH4 is 35°C, TH5 is 50°C, TH6 is 60°C 0x3 = TH4 is 40°C, TH5 is 55°C, TH6 is 60°C 0x4 = TH4 is 35°C, TH5 is 40°C, TH6 is 50°C 0x5 = TH4 is 35°C, TH5 is 45°C, TH6 is 50°C 0x6 = TH4 is 40°C, TH5 is 45°C, TH6 is 60°C 0x7 = TH4 is 40°C, TH5 is 50°C, TH6 is 60°C
1:0	TS_VSET_WARM	R/W	0x1	Reset by: REG_RESET	TS_WARM Voltage Setting 0x0 = Set VREG to VREG-300mV 0x1 = Set VREG to VREG-200mV (default) 0x2 = Set VREG to VREG-100mV 0x3 = VREG unchanged

6.6.2.17 REG0x1C_NTC_Control_2 Register (Address = 0x1C) [Reset = 0x3F]

REG0x1C_NTC_Control_2 is shown in Figure 6-30 and described in Table 6-22.

Return to the Summary Table.

NTC Control 2

Figure 6-30. REG0x1C_NTC_Control_2 Register

7	6	5	4	3	2	1	0
RESERVED	TS_VSET_SYM	TS_VSET_PREWARM		TS_ISET_PREWARM		TS_ISET_PRECOOL	
R-0x0	R/W-0x0	R/W-0x3		R/W-0x3		R/W-0x3	

Table 6-22. REG0x1C_NTC_Control_2 Register Field Descriptions

Bit	Field	Type	Reset	Notes	Description
7	RESERVED	R	0x0		RESERVED
6	TS_VSET_SYM	R/W	0x0	Reset by: REG_RESET	When this bit is set to 0, the voltage regulation for TS_PRECOOL and TS_COOL is unchanged. When this bit is set to 1, TS_PRECOOL uses the TS_VSET_PREWARM setting of TS_PREWARM and TS_COOL uses the TS_VSET_WARM setting of TS_WARM . 0x0 = VREG unchanged (default) 0x1 = TS_COOLx matches TS_WARMx
5:4	TS_VSET_PREWARM	R/W	0x3	Reset by: REG_RESET	Advanced temperature profile voltage setting for TS_PREWARM (TH4 - TH5) 0x0 = Set VREG to VREG-300mV 0x1 = Set VREG to VREG-200mV 0x2 = Set VREG to VREG-100mV 0x3 = VREG unchanged (default)
3:2	TS_ISET_PREWARM	R/W	0x3	Reset by: REG_RESET	Advanced temperature profile current setting for TS_PREWARM zone(TH4 - TH5) 0x0 = Charge Suspend 0x1 = Set ICHG to 20% 0x2 = Set ICHG to 40% 0x3 = ICHG unchanged (default)
1:0	TS_ISET_PRECOOL	R/W	0x3	Reset by: REG_RESET	Advanced temperature profile current setting for TS_PRECOOL zone (TH2 - TH3) 0x0 = Charge Suspend 0x1 = Set ICHG to 20% 0x2 = Set ICHG to 40% 0x3 = ICHG unchanged (default)

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6.6.2.18 REG0x1D_Charger_Status_0 Register (Address = 0x1D) [Reset = 0x00]

REG0x1D_Charger_Status_0 is shown in Figure 6-31 and described in Table 6-23.

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Charger Status 0

Figure 6-31. REG0x1D_Charger_Status_0 Register

7	6	5	4	3	2	1	0
RESERVED	ADC_DONE_STAT	TREG_STAT	VSYS_STAT	IINDPM_STAT	VINDPM_STAT	SAFETY_TMR_STAT	WD_STAT
R-0x0	R-0x0	R-0x0	R-0x0	R-0x0	R-0x0	R-0x0	R-0x0

Table 6-23. REG0x1D_Charger_Status_0 Register Field Descriptions

Bit	Field	Type	Reset	Notes	Description
7	RESERVED	R	0x0		Reserved
6	ADC_DONE_STAT	R	0x0		ADC Conversion Status (in one-shot mode only) Note: Always reads 0 in continuous mode 0x0 = Conversion not complete 0x1 = Conversion complete
5	TREG_STAT	R	0x0		IC Thermal regulation status 0x0 = Normal 0x1 = Device in thermal regulation
4	VSYS_STAT	R	0x0		VSYS Regulation Status (forward mode) 0x0 = Not in VSYSMIN regulation (BAT>VSYSMIN) 0x1 = In VSYSMIN regulation (BAT<VSYSMIN)
3	IINDPM_STAT	R	0x0		In forward mode, indicates that either IINDPM regulation is active or ILIM pin regulation is active 0x0 = Normal 0x1 = In IINDPM/ILIM regulation
2	VINDPM_STAT	R	0x0		VINDPM status (forward mode) 0x0 = Normal 0x1 = In VINDPM regulation
1	SAFETY_TMR_STAT	R	0x0		Fast charge, trickle charge and pre-charge timer status 0x0 = Normal 0x1 = Safety timer expired
0	WD_STAT	R	0x0		I2C watch dog timer status 0x0 = Normal 0x1 = WD timer expired

6.6.2.19 REG0x1E_Charger_Status_1 Register (Address = 0x1E) [Reset = 0x00]

REG0x1E_Charger_Status_1 is shown in Figure 6-32 and described in Table 6-24.

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Charger Status 1

Figure 6-32. REG0x1E_Charger_Status_1 Register

7	6	5	4	3	2	1	0
RESERVED			CHG_STAT		VBUS_STAT		
R-0x0			R-0x0		R-0x0		

Table 6-24. REG0x1E_Charger_Status_1 Register Field Descriptions

Bit	Field	Type	Reset	Notes	Description
7:5	RESERVED	R	0x0		Reserved

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Table 6-24. REG0x1E_Charger_Status_1 Register Field Descriptions (continued)

Bit	Field	Type	Reset	Notes	Description
4:3	CHG_STAT	R	0x0		Charge Status bits 0x0 = Not Charging or Charge Terminated 0x1 = Trickle Charge, Pre-charge or Fast charge (CC mode) 0x2 = Taper Charge (CV mode) 0x3 = Top-off Timer Active Charging
2:0	VBUS_STAT	R	0x0		VBUS status bits 000b = No qualified adapter 100b = Unknown Adapter (default IINDPM setting)

6.6.2.20 REG0x1F_FAULT_Status_0 Register (Address = 0x1F) [Reset = 0x00]

REG0x1F_FAULT_Status_0 is shown in Figure 6-33 and described in Table 6-25.

Return to the Summary Table.

FAULT Status 0

Figure 6-33. REG0x1F_FAULT_Status_0 Register

7	6	5	4	3	2	1	0
VBUS_FAULT_STAT	BAT_FAULT_STAT	SYS_FAULT_STAT	RESERVED	TSHUT_STAT	TS_STAT		
R-0x0	R-0x0	R-0x0	R-0x0	R-0x0	R-0x0		

Table 6-25. REG0x1F_FAULT_Status_0 Register Field Descriptions

Bit	Field	Type	Reset	Notes	Description
7	VBUS_FAULT_STAT	R	0x0		VBUS fault status, VBUS OVP and sleep comparator 0x0 = Normal 0x1 = Device not switching due to over voltage protection or sleep comparator
6	BAT_FAULT_STAT	R	0x0		BAT fault status, IBAT OCP and VBAT OVP 0x0 = Normal 0x1 = Device in battery over current protection or battery overvoltage protection
5	SYS_FAULT_STAT	R	0x0		VSYS under voltage and over voltage status 0x0 = Normal 0x1 = SYS in SYS short circuit or over voltage
4	RESERVED	R	0x0		Reserved
3	TSHUT_STAT	R	0x0		IC temperature shutdown status 0x0 = Normal 0x1 = Device in thermal shutdown protection
2:0	TS_STAT	R	0x0		The TS temperature zone. 0x0 = TS_NORMAL 0x1 = TS_COLD or TS resistor string power rail is not available. 0x2 = TS_HOT 0x3 = TS_COOL 0x4 = TS_WARM 0x5 = TS_PRECOOL 0x6 = TS_PREWARM 0x7 = TS pin bias reference fault

6.6.2.21 REG0x20_Charger_Flag_0 Register (Address = 0x20) [Reset = 0x00]

REG0x20_Charger_Flag_0 is shown in Figure 6-34 and described in Table 6-26.

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Charger Flag 0

Figure 6-34. REG0x20_Charger_Flag_0 Register

7	6	5	4	3	2	1	0
RESERVED	ADC_DONE_FLAG	TREG_FLAG	VSYS_FLAG	IINDPM_FLAG	VINDPM_FLAG	SAFETY_TMR_FLAG	WD_FLAG
R-0x0	R-0x0	R-0x0	R-0x0	R-0x0	R-0x0	R-0x0	R-0x0

Table 6-26. REG0x20_Charger_Flag_0 Register Field Descriptions

Bit	Field	Type	Reset	Notes	Description
7	RESERVED	R	0x0		Reserved
6	ADC_DONE_FLAG	R	0x0		ADC conversion flag (only in one-shot mode) 0x0 = Conversion not completed 0x1 = Conversion completed
5	TREG_FLAG	R	0x0		IC Thermal regulation flag 0x0 = Normal 0x1 = TREG signal rising threshold detected
4	VSYS_FLAG	R	0x0		VSYS min regulation flag 0x0 = Normal 0x1 = Entered or existed VSYS min regulation
3	IINDPM_FLAG	R	0x0		Indicates that either the IINDPM regulation loop or ILIM pin regulation loop has been entered. 0x0 = Normal 0x1 = IINDPM or ILIM regulation signal rising edge detected
2	VINDPM_FLAG	R	0x0		VINDPM flag 0x0 = Normal 0x1 = VINDPM regulation signal rising edge detected
1	SAFETY_TMR_FLAG	R	0x0		Fast charge, trickle charge and pre-charge timer flag 0x0 = Normal 0x1 = Fast charge timer expired rising edge detected
0	WD_FLAG	R	0x0		I2C watchdog timer flag 0x0 = Normal 0x1 = WD timer signal rising edge detected

6.6.2.22 REG0x21_Charger_Flag_1 Register (Address = 0x21) [Reset = 0x00]

REG0x21_Charger_Flag_1 is shown in Figure 6-35 and described in Table 6-27.

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Charger Flag 1

Figure 6-35. REG0x21_Charger_Flag_1 Register

7	6	5	4	3	2	1	0
RESERVED				CHG_FLAG	RESERVED		VBUS_FLAG
R-0x0				R-0x0	R-0x0		R-0x0

Table 6-27. REG0x21_Charger_Flag_1 Register Field Descriptions

Bit	Field	Type	Reset	Notes	Description
7:4	RESERVED	R	0x0		Reserved
3	CHG_FLAG	R	0x0		Charge status flag 0x0 = Normal 0x1 = Charge status changed
2:1	RESERVED	R	0x0		Reserved

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Table 6-27. REG0x21_Charger_Flag_1 Register Field Descriptions (continued)

Bit	Field	Type	Reset	Notes	Description
0	VBUS_FLAG	R	0x0		VBUS status flag 0x0 = Normal 0x1 = VBUS status changed

6.6.2.23 REG0x22_FAULT_Flag_0 Register (Address = 0x22) [Reset = 0x00]

REG0x22_FAULT_Flag_0 is shown in Figure 6-36 and described in Table 6-28.

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FAULT Flag 0

Figure 6-36. REG0x22_FAULT_Flag_0 Register

7	6	5	4	3	2	1	0
VBUS_FAULT_FLAG	BAT_FAULT_FLAG	SYS_FAULT_FLAG	RESERVED	TSHUT_FLAG	RESERVED		TS_FLAG
R-0x0	R-0x0	R-0x0	R-0x0	R-0x0	R-0x0		R-0x0

Table 6-28. REG0x22_FAULT_Flag_0 Register Field Descriptions

Bit	Field	Type	Reset	Notes	Description
7	VBUS_FAULT_FLAG	R	0x0		VBUS over-voltage or sleep flag 0x0 = Normal 0x1 = Entered VBUS OVP or sleep
6	BAT_FAULT_FLAG	R	0x0		IBAT over-current and VBAT over-voltage flag 0x0 = Normal 0x1 = Entered battery discharged OCP or VBAT OVP
5	SYS_FAULT_FLAG	R	0x0		VSYS over voltage and SYS short flag 0x0 = Normal 0x1 = Stopped switching due to system over-voltage or SYS short fault
4	RESERVED	R	0x0		Reserved
3	TSHUT_FLAG	R	0x0		IC thermal shutdown flag 0x0 = Normal 0x1 = TS shutdown signal rising threshold detected
2:1	RESERVED	R	0x0		Reserved
0	TS_FLAG	R	0x0		TS status flag 0x0 = Normal 0x1 = A change to TS status was detected

6.6.2.24 REG0x23_Charger_Mask_0 Register (Address = 0x23) [Reset = 0x00]

REG0x23_Charger_Mask_0 is shown in Figure 6-37 and described in Table 6-29.

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Charger Mask 0

Figure 6-37. REG0x23_Charger_Mask_0 Register

7	6	5	4	3	2	1	0
RESERVED	ADC_DONE_MASK	TREG_MASK	VSYS_MASK	IINDPM_MASK	VINDPM_MASK	SAFETY_TMR_MASK	WD_MASK
R/W-0x0	R/W-0x0	R/W-0x0	R/W-0x0	R/W-0x0	R/W-0x0	R/W-0x0	R/W-0x0

Table 6-29. REG0x23_Charger_Mask_0 Register Field Descriptions

Bit	Field	Type	Reset	Notes	Description
7	RESERVED	R/W	0x0		Reserved

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Table 6-29. REG0x23_Charger_Mask_0 Register Field Descriptions (continued)

Bit	Field	Type	Reset	Notes	Description
6	ADC_DONE_MASK	R/W	0x0	Reset by: REG_RESET	ADC conversion mask flag (only in one-shot mode) 0x0 = ADC conversion done does produce $\overline{INT}$ pulse 0x1 = ADC conversion done does not produce $\overline{INT}$ pulse
5	TREG_MASK	R/W	0x0	Reset by: REG_RESET	IC thermal regulation mask flag 0x0 = Entering TREG does produce $\overline{INT}$ 0x1 = Entering TREG does not produce $\overline{INT}$
4	VSYS_MASK	R/W	0x0	Reset by: REG_RESET	VSYS min regulation mask flag 0x0 = Enter or exit VSYSMIN regulation does produce $\overline{INT}$ pulse 0x1 = Enter or exit VSYSMIN regulation does not produce $\overline{INT}$ pulse
3	IINDPM_MASK	R/W	0x0	Reset by: REG_RESET	IINDPM or ILIM mask 0x0 = Enter IINDPM or ILIM does produce $\overline{INT}$ pulse 0x1 = Enter IINDPM or ILIM does not produce $\overline{INT}$ pulse
2	VINDPM_MASK	R/W	0x0	Reset by: REG_RESET	VINDPM mask 0x0 = Enter VINDPM does produce $\overline{INT}$ pulse 0x1 = Enter VINDPM does not produce $\overline{INT}$ pulse
1	SAFETY_TMR_MASK	R/W	0x0	Reset by: REG_RESET	Fast charge, trickle charge and pre-charge timer mask flag 0x0 = Fast charge, trickle charge or pre-charge timer expiration does produce $\overline{INT}$ 0x1 = Fast charge, trickle charge or pre-charge timer expiration does not produce $\overline{INT}$
0	WD_MASK	R/W	0x0	Reset by: REG_RESET	I2C watch dog timer mask 0x0 = I2C watch dog timer expired does produce $\overline{INT}$ pulse 0x1 = I2C watch dog timer expired does not produce $\overline{INT}$ pulse

6.6.2.25 REG0x24_Charger_Mask_1 Register (Address = 0x24) [Reset = 0x00]

REG0x24_Charger_Mask_1 is shown in Figure 6-38 and described in Table 6-30.

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Charger Mask 1

Figure 6-38. REG0x24_Charger_Mask_1 Register

7	6	5	4	3	2	1	0
RESERVED				CHG_MASK	RESERVED		VBUS_MASK
R-0x0				R/W-0x0	R-0x0		R/W-0x0

Table 6-30. REG0x24_Charger_Mask_1 Register Field Descriptions

Bit	Field	Type	Reset	Notes	Description
7:4	RESERVED	R	0x0		Reserved
3	CHG_MASK	R/W	0x0	Reset by: REG_RESET	Charge status mask flag 0x0 = Charging status change does produce $\overline{INT}$ 0x1 = Charging status change does not produce $\overline{INT}$
2:1	RESERVED	R	0x0		Reserved
0	VBUS_MASK	R/W	0x0	Reset by: REG_RESET	VBUS status mask flag 0x0 = VBUS status change does produce $\overline{INT}$ 0x1 = VBUS status change does not produce $\overline{INT}$

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6.6.2.26 REG0x25_FAULT_Mask_0 Register (Address = 0x25) [Reset = 0x00]

REG0x25_FAULT_Mask_0 is shown in Figure 6-39 and described in Table 6-31.

Return to the Summary Table.

FAULT Mask 0

Figure 6-39. REG0x25_FAULT_Mask_0 Register

7	6	5	4	3	2	1	0
VBUS_FAULT_MASK	BAT_FAULT_MASK	SYS_FAULT_MASK	RESERVED	TSHUT_MASK	RESERVED		TS_MASK
R/W-0x0	R/W-0x0	R/W-0x0	R/W-0x0	R/W-0x0	R-0x0		R/W-0x0

Table 6-31. REG0x25_FAULT_Mask_0 Register Field Descriptions

Bit	Field	Type	Reset	Notes	Description
7	VBUS_FAULT_MASK	R/W	0x0	Reset by: REG_RESET	VBUS over-voltage and sleep comparator mask flag 0x0 = Entering VBUS OVP or sleep does produce $\overline{INT}$ 0x1 = Entering VBUS OVP or sleep does not produce $\overline{INT}$
6	BAT_FAULT_MASK	R/W	0x0	Reset by: REG_RESET	IBAT over current and VBAT overvoltage mask flag 0x0 = IBAT OCP fault or VBAT OVP fault does produce $\overline{INT}$ 0x1 = Neither IBAT OCP fault nor VBAT OVP fault produces $\overline{INT}$
5	SYS_FAULT_MASK	R/W	0x0	Reset by: REG_RESET	SYS over voltage and SYS short mask 0x0 = System over-voltage or SYS short fault does produce $\overline{INT}$ 0x1 = Neither system over voltage nor SYS short fault produces $\overline{INT}$
4	RESERVED	R/W	0x0		Reserved
3	TSHUT_MASK	R/W	0x0	Reset by: REG_RESET	IC thermal shutdown mask flag 0x0 = TSHUT does produce $\overline{INT}$ 0x1 = TSHUT does not produce $\overline{INT}$
2:1	RESERVED	R	0x0		Reserved
0	TS_MASK	R/W	0x0	Reset by: REG_RESET	Temperature charging profile interrupt mask 0x0 = A change to TS temperature zone does produce $\overline{INT}$ 0x1 = A change to the TS temperature zone does not produce $\overline{INT}$

6.6.2.27 REG0x26_ADC_Control Register (Address = 0x26) [Reset = 0x30]

REG0x26_ADC_Control is shown in Figure 6-40 and described in Table 6-32.

Return to the Summary Table.

ADC Control

Figure 6-40. REG0x26_ADC_Control Register

7	6	5	4	3	2	1	0
ADC_EN	ADC_RATE	ADC_SAMPLE		ADC_AVG	ADC_AVG_INIT	EN_OOA	ITERM_2X
R/W-0x0	R/W-0x0	R/W-0x3		R/W-0x0	R/W-0x0	R/W-0x0	R/W-0x0

Table 6-32. REG0x26_ADC_Control Register Field Descriptions

Bit	Field	Type	Reset	Notes	Description
7	ADC_EN	R/W	0x0	Reset by: REG_RESET WATCHDOG	ADC Control The registers POR to all 0 's, then after that always retain the last measurement, and never clear. 0x0 = Disable (default) 0x1 = Enable

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Table 6-32. REG0x26_ADC_Control Register Field Descriptions (continued)

Bit	Field	Type	Reset	Notes	Description
6	ADC_RATE	R/W	0x0	Reset by: REG_RESET	ADC conversion rate control 0x0 = Continuous conversion (default) 0x1 = One shot conversion
5:4	ADC_SAMPLE	R/W	0x3	Reset by: REG_RESET	ADC sample speed 0x0 = 12 bit effective resolution 0x1 = 11 bit effective resolution 0x2 = 10 bit effective resolution 0x3 = 9 bit effective resolution (default)
3	ADC_AVG	R/W	0x0	Reset by: REG_RESET	ADC average control 0x0 = Single value (default) 0x1 = Running average
2	ADC_AVG_INIT	R/W	0x0	Reset by: REG_RESET	ADC average initial value control 0x0 = Start average using the existing register value (default) 0x1 = Start average using a new ADC conversion
1	EN_OOA	R/W	0x0	Reset by: REG_RESET	Out of Audio Mode 0x0 = Disable (default) 0x1 = Enable
0	ITERM_2X	R/W	0x0	Reset by: REG_RESET	Termination Current Range multiplier 0x0 = Termination current is ITERM setting (default) 0x1 = Termination current is 2x ITERM setting

6.6.2.28 REG0x27_ADC_Function_Disable_0 Register (Address = 0x27) [Reset = 0x00]

REG0x27_ADC_Function_Disable_0 is shown in Figure 6-41 and described in Table 6-33.

Return to the Summary Table.

ADC Function Disable 0

Figure 6-41. REG0x27_ADC_Function_Disable_0 Register

7	6	5	4	3	2	1	0
IBUS_ADC_DIS	IBAT_ADC_DIS	VBUS_ADC_DIS	VBAT_ADC_DIS	VSYS_ADC_DIS	TS_ADC_DIS	TDIE_ADC_DIS	VPMID_ADC_DIS
R/W-0x0	R/W-0x0	R/W-0x0	R/W-0x0	R/W-0x0	R/W-0x0	R/W-0x0	R/W-0x0

Table 6-33. REG0x27_ADC_Function_Disable_0 Register Field Descriptions

Bit	Field	Type	Reset	Notes	Description
7	IBUS_ADC_DIS	R/W	0x0	Reset by: REG_RESET	IBUS ADC control 0x0 = Enable (Default) 0x1 = Disable
6	IBAT_ADC_DIS	R/W	0x0	Reset by: REG_RESET	IBAT ADC control 0x0 = Enable (Default) 0x1 = Disable
5	VBUS_ADC_DIS	R/W	0x0	Reset by: REG_RESET	VBUS ADC control 0x0 = Enable (Default) 0x1 = Disable
4	VBAT_ADC_DIS	R/W	0x0	Reset by: REG_RESET	VBAT ADC control 0x0 = Enable (Default) 0x1 = Disable
3	VSYS_ADC_DIS	R/W	0x0	Reset by: REG_RESET	VSYS ADC control 0x0 = Enable (Default) 0x1 = Disable
2	TS_ADC_DIS	R/W	0x0	Reset by: REG_RESET	TS ADC control 0x0 = Enable (Default) 0x1 = Disable

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Table 6-33. REG0x27_ADC_Function_Disable_0 Register Field Descriptions (continued)

Bit	Field	Type	Reset	Notes	Description
1	TDIE_ADC_DIS	R/W	0x0	Reset by: REG_RESET	TDIE ADC control 0x0 = Enable (Default) 0x1 = Disable
0	VPMID_ADC_DIS	R/W	0x0	Reset by: REG_RESET	VPMID ADC control 0x0 = Enable (Default) 0x1 = Disable

6.6.2.29 REG0x28_IBUS_ADC Register (Address = 0x28) [Reset = 0x0000]

REG0x28_IBUS_ADC is shown in Figure 6-42 and described in Table 6-34.

Return to the Summary Table.

IBUS ADC

Figure 6-42. REG0x28_IBUS_ADC Register

15	14	13	12	11	10	9	8
IBUS_ADC							
R-0x0							
7	6	5	4	3	2	1	0
IBUS_ADC							RESERVED
R-0x0							R-0x0

Table 6-34. REG0x28_IBUS_ADC Register Field Descriptions

Bit	Field	Type	Reset	Notes	Description
15:1	IBUS_ADC	R	0x0		IBUS ADC reading Reported in 2 's Complement. When the current is flowing from VBUS to PMID, IBUS ADC reports positive value, and when the current is flowing from PMID to VBUS, IBUS ADC reports negative value. POR: 0mA (0h) Format: 2s Complement Range: -4000mA-4000mA (7830h-7FFFh), (0h-7D0h) Clamped Low Clamped High Bit Step: 2mA
0	RESERVED	R	0x0		Reserved

6.6.2.30 REG0x2A_IBAT_ADC Register (Address = 0x2A) [Reset = 0x0000]

REG0x2A_IBAT_ADC is shown in Figure 6-43 and described in Table 6-35.

Return to the Summary Table.

IBAT ADC

Figure 6-43. REG0x2A_IBAT_ADC Register

15	14	13	12	11	10	9	8
IBAT_ADC							
R-0x0							
7	6	5	4	3	2	1	0
IBAT_ADC						RESERVED	
R-0x0						R-0x0	

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Table 6-35. REG0x2A_IBAT_ADC Register Field Descriptions

Bit	Field	Type	Reset	Notes	Description
15:2	IBAT_ADC	R	0x0		IBAT ADC reading Reported in 2 's Complement. The IBAT ADC reports positive value for the battery charging current, and negative value for the battery discharging current. The IBAT ADC resets to zero when EN_CHG=0. POR: 0mA (0h) Format: 2s Complement Range: -7500mA-4000mA (38ADh-3FFFh), (0h-3E8h) Clamped Low Clamped High Bit Step: 4mA If polarity of battery current changes from charging to discharging or vice-versa during the ADC measurement, the conversion is aborted and the register reports code 0x8000 (which is code 0x2000 for IBAT_ADC field)
1:0	RESERVED	R	0x0		Reserved

6.6.2.31 REG0x2C_VBUS_ADC Register (Address = 0x2C) [Reset = 0x0000]

REG0x2C_VBUS_ADC is shown in Figure 6-44 and described in Table 6-36.

Return to the Summary Table.

VBUS ADC

Figure 6-44. REG0x2C_VBUS_ADC Register

15	14	13	12	11	10	9	8
RESERVED	VBUS_ADC						
R-0x0	R-0x0						
7	6	5	4	3	2	1	0
VBUS_ADC						RESERVED	
R-0x0						R-0x0	

Table 6-36. REG0x2C_VBUS_ADC Register Field Descriptions

Bit	Field	Type	Reset	Notes	Description
15	RESERVED	R	0x0		Reserved
14:2	VBUS_ADC	R	0x0		VBUS ADC reading POR: 0mV (0h) Range: 0mV-18000mV (0h-11B6h) Clamped High Bit Step: 3.97mV
1:0	RESERVED	R	0x0		Reserved

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6.6.2.32 REG0x2E_VPMID_ADC Register (Address = 0x2E) [Reset = 0x0000]

REG0x2E_VPMID_ADC is shown in Figure 6-45 and described in Table 6-37.

Return to the Summary Table.

VPMID ADC

Figure 6-45. REG0x2E_VPMID_ADC Register

15	14	13	12	11	10	9	8
RESERVED	VPMID_ADC						
R-0x0	R-0x0						
7	6	5	4	3	2	1	0
VPMID_ADC						RESERVED	
R-0x0						R-0x0	

Table 6-37. REG0x2E_VPMID_ADC Register Field Descriptions

Bit	Field	Type	Reset	Notes	Description
15	RESERVED	R	0x0		Reserved
14:2	VPMID_ADC	R	0x0		VPMID ADC reading POR: 0mV (0h) Range: 0mV-18000mV (0h-11B6h) Clamped High Bit Step: 3.97mV
1:0	RESERVED	R	0x0		Reserved

6.6.2.33 REG0x30_VBAT_ADC Register (Address = 0x30) [Reset = 0x0000]

REG0x30_VBAT_ADC is shown in Figure 6-46 and described in Table 6-38.

Return to the Summary Table.

VBAT ADC

Figure 6-46. REG0x30_VBAT_ADC Register

15	14	13	12	11	10	9	8
RESERVED			VBAT_ADC				
R-0x0			R-0x0				
7	6	5	4	3	2	1	0
VBAT_ADC							RESERVED
R-0x0							R-0x0

Table 6-38. REG0x30_VBAT_ADC Register Field Descriptions

Bit	Field	Type	Reset	Notes	Description
15:13	RESERVED	R	0x0		Reserved
12:1	VBAT_ADC	R	0x0		VBAT ADC reading POR: 0mV (0h) Range: 0mV-5572mV (0h-AF0h) Clamped High Bit Step: 1.99mV
0	RESERVED	R	0x0		Reserved

6.6.2.34 REG0x32_VSYS_ADC Register (Address = 0x32) [Reset = 0x0000]

REG0x32_VSYS_ADC is shown in Figure 6-47 and described in Table 6-39.

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VSYS ADC

Figure 6-47. REG0x32_VSYS_ADC Register

15	14	13	12	11	10	9	8
RESERVED			VSYS_ADC				
R-0x0			R-0x0				
7	6	5	4	3	2	1	0
VSYS_ADC							RESERVED
R-0x0							R-0x0

Table 6-39. REG0x32_VSYS_ADC Register Field Descriptions

Bit	Field	Type	Reset	Notes	Description
15:13	RESERVED	R	0x0		Reserved
12:1	VSYS_ADC	R	0x0		VSYS ADC reading POR: 0mV (0h) Range: 0mV-5572mV (0h-AF0h) Clamped High Bit Step: 1.99mV
0	RESERVED	R	0x0		Reserved

6.6.2.35 REG0x34_TS_ADC Register (Address = 0x34) [Reset = 0x0000]

REG0x34_TS_ADC is shown in Figure 6-48 and described in Table 6-40.

Return to the Summary Table.

TS ADC

Figure 6-48. REG0x34_TS_ADC Register

15	14	13	12	11	10	9	8
RESERVED				TS_ADC			
R-0x0				R-0x0			
7	6	5	4	3	2	1	0
TS_ADC							
R-0x0							

Table 6-40. REG0x34_TS_ADC Register Field Descriptions

Bit	Field	Type	Reset	Notes	Description
15:12	RESERVED	R	0x0		Reserved
11:0	TS_ADC	R	0x0		TS ADC reading as TS pin voltage in percentage of bias reference. Valid with TS pin bias reference active. POR: 0%(0h) Range: 0% - 98.3103% (0h-3FFh) Clamped High Bit Step: 0.0961%

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6.6.2.36 REG0x36_TDIE_ADC Register (Address = 0x36) [Reset = 0x0000]

REG0x36_TDIE_ADC is shown in Figure 6-49 and described in Table 6-41.

Return to the Summary Table.

TDIE ADC

Figure 6-49. REG0x36_TDIE_ADC Register

15	14	13	12	11	10	9	8
RESERVED				TDIE_ADC			
R-0x0				R/W-0x0			
7	6	5	4	3	2	1	0
TDIE_ADC							
R/W-0x0							

Table 6-41. REG0x36_TDIE_ADC Register Field Descriptions

Bit	Field	Type	Reset	Notes	Description
15:12	RESERVED	R	0x0		Reserved
11:0	TDIE_ADC	R/W	0x0		TDIE ADC reading Reported in 2 's Complement. POR: 0°C(0h) Format: 2s Complement Range: -40°C - 140°C (FB0h-118h) Clamped Low Clamped High Bit Step: 0.5°C

6.6.2.37 REG0x38_Part_Information Register (Address = 0x38) [Reset = 0x13]

REG0x38_Part_Information is shown in Figure 6-50 and described in Table 6-42.

Return to the Summary Table.

Part Information

Figure 6-50. REG0x38_Part_Information Register

7	6	5	4	3	2	1	0
RESERVED			PN			DEV_REV	
R-0x0			R-0x2			R-0x3	

Table 6-42. REG0x38_Part_Information Register Field Descriptions

Bit	Field	Type	Reset	Notes	Description
7:6	RESERVED	R	0x0		Reserved
5:3	PN	R	0x2		Device Part number 2h = BQ21128
2:0	DEV_REV	R	0x3		Device Revision

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7 Application and Implementation

Note

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes, as well as validating and testing their design implementation to confirm system functionality.

7.1 Application Information

A typical application consists of the device configured as an I²C controlled power path management device and a single cell battery charger for Li-Ion and Li-polymer batteries used in a wide range of smartphone and other portable devices. It integrates an input reverse-block FET (RBFET, Q1), high-side switching FET (HSFET, Q2), low-side switching FET (LSFET, Q3), and battery FET (BATFET Q4) between the system and battery. The device also integrates a bootstrap diode for the high-side gate drive.

7.2 Typical Application

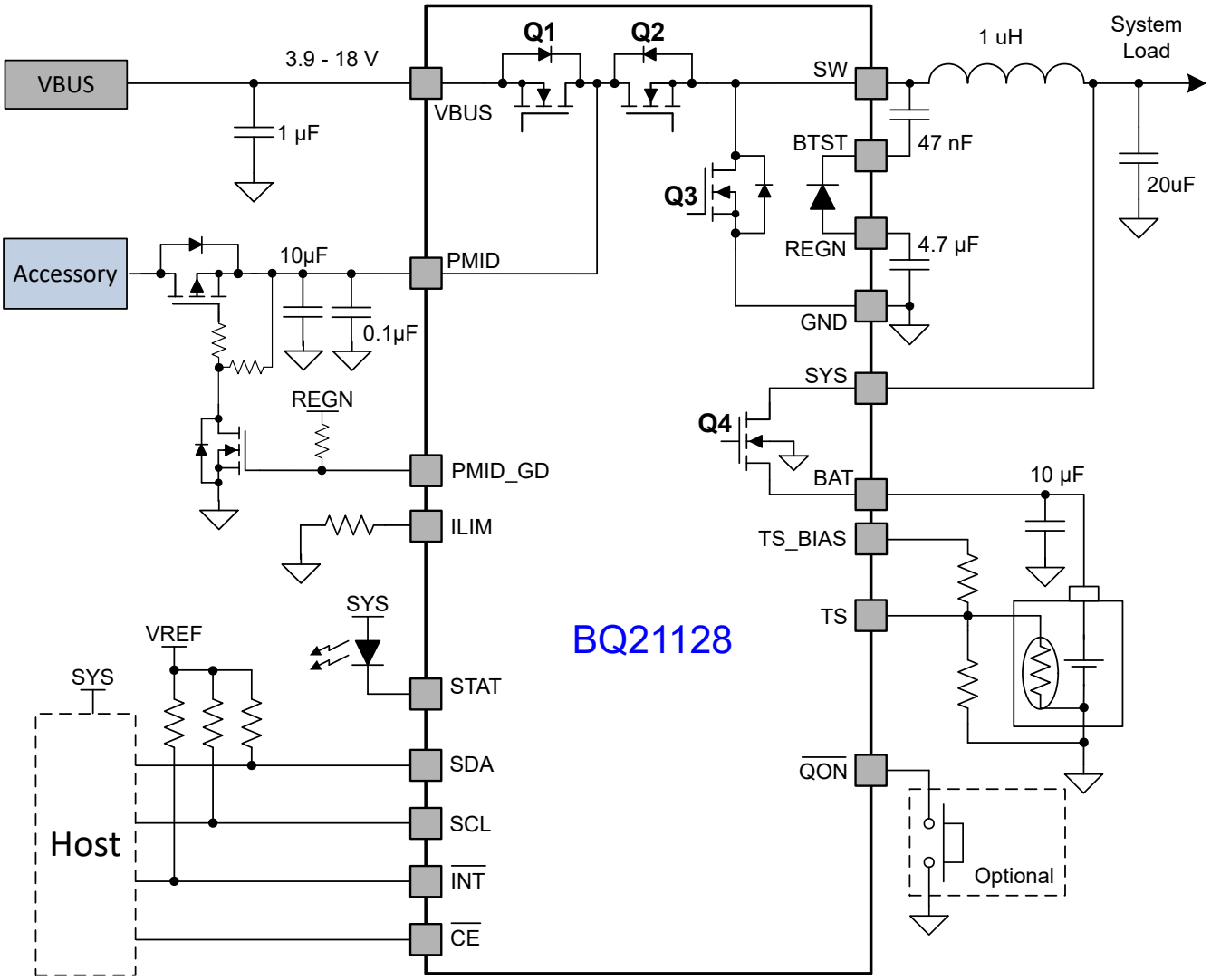


Figure 7-1. BQ21128 Application Diagram

7.2.1 Design Requirements

Table 7-1. Design Requirements

PARAMETER	VALUE
VBUS range	3.9 -18.0 V
Input current limit (REG0x06-0x07)	3200 mA
Fast charge current (REG0x02-0x03)	320 mA
Minimum system voltage (REG0x0E-0x0F)	3520 mV
Battery regulation voltage (REG0x04-0x05)	4200 mV

7.2.2 Detailed Design Procedure

7.2.2.1 Inductor Selection

The 1.5-MHz switching frequency allows the use of small inductor and capacitor values to maintain an inductor saturation current higher than the charging current (I_{CHG}) plus half the ripple current (I_{RIPPLE}):

$$I_{SAT} \geq I_{CHG} + (1/2)\, I_{RIPPLE} \tag{4}$$

The inductor ripple current depends on the input voltage (V_{VBUS}), the duty cycle ($D = V_{BAT}/V_{VBUS}$), the switching frequency (f_S) and the inductance (L).

$$I_{RIPPLE} = \frac{V_{IN} \times D \times (1 - D)}{fs \times L} \tag{5}$$

The maximum inductor ripple current occurs when the duty cycle (D) is approximately 0.5. Usually inductor ripple is designed between 20% and 40% of the maximum charging current as a trade-off between inductor size and efficiency.

7.2.2.2 Input Capacitor

Design input capacitance to provide enough ripple current rating to absorb input switching ripple current. The worst case RMS ripple current is half of the charging current when duty cycle is 0.5. If the converter does not operate at 50% duty cycle, then the worst case capacitor RMS current I_{Cin} occurs where the duty cycle is closest to 50% and can be estimated using Equation 6.

$$I_{CIN} = I_{CHG} \times \sqrt{D \times (1 - D)} \tag{6}$$

Low ESR ceramic capacitor such as X7R or X5R is preferred for input decoupling capacitor and should be placed as close as possible to the drain of the high-side MOSFET (PMID) and source of the low-side MOSFET (GND). Voltage rating of the capacitor must be higher than normal input voltage level. A rating of 25-V or higher capacitor is preferred for 15 V input voltage. 10-µF ceramic capacitor is suggested for typical of charging current.

7.2.2.3 Output Capacitor

Ensure that the output capacitance has enough ripple current rating to absorb the output switching ripple current. Equation 7 shows the output capacitor RMS current I_{COUT} calculation.

$$I_{COUT} = \frac{I_{RIPPLE}}{2 \times \sqrt{3}} \approx 0.29 \times I_{RIPPLE} \tag{7}$$

The output capacitor voltage ripple can be calculated as follows:

$$\Delta V_{SYS} = \frac{V_{SYS}}{8 \times L \times C_{SYS} \times f_{SW}^2}\left(1 - \frac{V_{SYS}}{V_{VBUS}}\right) \tag{8}$$

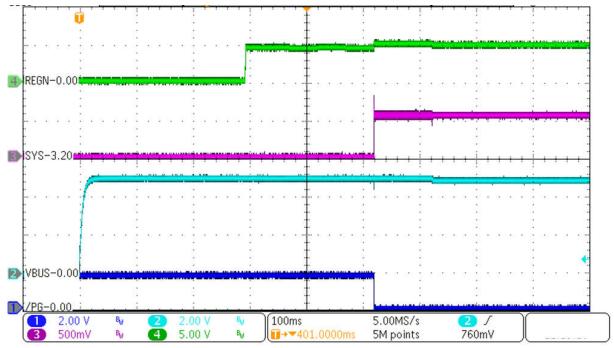
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At certain input and output voltage and switching frequency, the voltage ripple can be reduced by increasing the output filter LC.

The charger device has internal loop compensation optimized for ≥ 10-µF ceramic output capacitor. The preferred ceramic capacitor is 10-V rating, X7R or X5R.

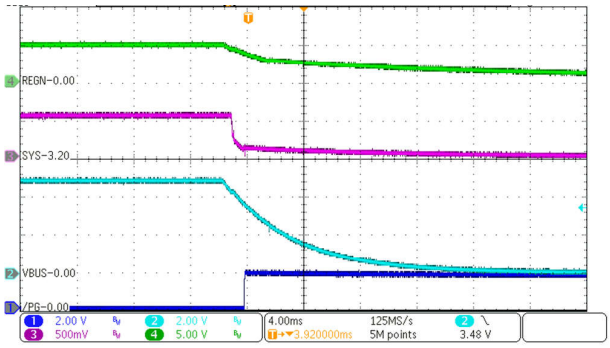
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7.2.3 Application Curves



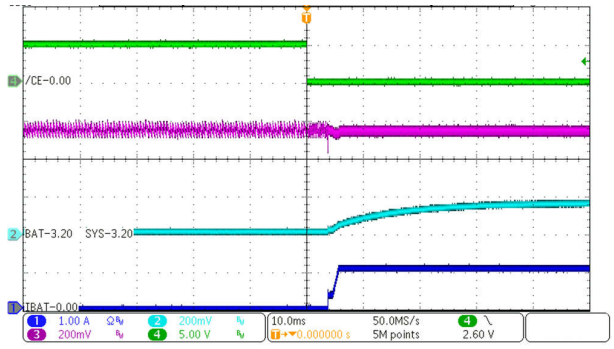
V_{VBUS} = 5 V
I_{CHG} = 1 A
V_{BAT} = 3.2 V

Figure 7-2. Power-Up with Charge Enabled



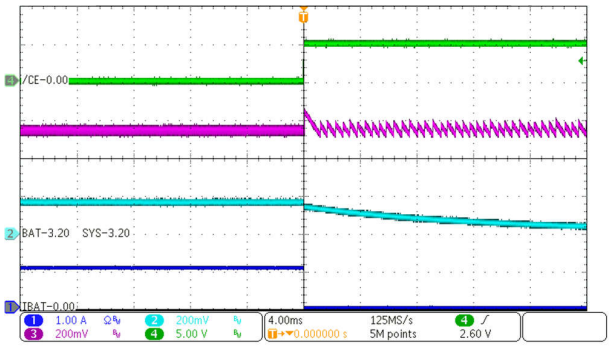
V_{VBUS} = 5 V
I_{CHG} = 1 A
V_{BAT} = 3.2 V

Figure 7-3. Power-Down



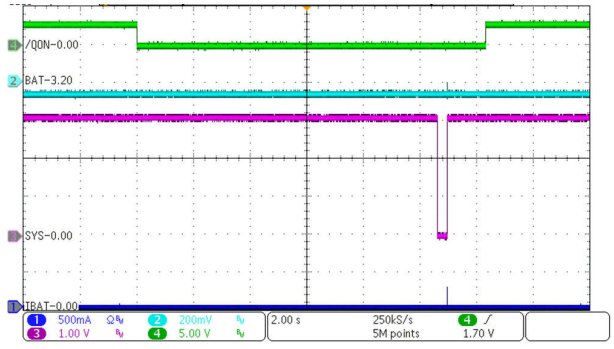
V_{VBUS} = 5 V
I_{CHG} = 1 A
V_{BAT} = 3.2 V

Figure 7-4. Charge Enable



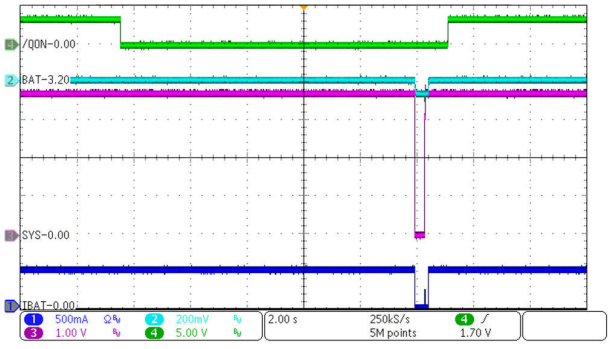
V_{VBUS} = 5 V
I_{CHG} = 1 A
V_{BAT} = 3.2 V

Figure 7-5. Charge Disable



V_{BAT} = 3.2 V

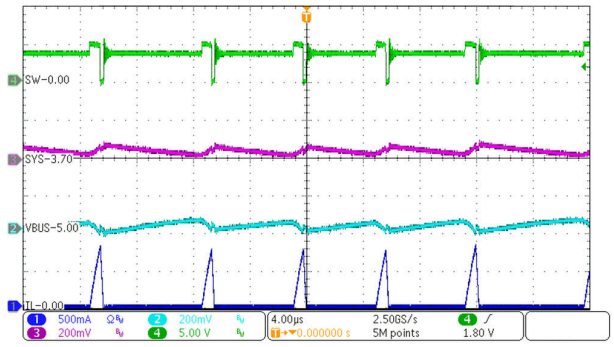
Figure 7-6. System Reset by QON without VBUS Present



V_{VBUS} = 5 V
I_{CHG} = 480mA
V_{BAT} = 3.2 V

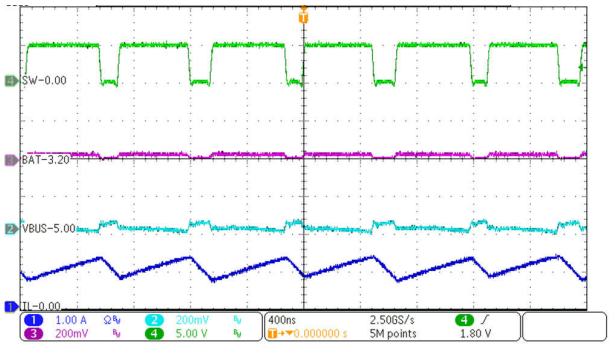
Figure 7-7. System Reset by QON with VBUS Present

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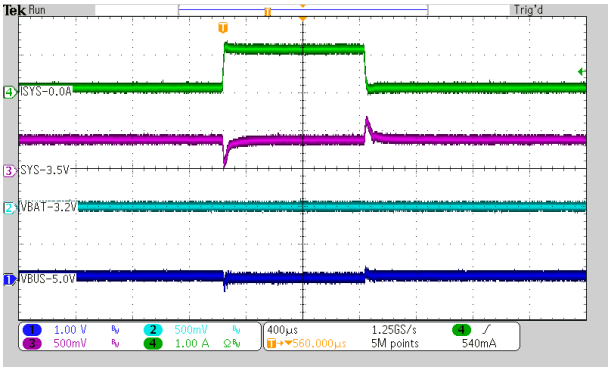
V_{VBUS} = 5 V
I_{SYS} = 50 mA
V_{BAT} = 3.2 V
Charge Disabled

Figure 7-8. PFM Switching in Buck Mode



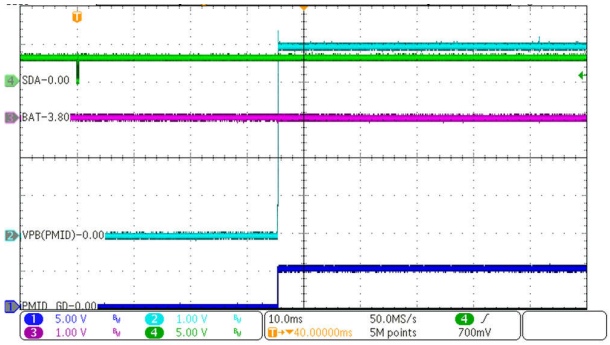
V_{VBUS} = 5 V
I_{CHG} = 1 A
V_{VBAT} = 3.2 V

Figure 7-9. PWM Switching in Buck Mode



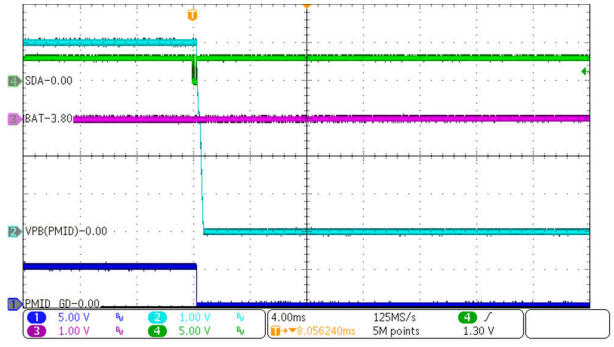
V_{VBUS} = 5 V
I_{SYS} from 0 A to 1 A
V_{BAT} = 3.2 V
Charge Disabled

Figure 7-10. System Load Transient



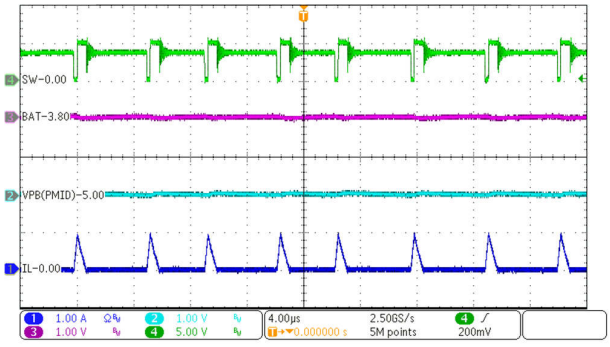
V_{BAT} = 3.8 V
I_{BOOST} = 100 mA
V_{BOOST} = 5.04 V

Figure 7-11. Boost Mode Power Up



V_{BAT} = 3.8 V
I_{BOOST} = 100 mA
V_{BOOST} = 5.04V

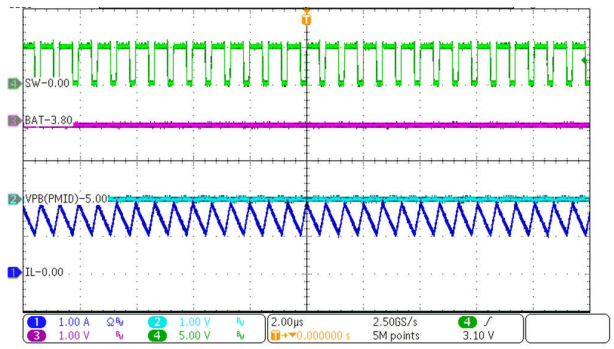
Figure 7-12. Boost Mode Power Down



V_{BAT} = 3.8 V
I_{BOOST}= 50 mA
V_{BOOST} = 5.04V

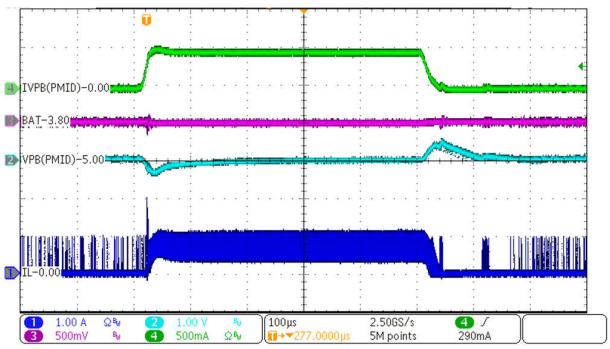
Figure 7-13. PFM Switching in Boost Mode

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V_{BAT} = 3.8 V V_{BOOST} = 5.04V
I_{BOOST}= 1 A

Figure 7-14. PWM Switching in Boost Mode



V_{BAT} = 3.8 V V_{BOOST} = 5.04 V
I_{BOOST} from 5 mA to 500 mA

Figure 7-15. Boost Mode Load Transient

7.3 Power Supply Recommendations

In order to provide an output voltage on SYS, the device requires a power supply between 3.9 V and 18.0 V input with at least 100-mA current rating connected to VBUS or a single-cell Li-Ion battery with voltage > $V_{BATUVLO}$ connected to BAT.

7.4 Layout

7.4.1 Layout Guidelines

The switching node rise and fall times should be minimized for lowest switching loss. Proper layout of the components to minimize high frequency current path loop (see Figure 7-16) is important to prevent electrical and magnetic field radiation and high frequency resonant problems. Follow this specific order carefully to achieve the proper layout.

1. For lowest switching noise during forward/charge mode, place the decoupling capacitor CPMID1 and then bulk capacitor CPMID2 positive terminals as close as possible to PMID pin. Place the capacitor ground terminal close to the GND pin using the shortest copper trace connection or GND plane on the same layer as the IC. See Figure 7-17.
2. For lowest switching noise during reverse mode, place the CSYS1 and CSYS2 output capacitors' positive terminals near the SYS pin. The capacitors' ground terminals must be via'd down through multiple vias to an all ground internal layer that returns to IC GND pin through multiple vias under the IC. See Figure 7-17.
3. Since REGN powers the internal gate drivers, place the CREGN capacitor positive terminal close to REGN pin to minimize switching noise. The capacitor's ground terminal must be via'd down through multiple vias to an all ground internal layer that returns to IC GND pin through multiple vias under the IC. See Figure 7-17.
4. Place the CVBUS and CBAT capacitors positive terminals as close to the VBUS and BAT pins as possible. The capacitors' ground terminals must be via'd down through multiple vias to an all ground internal layer that returns to IC GND pin through multiple vias under the IC. See Figure 7-17.
5. Place the inductor input pin near the positive terminal of the SYS pin capacitors. Due to the PMID capacitor placement requirements, the inductor's switching node terminal must be via'd down with multiple via's to a second internal layer with a wide trace that returns to the SW pin with multiple vias. See Figure 7-18. Using multiple vias ensures that the via's additional resistance is negligible compared to the inductor's dc resistance and therefore does not impact efficiency. The vias additional series inductance is negligible compared to the inductor's inductance.
6. Place the BTST capacitor on the opposite side from the IC using vias to connect to the BTST pin and SW node. See Figure 7-19.
7. A separate analog GND plane for non-power related resistors and capacitors is not required if those components are placed away from the power components traces and planes.
8. Ensure that the I2C SDA and SCL lines are routed away from the SW node.

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Additionally, it is important that the PCB footprint and solder mask for BQ21128 cover the entire length of each of the pins. GND, SW, PMID, SYS and BAT pins extend further into the package than the other pins. Using the entire length of these pins reduces parasitic resistance and increases thermal conductivity from the package into the board.

7.4.2 Layout Example

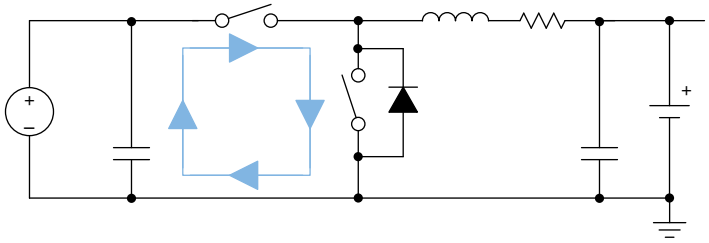


Figure 7-16. High Frequency Current Path

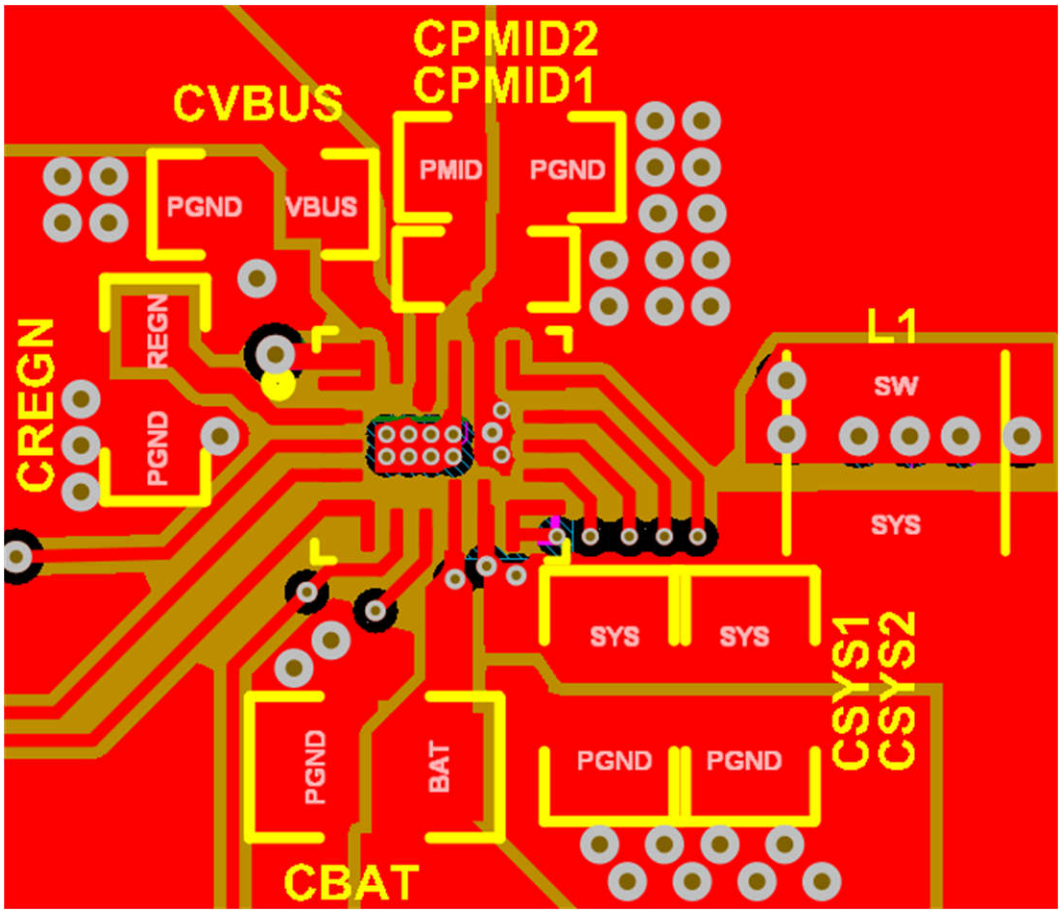


Figure 7-17. Layout Example: Top Layer (red) and All PGND Internal Layer 2 (brown)

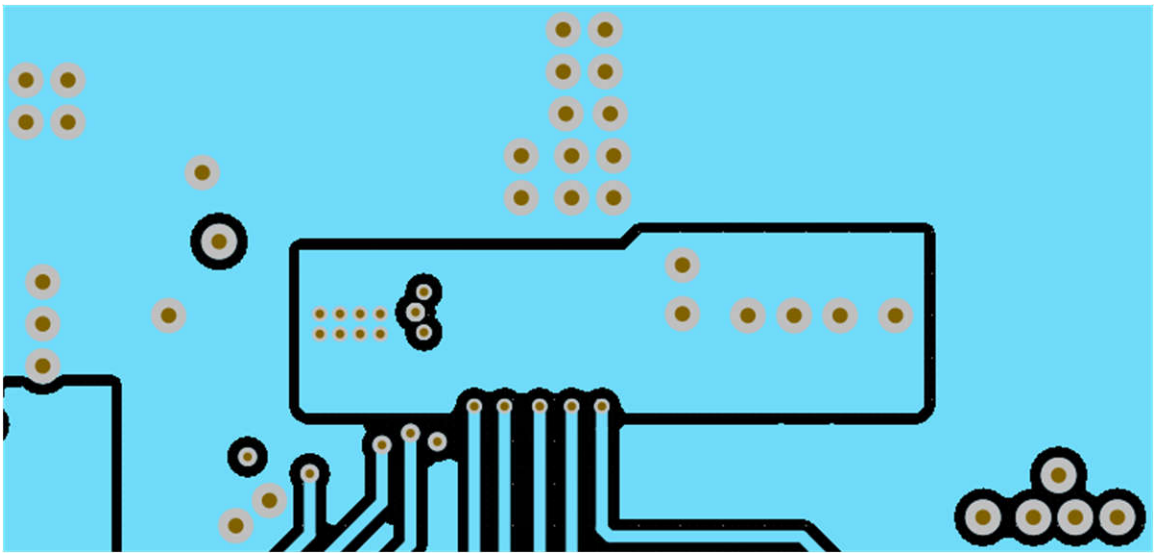


Figure 7-18. Layout Example: Inner Layer 3 (AGND pour; SW node pour; signal routing)

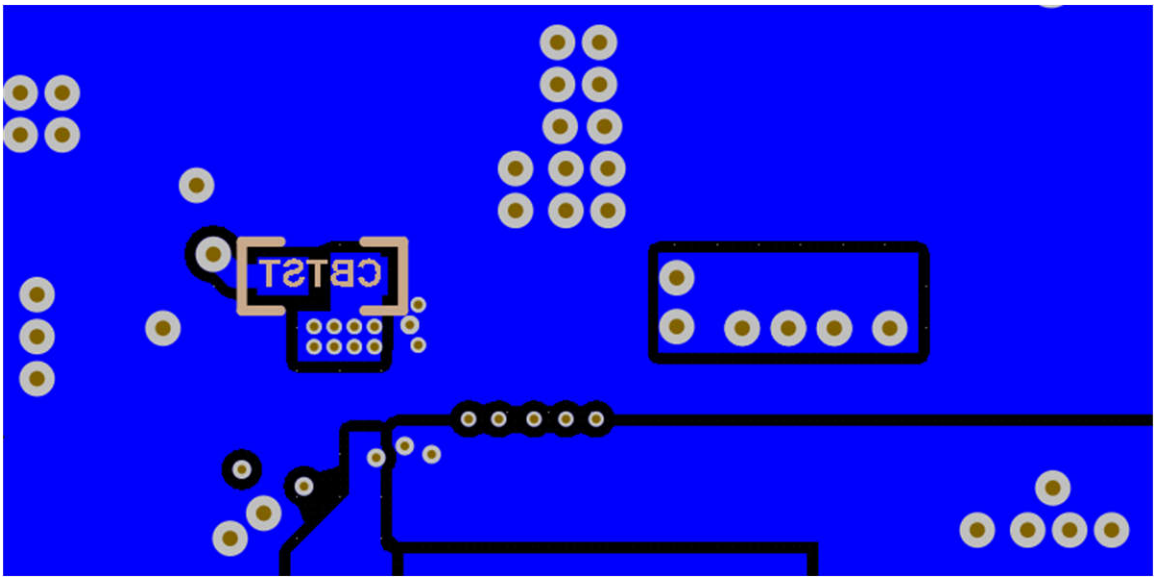


Figure 7-19. Layout Example: Bottom Layer X-Ray From Top (PGND pour; BTST capacitor; redundant SW, SYS and BAT pours)

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8 Device and Documentation Support

8.1 Device Support

8.1.1 Third-Party Products Disclaimer

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8.2 Documentation Support

8.2.1 Related Documentation

For related documentation see the following:

- *BQ25601 and BQ25601D (PWR877) Evaluation Module User's Guide*

8.3 Receiving Notification of Documentation Updates

To receive notification of documentation updates, navigate to the device product folder on ti.com. Click on *Notifications* to register and receive a weekly digest of any product information that has changed. For change details, review the revision history included in any revised document.

8.4 Support Resources

TI E2E™ support forums are an engineer's go-to source for fast, verified answers and design help — straight from the experts. Search existing answers or ask your own question to get the quick design help you need.

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8.6 Electrostatic Discharge Caution



This integrated circuit can be damaged by ESD. Texas Instruments recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

8.7 Glossary

TI Glossary	This glossary lists and explains terms, acronyms, and definitions.

9 Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

DATE	REVISION	NOTES
April 2026	*	Initial Release

10 Mechanical, Packaging, and Orderable Information

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.

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10.1 Package Option Addendum

10.1.1 Packaging Information

Orderable Device	Status (1)	Package Type	Package Drawing	Pins	Package Qty	Eco Plan (2)	Lead/Ball Finish(4)	MSL Peak Temp (3)	Op Temp (°C)	Device Marking(5) (6)
BQ21128RYKR	ACTIVE	WQFN-HR	RYK	18	3000				-40 to 85	BQ128

*IQC丝印检查
BQ128

(1) The marketing status values are defined as follows:
ACTIVE: Product device recommended for new designs.
LIFEBUY: TI has announced that the device will be discontinued, and a lifetime-buy period is in effect.
NRND: Not recommended for new designs. Device is in production to support existing customers, but TI does not recommend using this part in a new design.
PRE_PROD Unannounced device, not in production, not available for mass market, nor on the web, samples not available.
PREVIEW: Device has been announced but is not in production. Samples may or may not be available.
OBSOLETE: TI has discontinued the production of the device.

(2) Eco Plan - The planned eco-friendly classification: Pb-Free (RoHS), Pb-Free (RoHS Exempt), or Green (RoHS & no Sb/Br) - please check http://www.ti.com/productcontent for the latest availability information and additional product content details.
TBD: The Pb-Free/Green conversion plan has not been defined.
Pb-Free (RoHS): TI's terms "Lead-Free" or "Pb-Free" mean semiconductor products that are compatible with the current RoHS requirements for all 6 substances, including the requirement that lead not exceed 0.1% by weight in homogeneous materials. Where designed to be soldered at high temperatures, TI Pb-Free products are suitable for use in specified lead-free processes.
Pb-Free (RoHS Exempt): This component has a RoHS exemption for either 1) lead-based flip-chip solder bumps used between the die and package, or 2) lead-based die adhesive used between the die and leadframe. The component is otherwise considered Pb-Free (RoHS compatible) as defined above.
Green (RoHS & no Sb/Br): TI defines "Green" to mean Pb-Free (RoHS compatible), and free of Bromine (Br) and Antimony (Sb) based flame retardants (Br or Sb do not exceed 0.1% by weight in homogeneous material)

(3) MSL, Peak Temp. -- The Moisture Sensitivity Level rating according to the JEDEC industry standard classifications, and peak solder temperature.

(4) Lead/Ball Finish - Orderable Devices may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead/Ball Finish values may wrap to two lines if the finish value exceeds the maximum column width.

(5) There may be additional marking, which relates to the logo, the lot trace code information, or the environmental category on the device

(6) Multiple Device markings will be inside parentheses. Only on Device Marking contained in parentheses and separated by a "~" will appear on a device. If a line is indented then it is a continuation of the previous line and the two combined represent the entire Device Marking for that device.

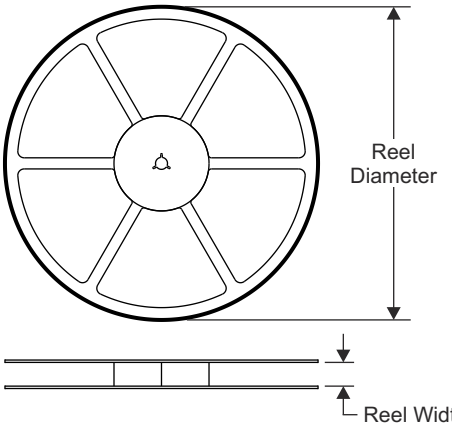
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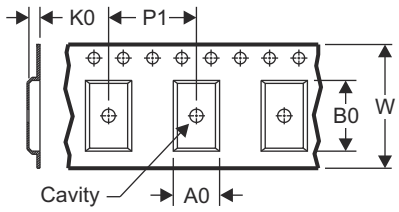
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10.2 Tape and Reel Information

REEL DIMENSIONS

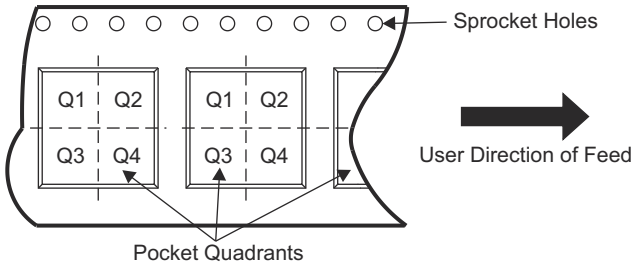


TAPE DIMENSIONS



A0	Dimension designed to accommodate the component width
B0	Dimension designed to accommodate the component length
K0	Dimension designed to accommodate the component thickness
W	Overall width of the carrier tape
P1	Pitch between successive cavity centers

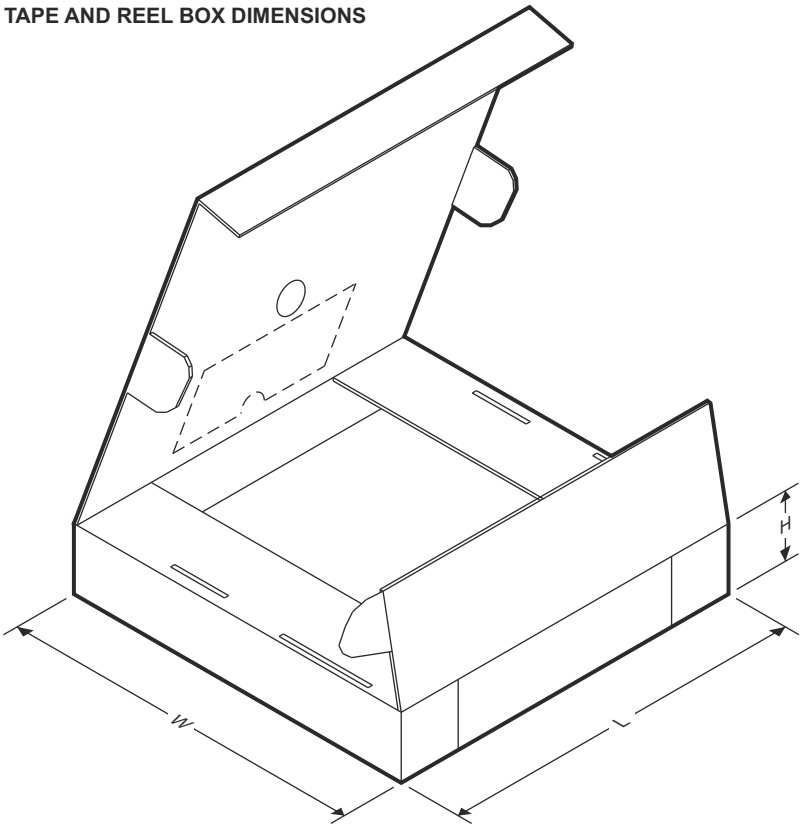
QUADRANT ASSIGNMENTS FOR PIN 1 ORIENTATION IN TAPE



Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
BQ21128RYKR	WQFN-HR	RYK	18	3000	330.0	12.4	5.25	6.25	1.15	8.0	12.0	Q2

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TAPE AND REEL BOX DIMENSIONS



Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
BQ21128RYKR	WQFN-HR	RYK	18	3000	210.0	185.0	35.0

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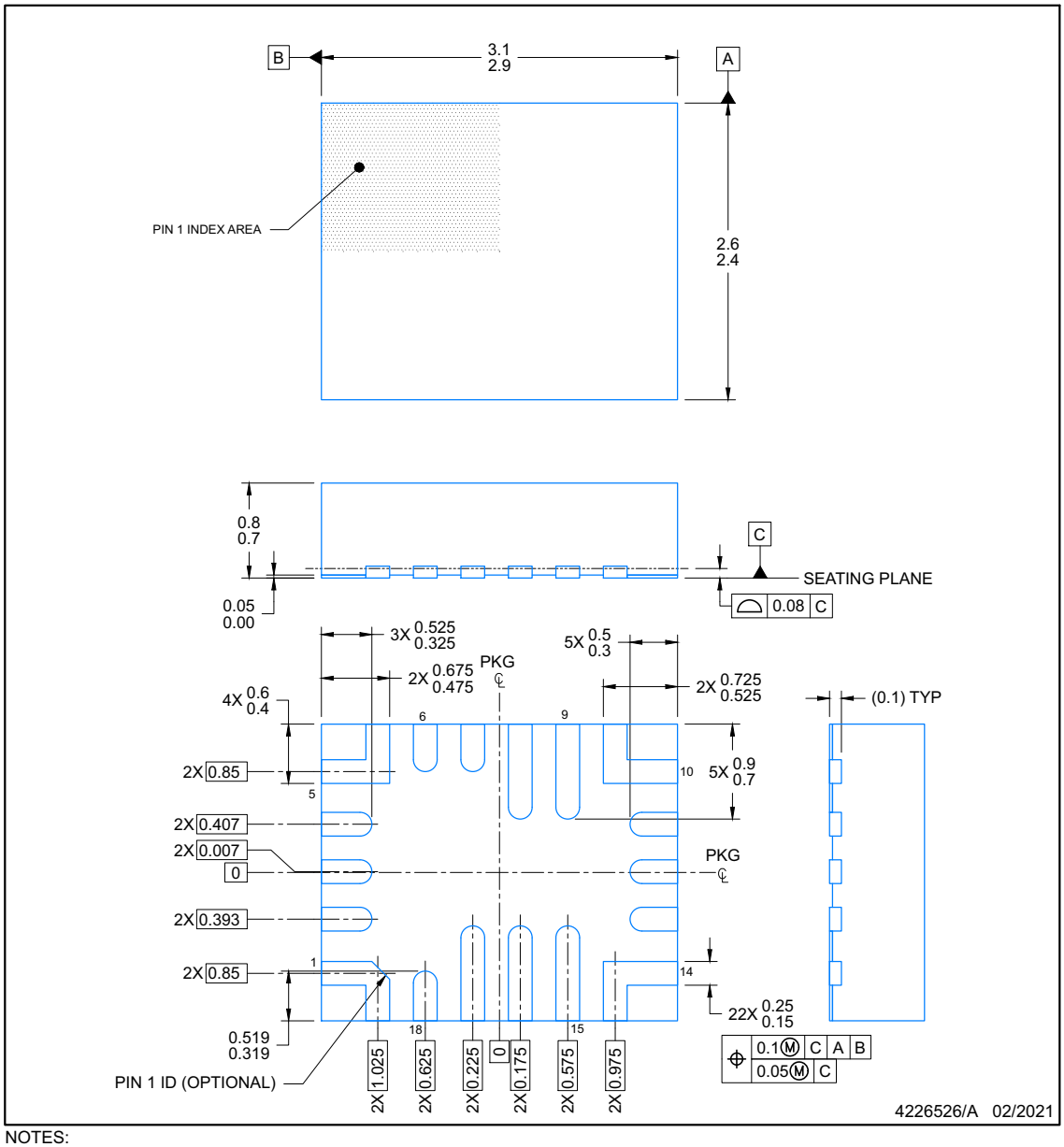
10.3 Mechanical Data

RYK0018A

PACKAGE OUTLINE

WQFN-HR - 0.8 mm max height

PLASTIC QUAD FLATPACK- NO LEAD



NOTES:

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.

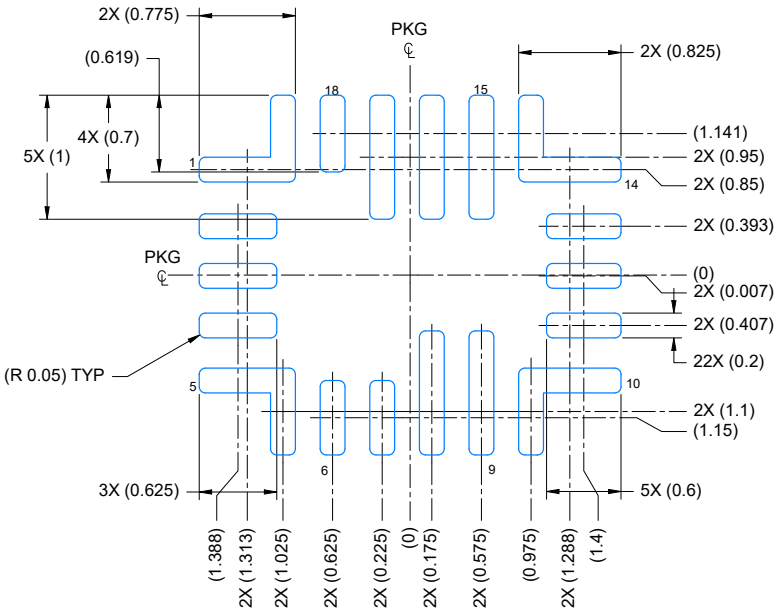
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EXAMPLE BOARD LAYOUT

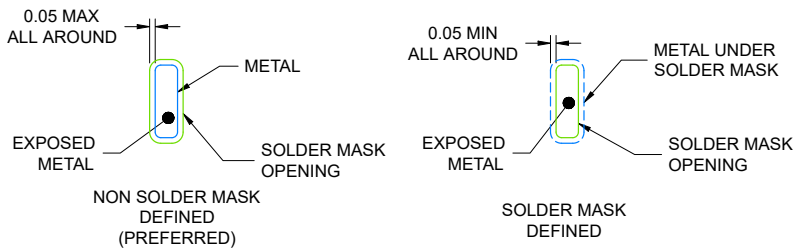
RYK0018A

WQFN-HR - 0.8 mm max height

PLASTIC QUAD FLATPACK- NO LEAD



LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE: 20X



SOLDER MASK DETAILS

4226526/A 02/2021

NOTES: (continued)

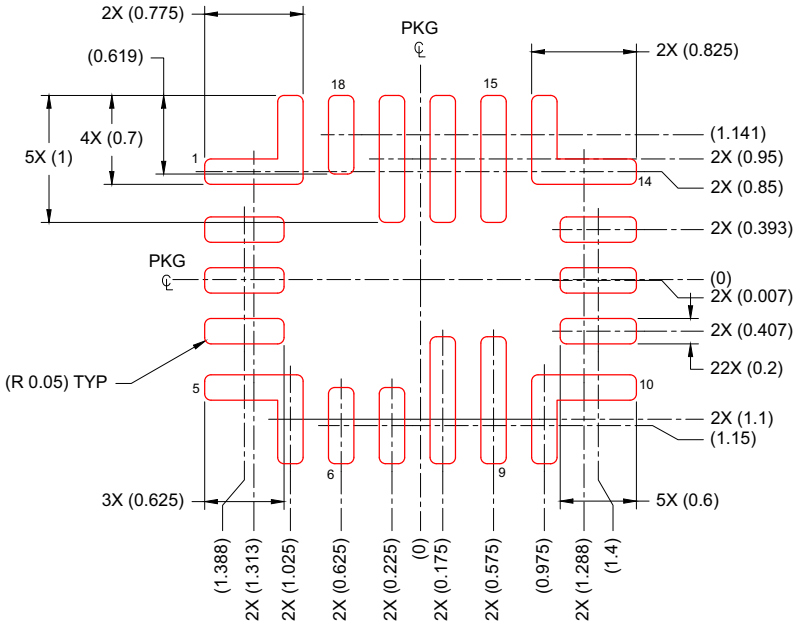
3. For more information, see Texas Instruments literature number SLUA271 (www.ti.com/lit/slua271) .
4. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

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EXAMPLE STENCIL DESIGN

RYK0018A — **WQFN-HR - 0.8 mm max height**

PLASTIC QUAD FLATPACK- NO LEAD



SOLDER PASTE EXAMPLE
BASED ON 0.100 mm THICK STENCIL
SCALE: 20X

4226526/A 02/2021

NOTES: (continued)

5. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.

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